IGNITE MACHINERY RENTALS AND MINING LTD

FEASIBILITY STUDY REPORT FOR THE APPLICATION OF MINING LEASE (ML)

OWORAM GOLD PROJECT

WEST AKIM MUNICIPAL DISTRICT

EASTERN REGION-GHANA

COMPILED&SUBMITTED BY:
IGNITE MACHINERY RENTALS AND MINING LTD

JANUARY, 2026

Application or Lease Registration Code Number:_ Year submitted: 2026

Minerals Commission - Mineral Titles Department

Form MISC: FR - Feasibility Report

O F F I C I A L U S E		**Received at:** ______________________________ **On:** ______________________________ **Authorizing officer:** ______________________________ **Report is confirmed complete:** ______________________ **On:** ______________________________ **Reviewing officer:** ______________________________	

Type of application or lease for which the bankable feasibility is submitted:

- ✓ Application for a Mining Lease
- Application for a Restricted Mining Lease
- Existing Mining Lease
- Existing Restricted Mining Lease

LEASE APPLICANT / HOLDER DETAILS

1. Name of Applicant / Lease Holder: **Ignite Machinery Rentals and Mining Ltd**

2. Address of Company:

3. Residential Address

4. Location of lease area: **West Akim Municipal**

5. Locality: **Oworam**

6. Contact Person:

4. Contact numbers:

7. Email:

COMPILED BY:

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SIGNATURE:

Other Technical Experts Services Sorted in this report:

Abayateye Andrews ……………Senior Exploration Geologist

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Yaw Ankomah……………………. Minerals and Mining Expert

Mary Kukua Tagoe………………. Financial Analyst

EXECUTIVE SUMMARY

IGNITE MACHINERY RENTALS AND MINING LTD is a private registered company in Ghana, seeking to apply for a mining lease for the Oworam concession in the West Akim Municipal of the Eastern Region of Ghana. The concession size is approximately 32.57 sq km (155 Blocks) and can be located on Field No.0501A2. The company has identified both hard rock and alluvial gold zone and intends to develop the potential of gold deposits in this property and its surrounding areas through systematic alluvial exploration methods.

The Oworam project area is situated within the Eastern Region of Ghana. The concession sits within the northwest portion of the West Akim Municipal of the Eastern Region. The West Akim Municipal lies between longitudes 00 25′ West and 00 47′ West and latitudes 500 40′ North and 600.0′ North. It shares boundaries with Denkyembour District to the North, Birim South District to the West; Agona, Awutu-Efutu-Senya and Ga Districts to the South and Suhum Municipal and Upper West Akim District to the East. The Municipal capital, Asamankese, is about 75 km. North-West of Accra.

Access to the project area is convenient, with a well-travelled asphalted road from the Accra-Nsawam Road through notable towns such as Kraboa coltar, Akwadum through Kwaboanta Nkwanta to the Oworam concession.

The land is generally undulating with heights ranging between 60 meters and 460 meters above sea level. The highest point is around the Atewa Range, located between Pabi-Wawase and Asamankese in the Northern part of the Municipality, most of which is occupied by the Atewa Range Extension Forest Reserve. The Municipality is well drained by rivers like Ayensu, Ntoasu, Abukyen, Akora, Supon, Obotwene/Ansing, and Adeiso among others. These rivers flow in the dendritic pattern. Most of the rivers have their sources in the Atewa Range and a few take their sources from the Eastern part. The rivers are mostly perennial due to the double rainfall season which feeds them annually.

Geologically the concession is located within the Kibi-Winneba Belt of the Birimian Supergroup and partly into the Suhum basin. The concession is located within southeastern margin parts of the Kibi area of the belt. The area is characterized by rocks of the Birimian supergroup composed of biotite (-/+hornblende, muscovite) granitoid undifferentiated, volcano sediment upper greenschist to Amphibolite facies with metasomatized granitoid intrusions, basaltic flow and minor interbedded volcaniclastics, two- mica granite and minor granodiorite (Eburnean plutonic suite) and Tamnean plutonic granitoid biotite gneiss.

Through intensive exploration work on the concession, the company's exploration team has delineated various gold anomalous zones to commence its commercial mining activities, and aimed at extracting the approximately **127,965.23 ounces** of gold within the **4,975,128.00 m^3** volume of ore defined in its

8.0 sqkm area at **0.80g/m**^{3} within the AYENSU Grid within the Oworam concession over a period of **7 years** after commencement of production. The company intends to operate **160m^3/h** fixed washing plant capacity.

Table of Contents

LIST OF FIGURES

LIST OF TABLES

TERMS OF REFERENCE

It's worth noting that Ignite Machinery Rentals and Mining Ltd field exploration activities were neither managed or conducted by Explorgeo Ltd. However, as part of their collaboration, Explorgeo received data from Ignite Machinery Rentals and Mining Ltd on various exploration programmes under a Non-Disclosure Agreement (NDA). This data is essential for Explorgeo to perform resource and reserve assessments, a crucial step in the preparation of the feasibility report for the Oworam Gold Project.

To ensure the accuracy and credibility of the data, Explorgeo conducted a verification process, which included field visitations and a general assessment of the delineated mineralized trends. Where possible, Explorgeo Ltd has verified this information from independent sources or implemented methods to validate results. Explorgeo, its staff, and its directors accept no liability for any losses arising from reliance upon the information presented in this Report. These measures are essential to provide a comprehensive and reliable basis for the preparation of the feasibility report.

DISCLAIMER

In the disclosure of information relating to legal, technical data, title and related issues, Explorgeo relied on information provided by Ignite Machinery Rentals and Mining Ltd and disclaims responsibility for such information. The disclosure represents no legal opinion of Explorgeo Ltd nor its representatives. In the disclosure of information relating to socio–political, environment and related issues, Explorgeo has relied on information obtained from public sources such as the Minerals Commission or other related departments of interest.

FORWARD-LOOKING STATEMENTS

This feasibility report was prepared by Ignite Machinery Rentals and Mining Ltd. The report includes certain “forward-looking statements”. forward- looking statements includes words to the effect that the Company or management expects a stated condition or result to occur. Forward-looking statements may be identified by such terms as “Projected,” believes”, “anticipates”, “expects”, “estimates”, “may”, “could”, “would”, “will”, or “plan”.Since forward-looking statements are based on assumptions and address future events and conditions, by their very nature they involve inherent risks and uncertainties. This list is not exhaustive of the factors that may affect any of the Company's forward-looking statements. These and other factors should be considered carefully, and readers should not place undue reliance on the Company’s forward-looking statements.

CHAPTER 1

1.0 INTRODUCTION

Ignite Machinery Rentals and Mining Ltd is applying for mining lease for the Oworam concession located in the West Akim Municipal of the Eastern Region. The concession size is approximately 32.57 km² (155 Blocks) and can be located on Field Sheet Numbers (Nos).0501A2. The company has delineated various alluvial and hard rock gold zones and intends to develop the potential of gold in this property and the surrounding areas through systematic exploration methods.

The concession is located at the northeast margin of the Kibi-Winneba belt of the Birimian. The Kibi – Winneba gold belt lies between the Cape coast basin to the West and the Suhum basin to the East. The belt is known for its gold mineralization, with artisanal workings and mining occurring along the margins of the greenstone belt and other geological contacts. The belt is known to have similar characteristics with the Ashanti gold belt and the closest to the Pan African mobile belt and is composed of volcanic belts and granitic intrusions.

Relevant gold mineralization is characterized by auriferous quartz-carbonate tensional vein arrays or stockworks hosted in either granodiorite, diorite and mafic- to intermediate metavolcanic layers or undifferentiated sills/dykes. This is geologically analogous to other intrusive- or volcanic-hosted gold deposits of Ghana, including Golden Star Wassa Mine in the Ashanti Belt, Kinross Gold's Chirano and Newmont Mining's Subika deposits in the Sefwi gold belt. These tensional vein arrays are generated where the abovementioned competent host lithologies are unable to absorb preferential strain, leading to fracturing and auriferous mineralization. Such sites are typically localized along host-metasediment contacts that define F1 and F2 limbs, fold hinges or apparent flexures. Auriferous vein arrays are penetrative into metasediments, but with a very limited extent. Hydrothermal alteration adjacent to the quartz-carbonate veins is highly variable, but in heavily veined (stockworks) granitoid the assemblage is characterized by moderate to strong quartz, carbonate, chlorite and sericite alteration. This is also associated with patchy to pervasive sulphidisation in the form of disseminated pyrrhotite, pyrite, and arsenopyrite (+/-sphalerite). Variations in alteration intensity in granitoid hosts is reflected in gold grade, with higher intensity alteration leading to higher Au grades, particularly encompassed in pervasive stockworks.

The company has embarked on some pitting and sampling program covering the Oworam concession area to evaluate and determine the grade and size potential of the alluvial resource for the whole concession. Alluvial gold exploration is tricky due to extreme variability of gold grades along the terrace beds and nuggettisation of gold. Free gold bearing gravel beds and soils are artisanally mined by locals; hence successful mechanical mining is always a challenge. Factors such as political and industrial disruption, currency fluctuation and interest rates could have an impact on future operations, and potential revenue streams can also be affected by these factors. Many of these factors are, and will be, beyond the control of any operating entity. Ignite Machinery Rentals and Mining Ltd have adopted a detailed systematic gold investment model for profits and security.

1.1 LOCATION

The Oworam concession is in West Akim Municipal of the Eastern Region of Ghana. The West Akim Municipal lies between longitudes 00 25′ West and 00 47′ West and latitudes 500 40′ North and 600.0′ North. It shares boundaries with Denkyembour District to the North; Birim South District to the West; Agona, Awutu-Efutu-Senya and Ga Districts to the South and Suhum Municipal and Upper West Akim District to the East. The total land area of the Municipality is estimated to be 559 km2. The Municipal capital, Asamankese, is about 75 km. North-West of Accra. The Municipal covers an area of about 559.9sq. km and comprises about 150 settlements including Asamankese (the Municipal Capital), Osenase, Owuram, Brekumanso, and others.

Geologically, it is located within the northern part of the Kibi- Winneba Belt of the Birimian Supergroup at the southeastern margin of the belt around the Kibi area. The concession is approximately 32.57 km² (155 Blocks) in size and can be found on topographic field sheet number 0501A2. The concession is defined by coordinates below: -

Table 1.1: Oworam Concession Pillar Coordinates.

OWURAM PILLAR CO ORDINATES		
PILLAR ID	UTM_Easting	UTM_Northing
P1	758349.9	658990.9
P2	759734.5	658996.8
P3	759749.7	658527.9
P4	766127.2	658579.3
P5	766162.6	657062.7
P6	772234.1	657067.4
P7	772234.4	654905.6
P8	765752.9	654873.7
P9	765742.5	656243.7
P10	758361.5	656225.1

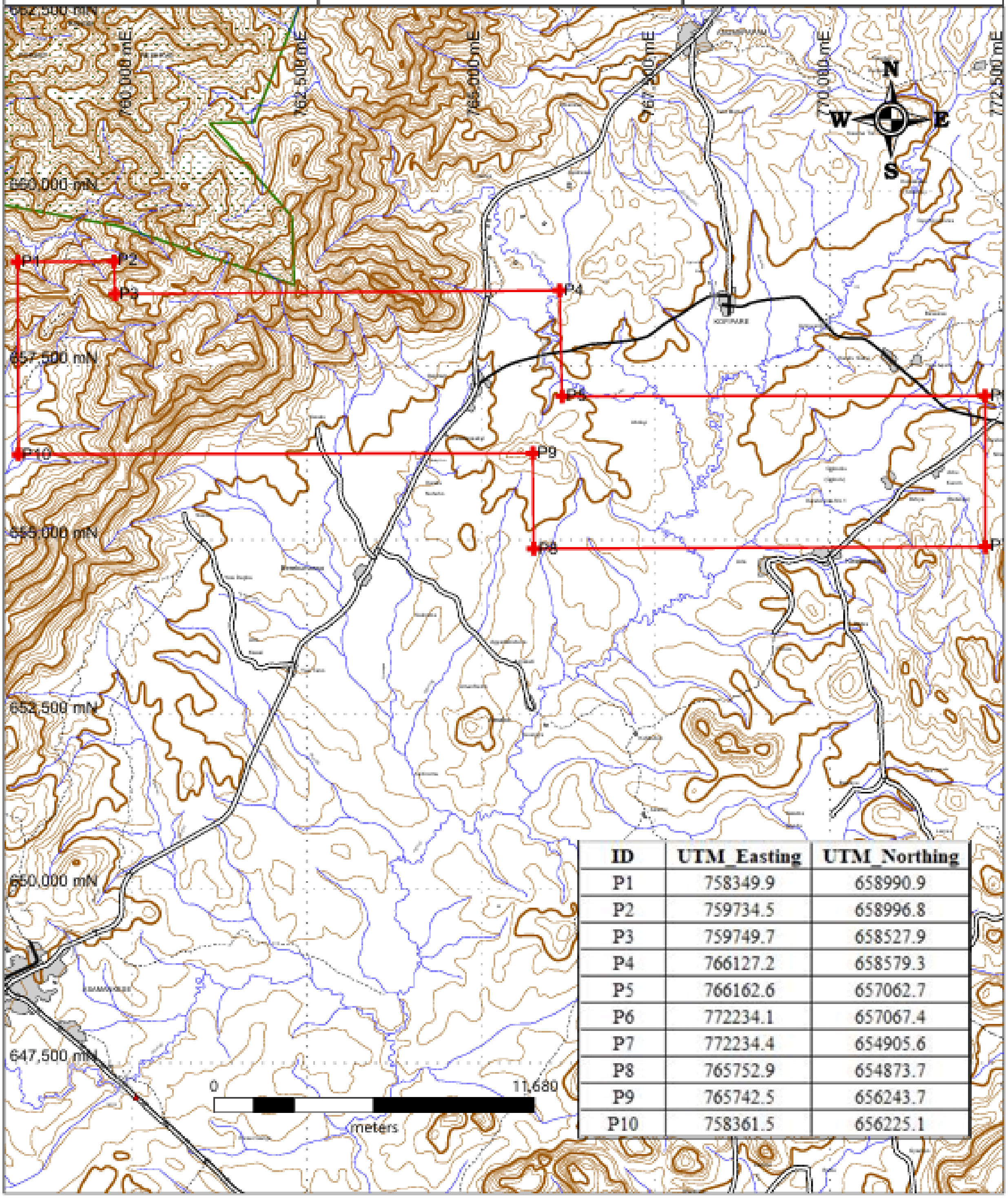
Figure 1.1: Map of Ghana showing location of Oworam concession.

PLAN OF LAND
LARGE SCALE MINING CONCESSION
FOR: IGNITE MACHINERY RENTALS AND MINING LTD

-shewn edged red-

Scale: 1:72,430 Size: 32.57 sq km

LOCALITY	DISTRICT	REGION
OWORAM	WEST AKIM	EASTERN



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P7	772234.4	654905.6
P8	765752.9	654873.7
P9	765742.5	656243.7
P10	758361.5	656225.1

Figure 1.2: Oworam Concession Map.

1.2 GENERAL DESCRIPTION

1.2.1 Accessibility

The concession is easily accessible from the Greater Accra region to the Eastern Region. The concession is connected through the Accra -Kumasi highway through Nsawam-Kraboa Coltar-Akwadum road via Kwaboanta Nkwanta then to Oworam. Travel time is fast and is about 75 km and travel time is about 1 hr 15 min from Accra the Capital of Ghana.

Access within the concession and the district is generally fair with good road networks connecting various parts of the districts.

1.2.2 Infrastructure

The geographical location of the Municipality would foster trade and other economic relations with adjoining districts. The Municipality can also take advantage of its closeness to key Central Business Districts such as Kibi, Kade, Adeiso and Accra to engage in some trading activities.
This implies that road conditions must be developed to promote trading activities. Also, the location of the Municipality would enhance administrative work as information and ideas can be exchanged with the other districts.
Most facilities including guest houses, restaurants, hotels, health facilities, gas stations and electricity from the national grid are readily available to the concession.

The municipality can boast of several public and private schools from the basic through to tertiary schools catering to the educational needs of their residents and made up of both public and private. Such as senior high school and other Tvet educational facilities to cater for the needs of residents.

The municipal has a robust healthcare infrastructure facility across public, private, and CHAG sectors, including hospitals, health centers, clinics, maternity homes, and CHPS compounds.

The safety and security of its residents is safeguarded through the presence of security agencies. These include the police force, fire service, immigration officials, and forestry authorities, working collaboratively to maintain law and order, respond to emergencies, and protect the natural environment within the municipality

There are market centers, filling stations and other recreational facilities available at Asamankese and other surrounding areas to cater for the needs of the locals.

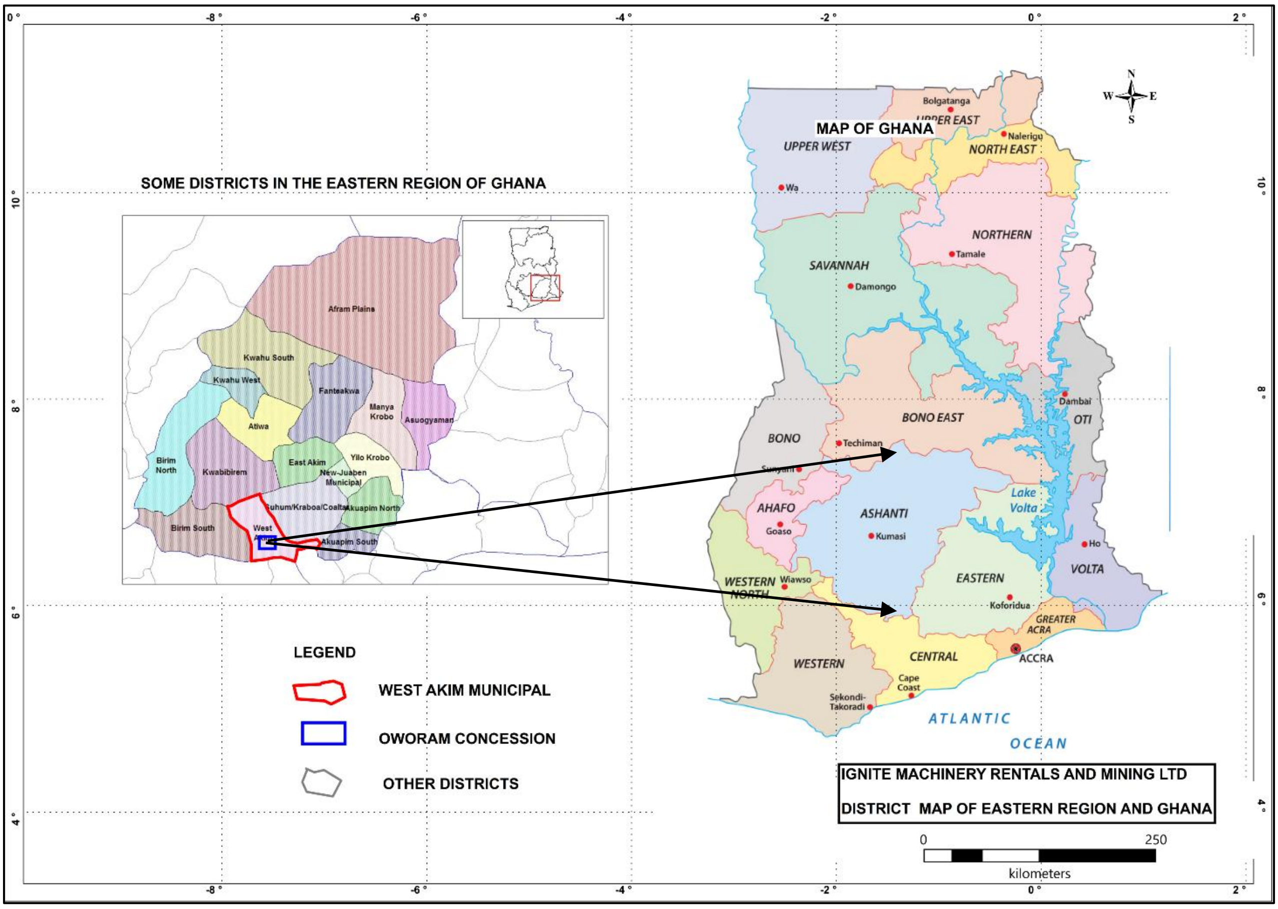


Figure 1.3: District map of the Eastern Region and the Oworam Concession.

1.3 GEOGRAPHIC SETTING

1.3.1 Relief

The land is generally undulating with heights ranging between 60 meters and 460 meters above sea level. The highest point is around the Atewa Range, located between Pabi-Wawase and Asamankese in the Northern part of the Municipality, most of which is occupied by the Atewa Range Extension Forest Reserve.

1.3.2 Drainage

The Municipality is well drained by rivers like Ayensu, Ntoasu, Abukyen, Akora, Supon, and Obotwene/Ansing among others. These rivers flow in a dendritic pattern. Most of the rivers have their sources in the Atewa Range and a few taking their sources from the Eastern part. The rivers are mostly perennial due to the double maximum rainfall which feeds them.

1.3.3 Climate: Rainfall, Temperature and Humidity

The Municipality lies within the wet-semi-equatorial climatic zone which receives rainfall between 1,238mm and 1,660mm. This is characterized by a double maxima rainfall pattern with which supports all-year round plant growth. The major rainy season is from March to June with the heaviest rainfall in June. The second rainfall season is from September to October. The average temperature ranges between 25.20C minimum and 27.90C maximum. Relative humidity is highest during the rainy season of about 80-95% and 55-80% during the dry season.

1.3.4 Vegetation

The Municipality falls within the semi-deciduous forest. The vegetation is mainly characterized by tall trees with evergreen undergrowth and contains valuable economic trees like Odum, Wawa, Sapele, Obeche, Onyina, Emire and others. Scattered particles of secondary forest are characteristic of the vegetation as a result of indiscriminate farming, lumbering, building and mining activities. The West Akim district has about 42 km2 of the Atewa Range Extension Forest Reserve.

CHAPTER 2

2.0 TECHNICAL INFORMATION

2.1 GEOLOGICAL SETTING

Ghana lies within the eastern margin of the West African Craton (WAC). The West African Craton stabilized in the early Proterozoic (2000 Ma) during the Eburnean Orogeny. The West African Craton (WAC) extends across western Africa and consists of two Archean nuclei in the north-western and south-western parts. The two nuclei consist of Reguibat and Leo-Man Shields in the north and south of the craton respectively. Ghana falls within the Leo-Man Shield Archean rocks consisting of oldest banded gneiss overlain by greenstone belts with (ultra) mafic metavolcanics, banded ironstone, phyllite, greywacke and quartzite (e.g. Sula & Nimini belts in Sierra Leone) and intruded by 2.97-2.78 Ga granite (Attoh & Ekweme, 1997).

The West African Craton is the second largest region in Africa where lower Proterozoic rocks are extensively preserved. These early Proterozoic rocks comprise extensive belts of metamorphosed volcanic and sedimentary rocks exposed in Ghana, Burkina Faso, Niger and Cote d'Ivoire. In Ghana these rocks are referred to as the Birimian Supergroup System. The West African Shield represents approximately 45% of the exposed geology of Ghana, being largely restricted to the northern, western and south-western portions of the country. The shield area is confined to the southeast by a Proterozoic mobile zone and the central-eastern portion is largely veneered by late Proterozoic to early Palaeozoic sediments of the Voltaian Basin.

In general, Ghana which is located southeast of the West African Craton can be divided into several distinct terranes as:

i) An early Proterozoic terrane (Birimian System) which hosts most of the country's mineral deposits and occupies the western and northernmost part of the country.

ii) The Tarkwaian System, a distinctive sequence of clastic sediments within the Ashanti, Bui, and Bole-Navrongo Belts; iii) The Voltaian Basin, in which are preserved the late Precambrian to Paleozoic sediments that mantle the craton.

iv) The Dahomeyan System, occupying the easternmost part of Ghana.

v) A Pan-African mobile belt, the Togo and Buem Formations, separated from the Birimian terrane by a prominent topographic feature known as the Akwapim - Togo range.

vi) Phanerozoic sedimentary rock and

vii) Intrusive rocks.

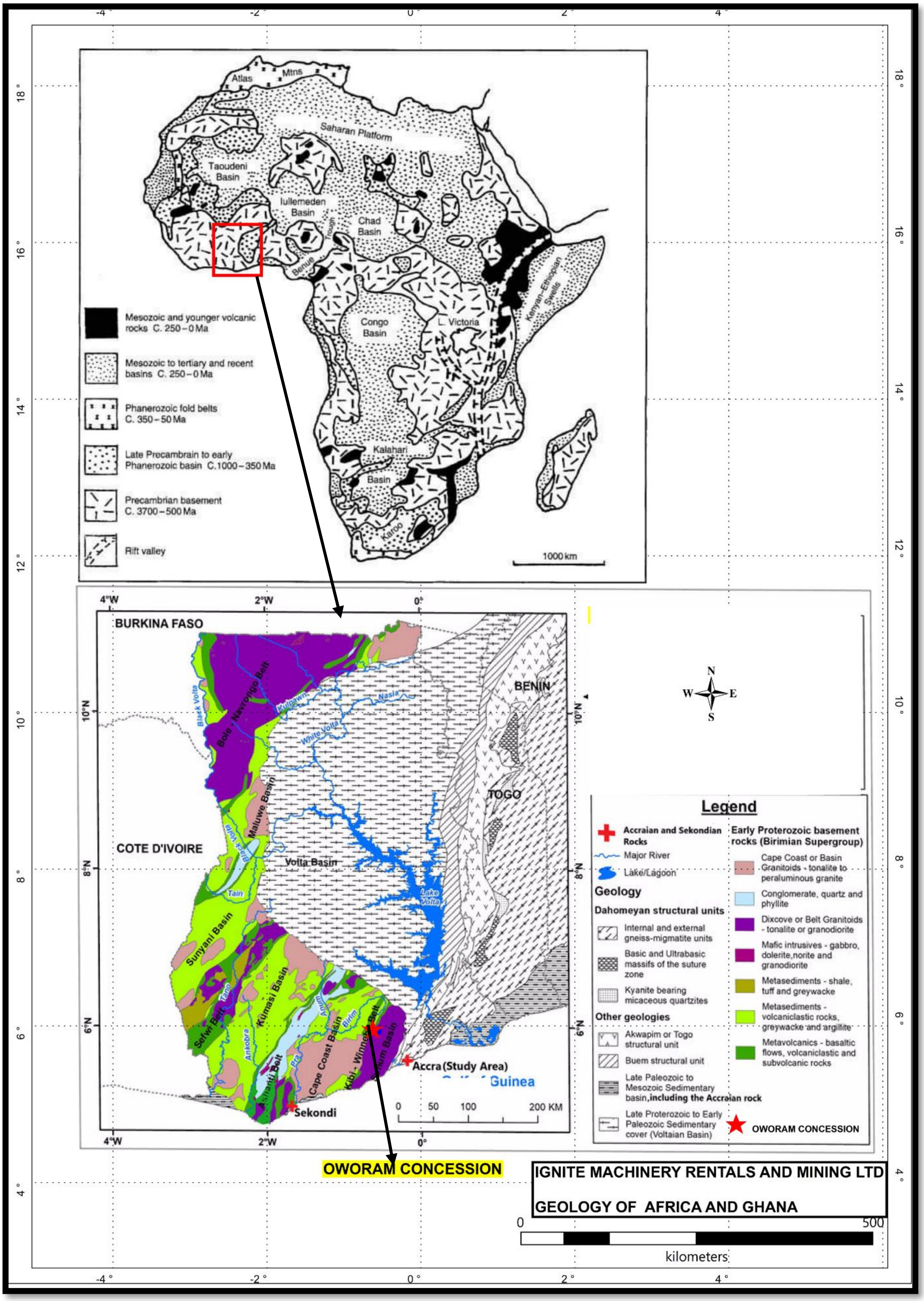


Figure 2.1: Geology of West Africa and Ghana.

2.2 REGIONAL GEOLOGY

The Concession falls within the Birimian System at the eastern margin of the West African Craton in the Leo-Man Shield formed at the southern part of the West African Craton. The Birimian group consist of five parallel approximately evenly spaced gold belts consisting of low-grade metamorphic units and tholeiitic lavas. The volcanic belts are interbedded with volcanoclastic sediments within the volcanic flows. The gold belts are separated by basins with isoclinically folded wackes, argillitic sediments and granitoids. The supra-crustal rocks are highly deformed. However, the sedimentary rocks are particularly characterized by extensive folding. The lavas are mainly basaltic composition, though andesitic, dacitic and rhyolitic rocks are also present. Some pattern of facies distribution is shown by the Birimian sedimentary basins from the margins towards the basin centres. The transition zone between the volcanic belts and the sedimentary basins is marked by chemical facies, which has of late been found to be the site of much of gold mineralization in Ghana.

The Eburnean orogeny that stabilised the WAC and affected the birimian rocks occurred in two discrete cycles of orogenesis. An earlier cycle, the Eburnean I orogeny, was associated with eruption of the Birimian mafic volcanic rocks, intrusion of granitoids and associated metamorphism, between ca 2200 and 2150 Ma. The later, Eburnean II, orogeny involved deformation, metamorphism and intrusion of all Birimian and Tarkwaian rocks between ca 2116 Ma and 2088 Ma.

The Birimian System has been intruded by two distinctive granitoid types. The larger basin-type granitoids (and gneisses) are muscovite and/or biotite-rich and are distinctly foliated and deformed, providing a pre-tectonic appearance. The smaller belt-type (arc related) granitoids are hornblende-rich, lack the characteristic foliation of the former and are generally interpreted to be syn or post-tectonic. Despite their appearance, the belt-type granitoids are dated as being 60 to 90 million years older than the larger basin-type granitoids.

The concession area falls within the northern part of the Kibi-Winneba Gold Belt of the Birimian, at the southeastern margin of Kibi belt area and transitions into the western margin of Suhum basin to the east.

Both the volcanic belts and sedimentary basins are intruded by three types of granitoids differing in age, mineralogy and chemistry, namely:

- Granitoids in the sedimentary basins - the basin type - are dominated by two-mica granites

- Granitoids associated with the volcanic belts - the belt type - are dominated by hornblende-bearing granites

- The late K-rich granitoids (post-Tarkwaian) comprising the Bongo, Tongo and Banso granitoids

Despite their appearance, the belt-type granitoids are dated as being 60 to 90 million years older than the larger basin-type granitoids.

The presence or absence of a foliation is not sufficient criterion to establish timing relationships in granitoids (Paterson et al.,1989). In particular, amphibole bearing granitoids have demonstrated to be less likely to develop a foliation during deformation than biotite-rich granitoids (Vernon and Flood, 1988).

Locally, the Kibi Winneba Belt is the easternmost of the Birimian Greenstone Belts in southern Ghana. It is located east of, and parallel to, the well-endowed and world-renowned Ashanti Gold Belt, which hosts many of Ghana's active producing gold mines. The north-east trending Kibi Belt, i.e the northern continuation of the Kibi Winneba Belt, is approximately 60 km long and 20 km wide. Its southern limit is truncated by a large granitoid batholith, whereas its northern extent is overlain by younger, flat lying sediments of the Pan-African Voltaian Supergroup. Locally, the geology of the Kibi Belt is poorly established, compared to the Ashanti Gold Belt, largely due to the poor exposures, limited government survey mapping and the lack of formal exploration activities. The general Kibi Project area geology is summarized from Griffis (1998) and Griffis et al (2002).

The Birimian belt-basin development was followed at around 2.1 million by the Eburnean Tectono-thermal event, a single -stage progressive SE-NW compressional and regional metamorphic episode, expressed by foliation and shear development oriented between 025^o and 050^o. Metamorphic grade is generally lower greenschist facies with higher-grade facies developed locally, particularly proximal to intrusive bodies. Isoclinal folding (foliation-oriented NNE-SSW) is regionally well developed in the argillaceous facies. Multiple episodes of strike-slip and overthrust faulting from the northwest (and perhaps southeast in part) are more apparent in the arc-related birimian metavolcanics and Tarkwaian series.

The spatial distribution of gold mineralisation in Ghana is closely governed by north to northeast trending belts of metavolcanic rocks, which range from 15km - 40km in width. Almost without extension exception, the major gold deposits lie at or near close to the margin of the belt in close proximity to the strongly deformed belt-basin birimian sequences.

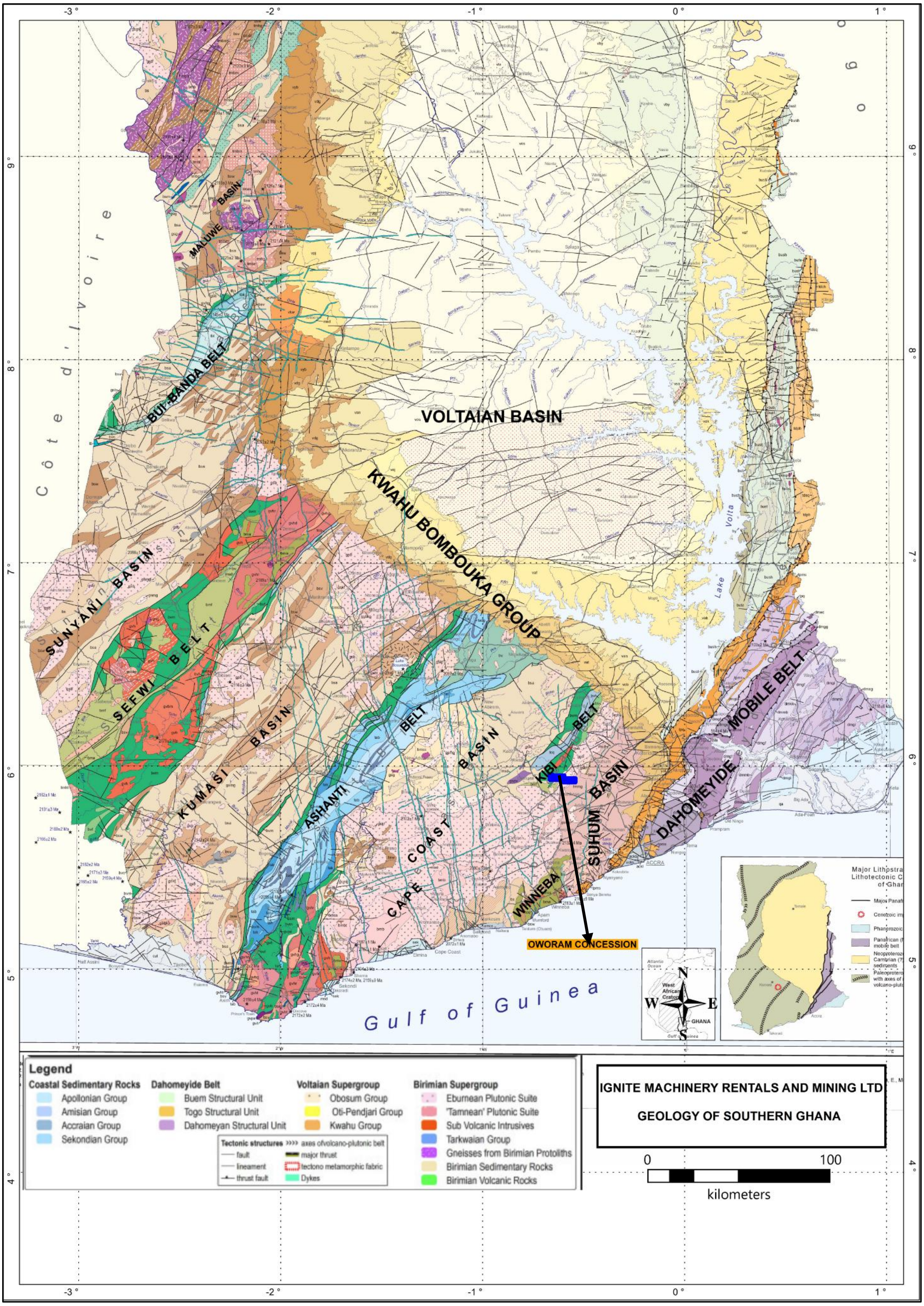


Figure 2.2: Regional Geology Map.

2.3 LOCAL GEOLOGY

The Oworam concession falls within the southeastern margin of the Kibi area of the Kibi-Winneba belt of the Birimian of Ghana. The belt is characterized by the Birimian metavolcanic and metasedimentary rocks with prevalence of intrusive granitoids.

Locally the concession area is underlain by north east trending greenschist to amphibolite facies of the Birimian supergroup comprising basaltic flow/subvolcanic rock and minor interbedded volcaniclastics from the northwest followed by Sediment/volcano-sediment, undifferentiated with upper greenschist to amphibolite facies metamorphic, metasomatized +/- lit-par-lit granitoid intrusions, Eburnean Plutonic Suite:- non metamorphic Biotite (+/- hornblende, +/- muscovite) granitoid, undifferentiated, Tamnean' Plutonic Suite granitoid gneiss, biotite gneiss and the upper north corner is occupied by Two-mica or muscovite granite and minor granodiorite, locally leucogranite.

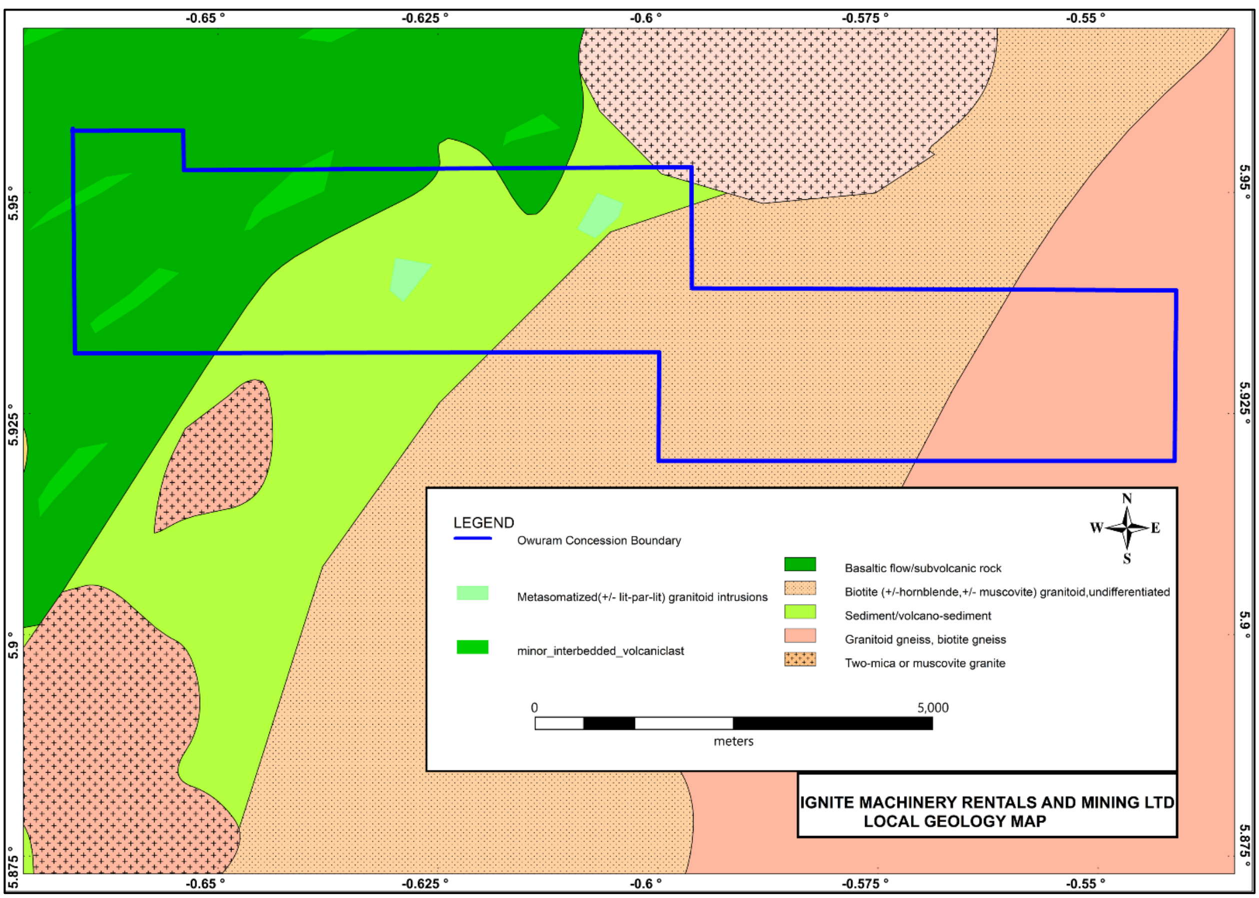


Figure 2.3: Local Geology Map.

2.4 MINERALIZATION

The Birimian rocks in Ghana are the primary source of gold mineralization trends. Gold in the Birimian rocks is usually controlled by shear systems and quartz vein stockworks. Pathfinder minerals are usually sulphides, with arsenopyrites and pyrites being the principal pathfinder minerals.

Mineralization occurs as quartz vein type and disseminated type of sulphide mineralization. Along contacts zones between the Birimian rocks and the granitoids are potential zones of gold mineralization and rare metals from the mesothermal fluids of the intrusive.

2.4.1 Laterites and oxide caps Mineralisation

These are deposits formed by in situ weathering of enriched hard rock sources and large veins yet to be eroded. These are found in saprolite and sap rock with gradational contact with the hard rock are less prolific compared to the alluvial deposits. This auriferous bauxite cap suggests that the underlying rocks are auriferous and could be the source of alluvial gravels. Locally, presence and extent of artisanal mining can be attributed to the substantial enrichment in alluvial gravels and oxide caps.

2.5 STRUCTURE AND DEFORMATION

Deformation related to the Eburnean II orogeny dominates the structural geology of southwest Ghana. Five successive phases (D1 to D5) are recognized.

D1 resulted in the formation of a weakly developed bedding parallel cleavage (S1) in the Birimian sedimentary units, and minor folds related to early northwest-southeast compression, thrust faulting, and basin inversion.

D2 formed major thrust faults, gently plunging, tight to isoclinal doubly plunging folds and a second cleavage (S2). The S2 cleavage has a northeast strike and is sub-vertical to steeply northwest dipping. The D2 thrust faults involved significant displacements and part, or all, of the shortening in Southwest Ghana occurred during D2 along these faults.

S3 and F3 axial planes generally strike northeast and dip between five degrees to the southeast and forty degrees to the northwest. D3 is recognized as a relatively minor deformation event and did not modify substantially the structural architecture of the Birimian Supergroup.

Steeply plunging, upright folds (F4) with axial planes that strike close to east-west and an associated axial planar cleavage (S4) overprint D3 structures. The F4 folds plunge moderately to steeply northeast, and the associated axial planar cleavage (S4) strikes east-northeast and dips steeply to the north-northwest. In some F4 hinges, bedding (S0), S1 and S2 are transposed into S4, defining east- striking zones of weakly graphitic schists.

D5 is characterized by a reactivation of D2 faults and represents the last major deformation event in Southwest Ghana. The onset of D5 deformation may overlap with the waning stages of D4.

Slicken lines on faults and surfaces of internal veins generally plunge gently towards the southwest. This geometry indicates that the D5 reactivation was largely strike slip with a small component of dip slip. Asymmetric dilatant breccias, quartz vein arrays, and boudinage quartz veins localized along these faults, in the Obuasi deposits imply a component of sinistral movement during D5. None of the faults appear to have been folded during D4, despite evidence that most formed prior to D4. These apparent contradictions imply that the F4 folds and D5 sinistral strike slip offset on the adjacent faults may both be products of D4 rather than separate D4 and D5 events.

The Birimian System hosts most of the gold and diamond deposits in the country, hence they have been subjected to considerable study.

CHAPTER 3

3.1 REGOLITH AND HARD ROCK OROGENIC GOLD DEPOSITS

The company plans to advance the deposit sources within the concession into the regolith and hard rock. This will involve a systematic approach including geological, geochemical and geophysical methods from field mapping to drilling stage.

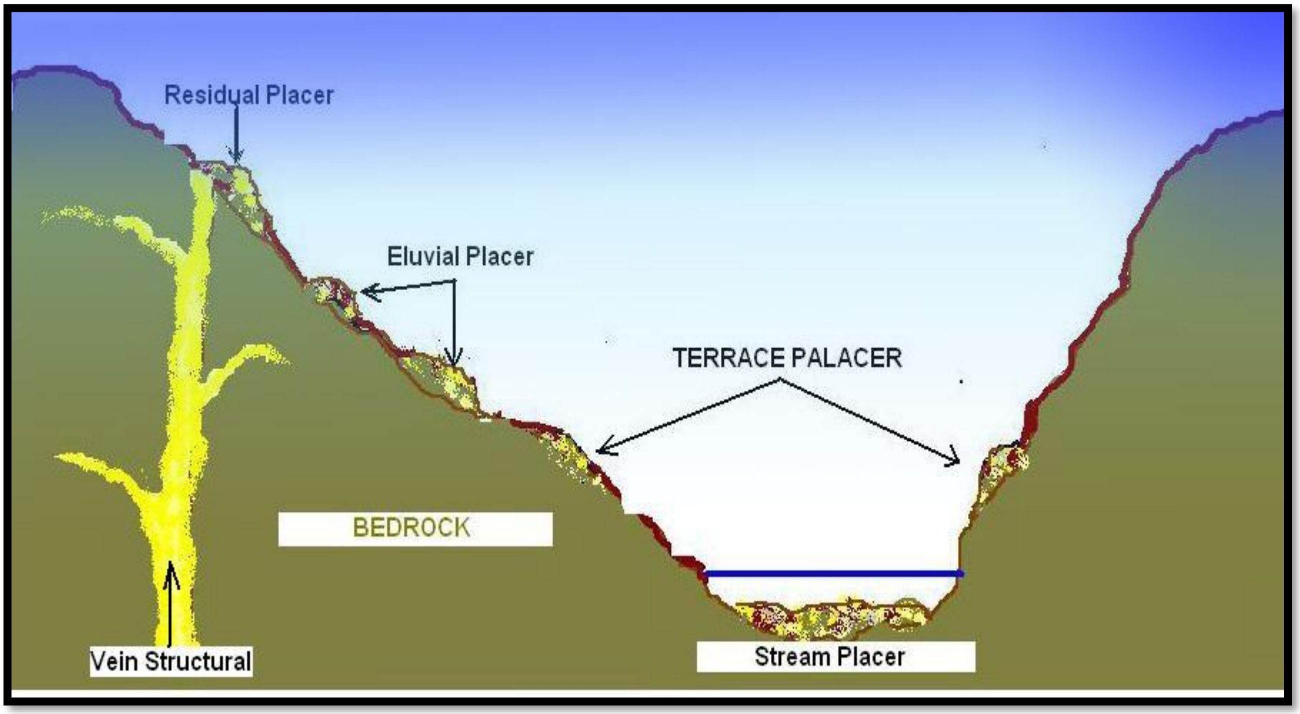
Three broad exploration stages include:

- Area Selection
- Target Definition
- Ore deposit Delineation

Area selection typically involves using broad geological concepts, regional geological mapping and syntheses, airborne geophysical surveys, and regional geochemistry (such as systematic sampling of sediments in drainage channels, or sampling weathering crusts). Favorable areas are selected based on interpretation of these inputs.

Target definition involves more detailed geological investigations of such favorable areas, further geophysics (which can include surveys carried out on the ground), closer spaced sampling of regolith materials for geochemistry, and initial drilling to provide subsurface sampling. The drill samples commonly come to the surface by air blast in the form of gritty powders and chips of subsurface regolith, important materials for geochemical analyses). Targets are located and ranked based on assessment of their characteristics.

Deposit delineation centres on investigation of the located targets, particularly using drilling, including diamond drilling (the most expensive form of drilling) where solid core samples of bedrock are produced. This stage also involves detailed geophysics (which may be done down drill holes) and geochemistry. Geological and geochemical investigations typically involve microscope studies of textures and minerals present and can include sophisticated studies using isotope chemistry and investigating micron-scale fluid inclusion in transparent minerals such as quartz. Detailed feasibility studies for intended mining are then undertaken for the most promising deposits.



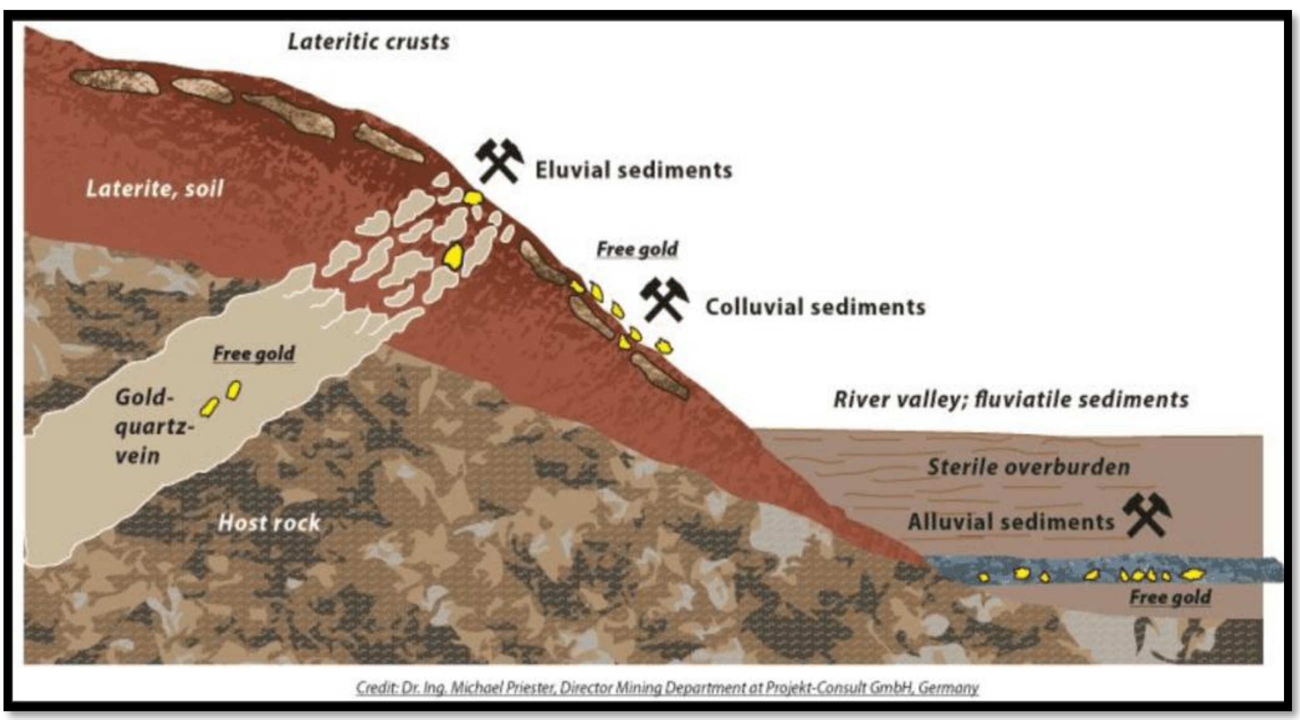


Figure 3.1: Potential for Gold Sources.

3.2 OROGENIC GOLD POTENTIAL OF THE OWORAM CONCESSION

3.2.1 Interpretation of geophysical and structural data

Review of technical data has indicated that the area has a complex structural system based on mapping and regional geophysical survey. The data interpretation proved that early mapping of the region recognized the existence of relatively long, narrow structural system and trending contacts that are mineralized.

3.2.2 Magnetic Image- Total Derivative:

Aeromagnetic geophysical data over the survey area was carefully analyzed and subsequent interpretations suggested several high-conductivities spots as well as mid to low conductive zones that roughly coincided with the structural data of the area and follows a NE -SW successive trends with the concession area.

The Data sets indicated that the high-resolution total aeromagnetic geophysical survey partly corroborated the presence of NW-SE (S3) partly and partly the NE -SW trending (S1, S2) shear. Minor trending lithological contact structures (L1, L2) and S1, S3 intersecting in the targeted area are interpreted as a shear zone. These major zone S3 trend towards from NW to SE cutting across other lithological contacts. They are prioritized as target for detailed exploration.

Furthermore, geological interpretations made from geophysical data appeared to suggest that the concession is underlain dominantly by upper greenschist to Amphibolite facies comprising volcano sediment rock and granitoid intrusions (stratigraphically, Birimian Supergroup), biotite (+/- hornblende, +/- muscovite) granitoid(undifferentiated) and basaltic flow or subvolcanic rock with minor interbedded volcaniclastics of the Eburnean Plutonic Suite.

Refer figure below: -

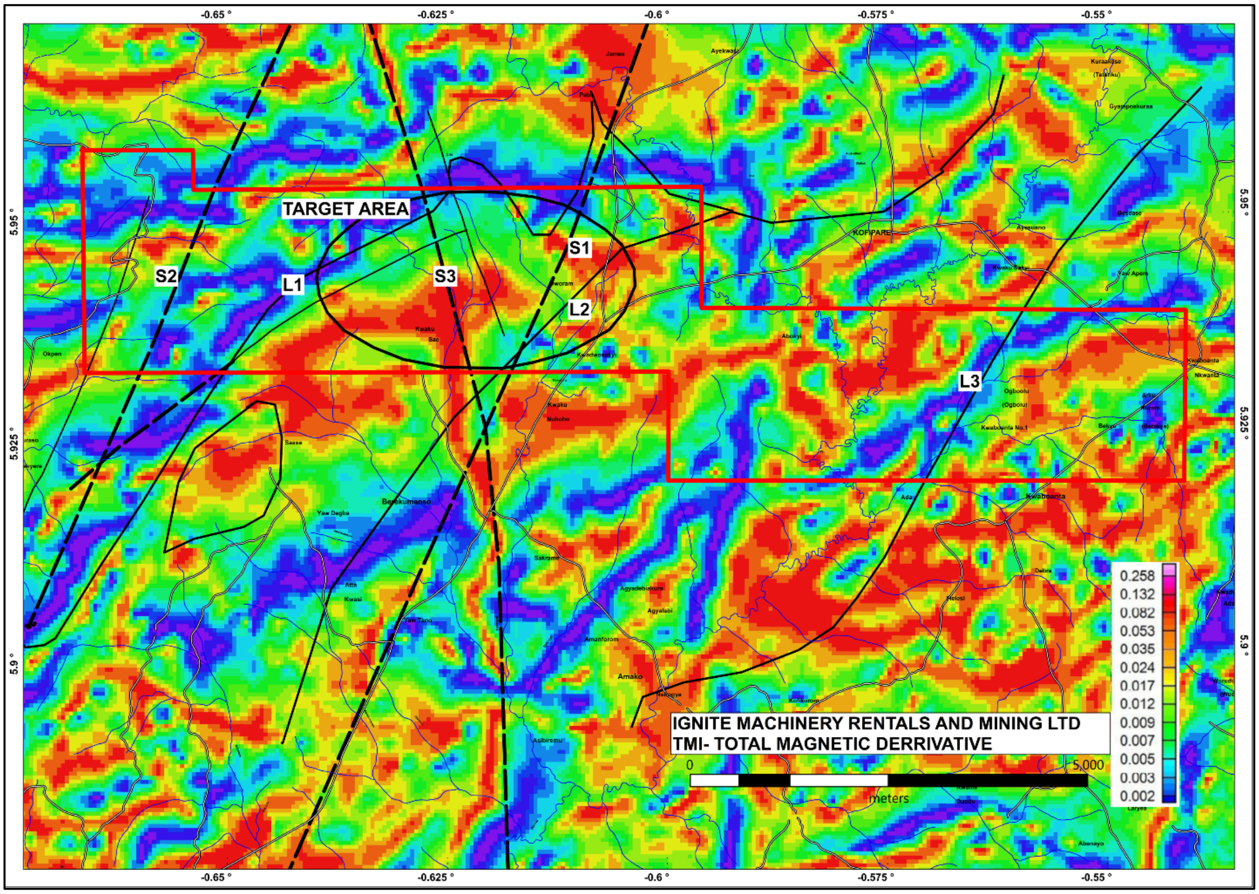


Figure 3.2: Total Magnetic Image

3.2.3 First Vertical Derivative (1VD)-magnetic:

The first vertical derivative (1VD) map of the Oworam concession has been useful in sharpening short wavelength magnetic anomalies. The derivative was made useful in elevating the structural feature visibility. The structures labelled in the figure below as “S1and S2” clearly shows a contact between supposed upper greenschist to Amphibolite facies comprising basaltic flow or subvolcanic rock and minor interbedded volcaniclastics (stratigraphically, Birimian Supergroup) and Biotite (+/-hornblende, +/- muscovite) granitoid (undifferentiated) of the Eburnean Plutonic Suite, and volcano sediment with metasomatized granitoid intrusions.(S3) structure reveals a major N-S trend structure from the south towards north of the concession.

The 1VD indicates that the main mineralized structure (Target) of NW-SE trend major fault (S3) which is crosscut by lithological contact (L1, L2).

The 1VD image confirms the total derivative image on the NW-SE general trend of potential mineralization-target. See the figure below: -

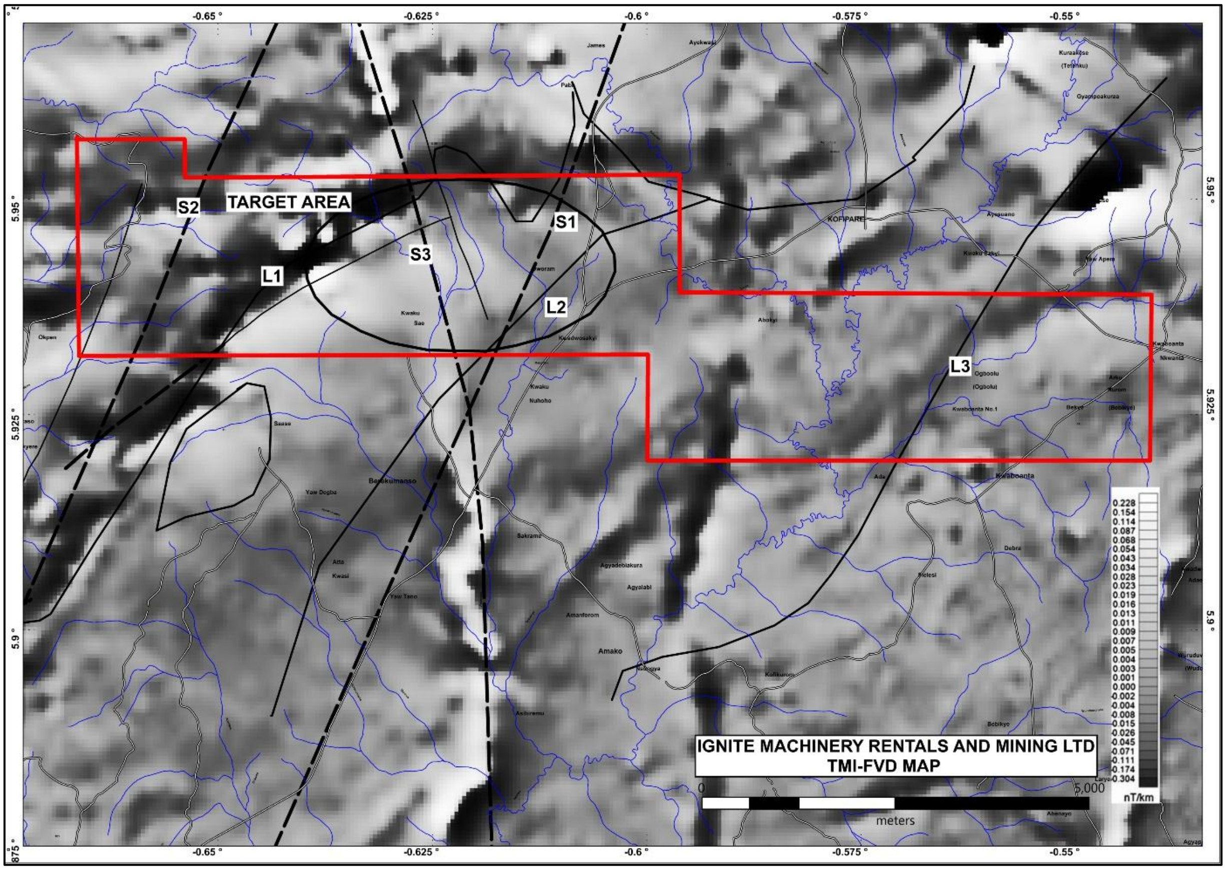


Figure 3.3: First Vertical Derivative (1VD)-magnetic-Oworam concession

3.2.4 Ternary Airborne radiometric:

The Ternary radiometric image correlated with the pattern and trend of the geological units extracted from regional geology. The image helped to identify zones that have high concentrations of potassium and thorium, which is generally considered very immobile. This is because of the role they play in the delineation of gold mineralized zones since gold deposits show increases in (K) and (Th), which suggest that (Th) was mobilized in hydrothermally altered systems (Silva et al., 2003).

The signatures associated with the radiometric data, juxtapose with the possible host rocks at Oworam were studied to derive relationship with host rocks alteration, pattern and trends of the geological units. The area showed moderate potassium(K), uranium (U) and thorium (Th) distribution throughout the concession and around targeted area towards west, in a NE-SW pattern. This indicates that these elements were mobilized in hydrothermally altered systems and indicates possible mineralization of these areas.

Airborne radiometric image Integrated into the structural occurrences and localized geological data assessed relationships between radiometric, lithotypes, structure(s) and mineralization. The resultant interpretation suggested strong indications of shearing- quartz vein type gold mineralization within some parts of the Oworam concession. See the figure below.

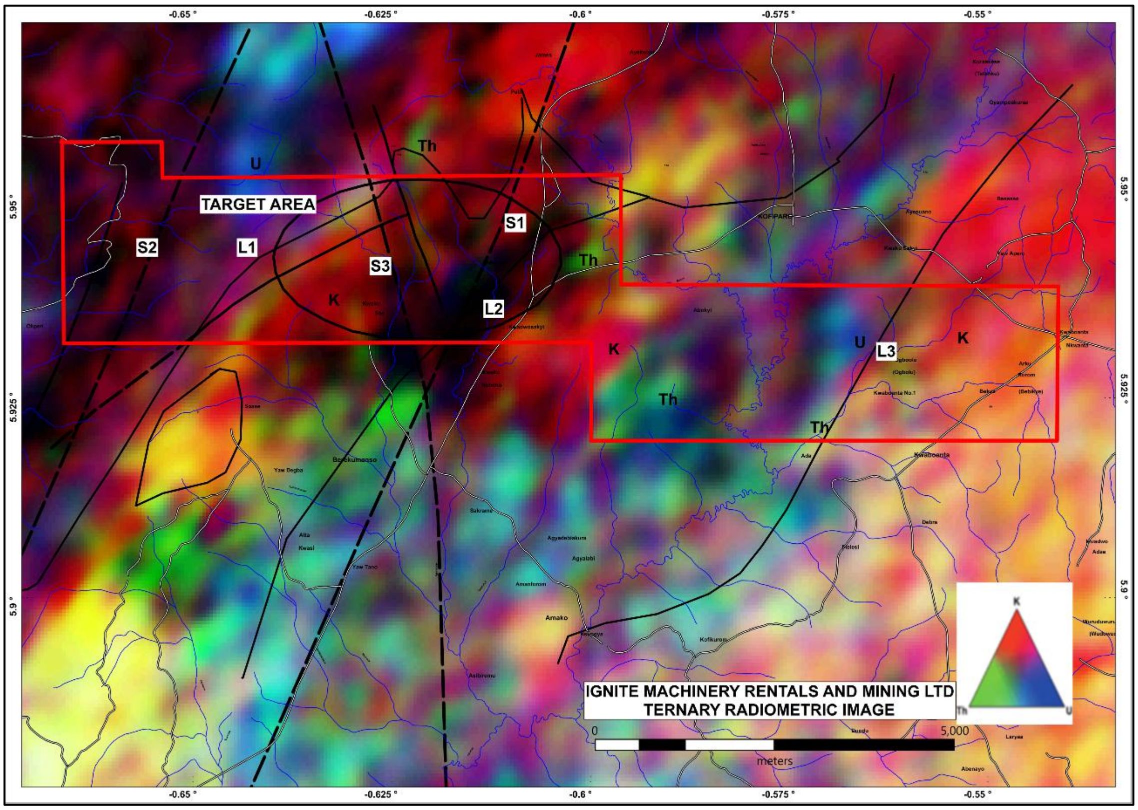


Figure 3.4: Ternary Airborne radiometric

3.2.5 Analytical Signature (Ansig-TMI)

The Analytical Signature conductivity map related to the area shows that low conductance values are concentrated mostly across the Concession in NE-SW trend through the southeastern portion of the area and trend narrowly across through the middle along a contact as well. With reference to the figure below the medium conductance signatures spots trend N-S across the middle and to the NE around the western part. The N-S trends define occurrence of vein contacts. The major N-S trend continues middle the south from the north indicating an apparent anomalous zone. The superposition of trends in the figure below shows the relevant responses of various geological structures. It also proves the efficiency of the airborne analytical magnetic method in the characterization of conductivity bodies. For the high conductivity values, anomalies are often not continued but keep the same trend. The presence of good conductor bodies is observed, and their signal response persists in the order of conductivity. Refer to the figure below:

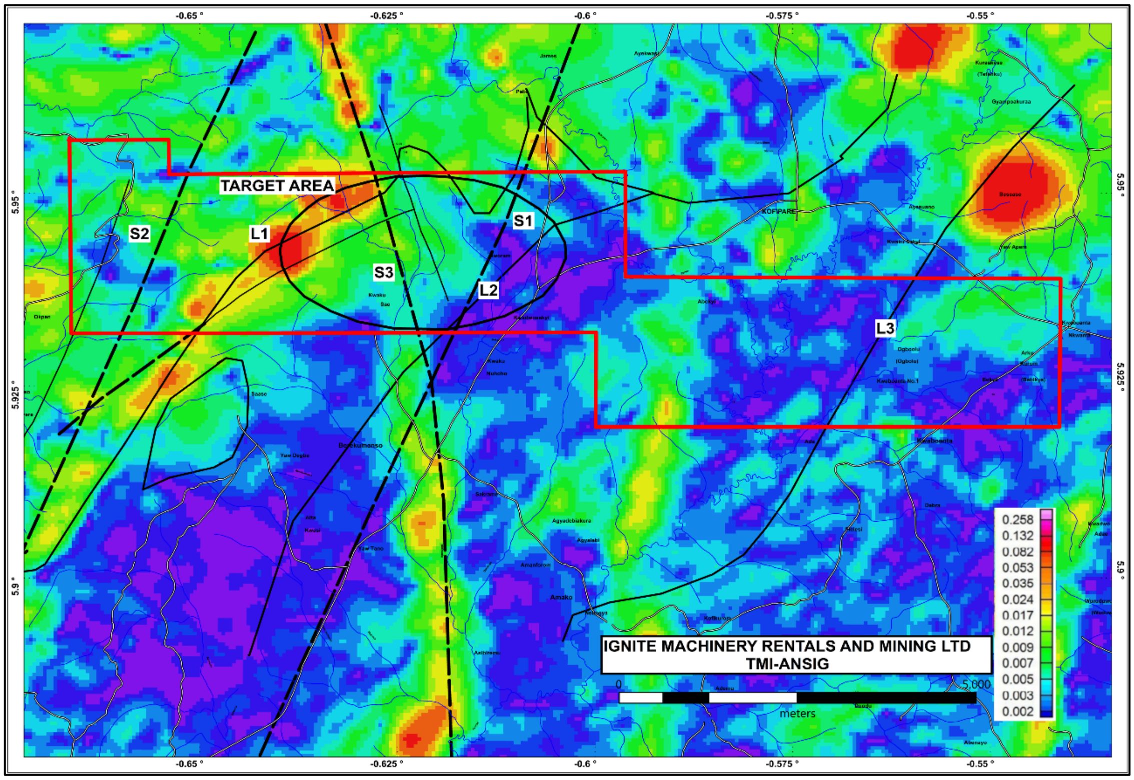


Figure 3.5: Analytical Signature (Ansig-TMI)

3.2.6 Structural analysis interpretations

Structural data was acquired to confirm structural correlation with the geophysics images and the lithology in general. The structural map has shown that there are inferred shear structures within the target area. These shear structures trends NE-SW within the mid portion to the northern portion of the Oworam concession. There may be possible cross cutting structures around the targeted area. Refer to the figure below.

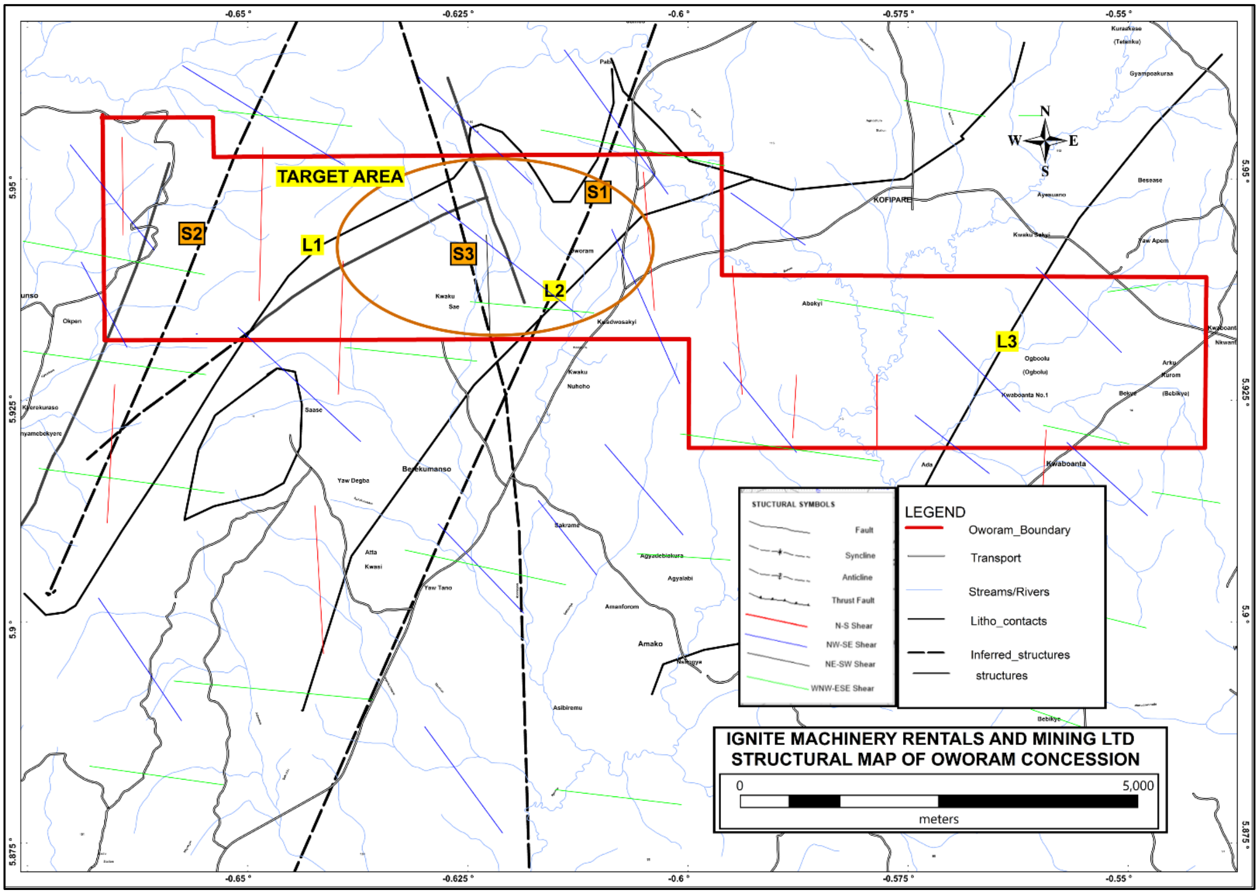


Figure 3.6: Structural map showing main target

3.3 HARD ROCK MINERAL RESOURCE POTENTIAL

Analysis and interpretation of data from the structural, geological, geochemical and geophysical perspective, strongly suggest that the gold is associated with volcanics that are highly sheared with quartz veins, carbonated and sulphide-rich (disseminated). The mineralized zones are likely to be within contact of the subvolcanic rock and minor interbedded volcaniclastics (stratigraphically, Birimian Supergroup) and Sediment/volcaniclastic sediment, undifferentiated, locally mica schist:-Sedimentary-Volcano Sedimentary Group ('Sedimentary Basin') intruded in the mid by Biotite (+/- hornblende, +/- muscovite) granitoid(undifferentiated) of the Eburnean Plutonic Suite, and non-metamorphic ,K-feldspar-rich granitoid, mainly granite and monzonite ('Banso'/'Bongo' type).

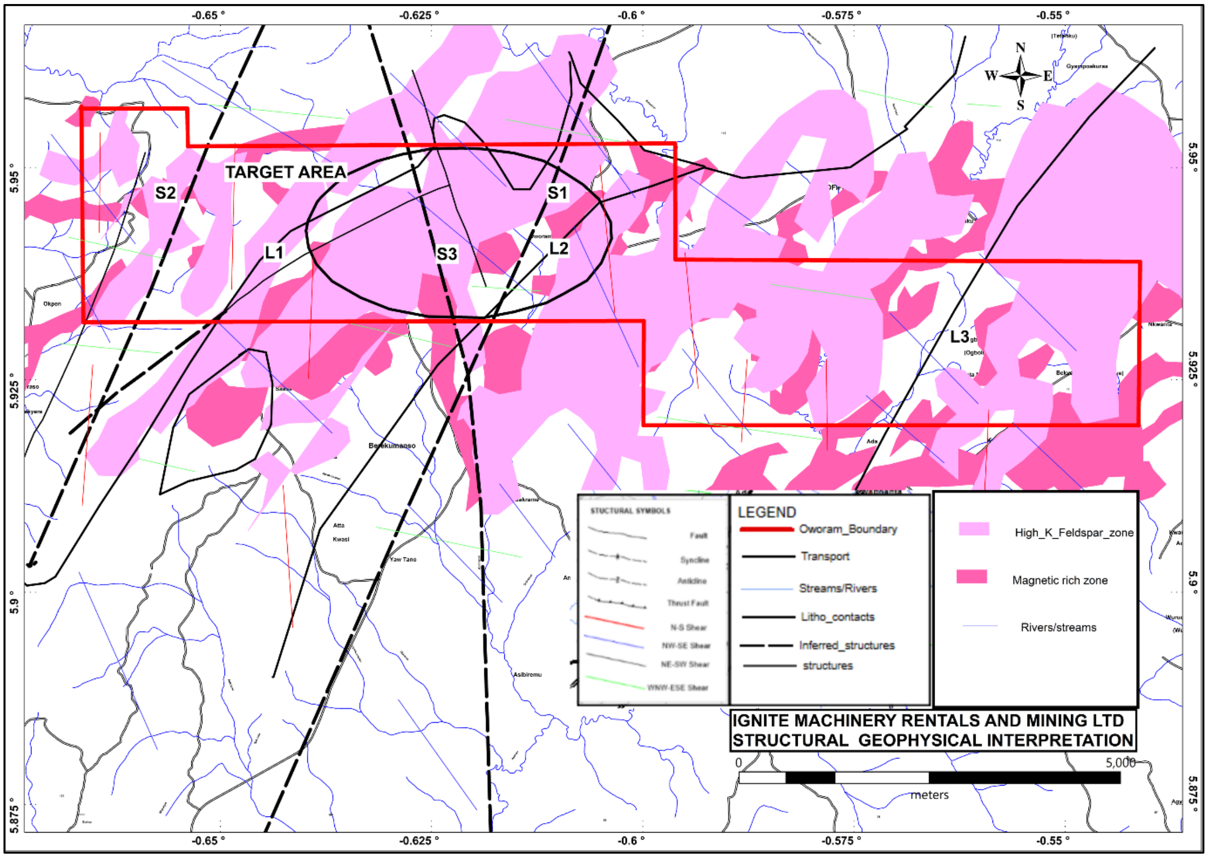


Figure 3.7: Litho, structural and geophysical interpretation of data.

3.4 ALLUVIAL GOLD DEPOSIT PROCESSES

Alluvial gold deposits are formed in active drainages that can be classified into four types: gulches, creeks, rivers and gravel plains. **Gulches** are the headwater drainages of alluvial gold provinces. They typically have steep gradients, varying from over 200m/ km down to about 25m/km**. Creeks or streams** often contain the first bodies of auriferous alluvium of any significant size and are characterised, in addition, by abrupt decreases in gradients as compared with the gulches which often feed them. Typically, creek gradients range from 30 to 6m/km. Separation of rivers from streams may be somewhat arbitrary. However, auriferous **rivers** can be considered as forming by the coalescence of streams and can contain large productive gold deposits. They have gradients ranging from about 10 down to about 2m/km. Where a river washes out into an unconfined valley or coastal plain, it's carrying capacity is suddenly reduced and a **gravelly plain** form. Gradients of gravel plains are typically 3 to less than 1m/km.

Alluvial gold deposits form over time where a river runs or has previously run through ground which is rich in gold. The erosive power of the water removes the surrounding rock due to its comparative low density while the heavier gold resists being moved. Alluvial gold usually takes the form of dust, thin flakes or nuggets.

The first stage in alluvial gold mining is to take the dredged or excavated feed material and separate the small sand faction (where the gold is found) from the larger mineral and rock fraction. In a typical alluvial mining process physical separation method such as screening and gravity separation are employed to separate the gold from the mineral fraction.

Our proven wet processing solutions are modular and can easily integrate with an existing alluvial mining operation, allowing you to improve the quality of the output material which will then go through further density separation to catch the heavier gold particles. Alluvial gold dredging along the river systems also revealed the presence of alluvial diamonds which were derived from more extensive deposits in some river systems e.g. the Pra River.

Generally, alluvial gold deposits primarily have a bedrock source of gold, a zone of transportation, and a repository. However, the alluvial regime is a dynamic and extremely complex system subject to paleoclimatic, eustatic, tectonic and other distorting long-term influences. Sometimes even simple features are often not adequately understood. Alluvial gold is recovered by gravity techniques, it is essential that source areas contain an adequate supply of coarse gold that can be transported into and down drainages.

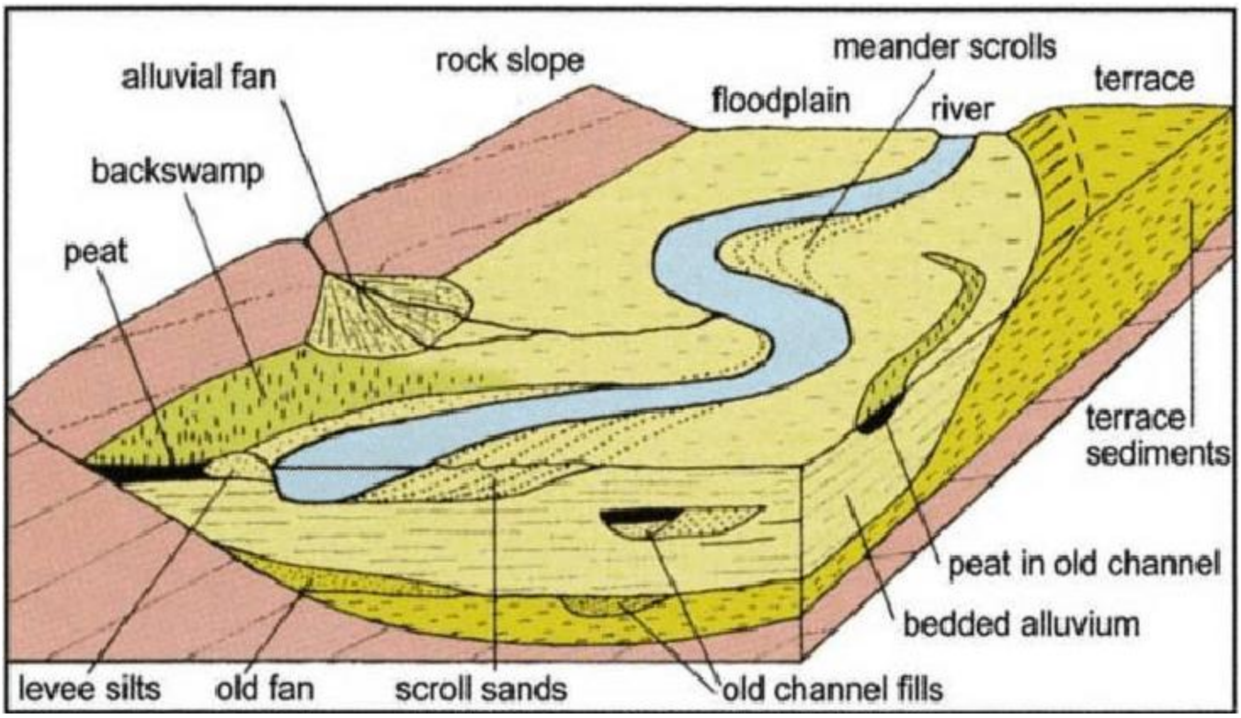


Figure 3.8: Alluvial deposit processes.

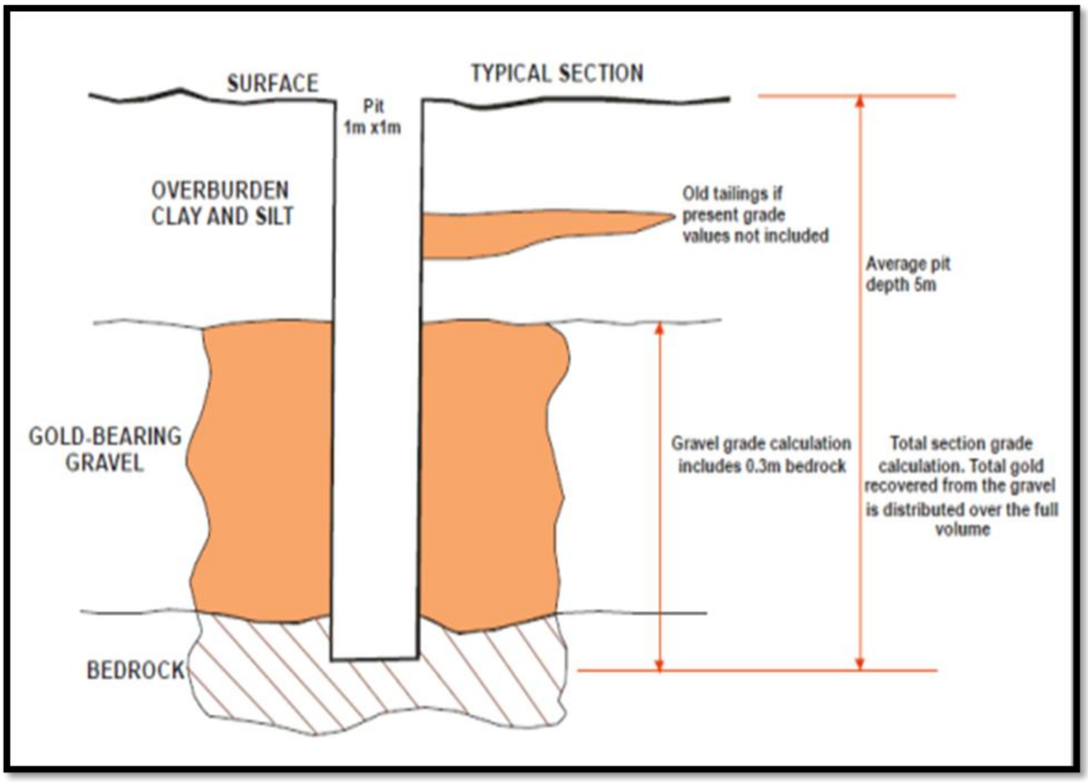


Figure 3.9: A Typical Vertical Section of an Alluvial Profile.

3.5 ALLUVIAL GOLD OCCURRENCES IN GHANA

The metallogenic province of Ghana is associated with the Proterozoic rocks of the Birimian and Tarkwaian System cropping out of the northern-western-south-western parts of Ghana. All mining and exploration activities are concentrated within this province. Hard rock, residual and alluvial placer gold are found within this province, but the alluvial type is by far the most common. The most favourable sites or target areas for alluvial gold concentrations are the flood plains, buried valleys, present riverbeds and the channels and terraces of rivers. In Ghana, the best-known localities of alluvial gold concentrations occur along the Pra, Offin, Ankobra, Mansi, Awere, Anum, Bonte, Birim and Tano River Flood Plains.

Ghana has about 20 large-scale mining companies producing gold, diamonds, bauxite and manganese and over 300 registered small-scale mining groups. Most of the registered small-scale mining groups operate at alluvial miners. Out numbering these registered companies are numerous illegal operators "Galamsey" exploiting alluvial deposits using artisanal methods. Some notable large-scale alluvial gold mines operating presently include the Dunkwa Goldfields Limited, Goldenrae Mining Company Limited, Bonte Gold Mines, Kibi Goldfields, Tepa Alluvial Gold Mine etc.

Dunkwa reserve is about 300 million cubic metres at the grade of 0.113g/m^3 Au with average depth to bedrock is 10metres. Estimated reserves for Goldenrae Mining Company as since 1991 stands at 9 million cubic metres at an average grade of 0.57g/m^{3}Au, the average depth to bedrock is 5metres with overburden thickness of 2metres. Bonte Gold Mine since 1990 stands at 4.6 million cubic metres at an average grade of 0.53g/m^3 Au, the gravel thickness is between 4 -5 metres and average width of alluvial plain is 280metres. The Tepa Alluvial Gold Mine is about 50-acre property in the Ahafo Ano District. At 2021, Kibi Goldfields Ore resources were estimated at more than 500,000 ounces of gold, have operated in the country consistently for the past decade and continue to focus on discovering, acquiring, and developing a portfolio of good mineral resources in Ghana.

Most of the historical gold production from Ghana before the 20th century came from a myriad of small streams and rivers draining through the areas with underlying oxide and primary gold deposits (Peters, 2013). Also, several of the major rivers have been mined with dredges starting in the early 1900s. Mineralization occurs at a time when there was high rainfall, minimal vegetation covers and at least moderate relief (Griffis et al., 2002).

The gold found in northern Ghana's alluvial deposits originates from primary gold mineralization within the Birimian and Tarkwaian rock formations. These alluvial deposits are commonly found along the flood plains of the Red and Black Volta rivers, and other rivers in the region. Alluvial gold mining in Ghana can be done through various methods, including shallow alluvial mining, deep alluvial mining, and hard rock (lode) mining. Alluvial gold exploration in northern Ghana has faced challenges, including a lack of local technical expertise, poor planning, and inadequate capital, but successful projects have demonstrated the potential for profitable ventures.

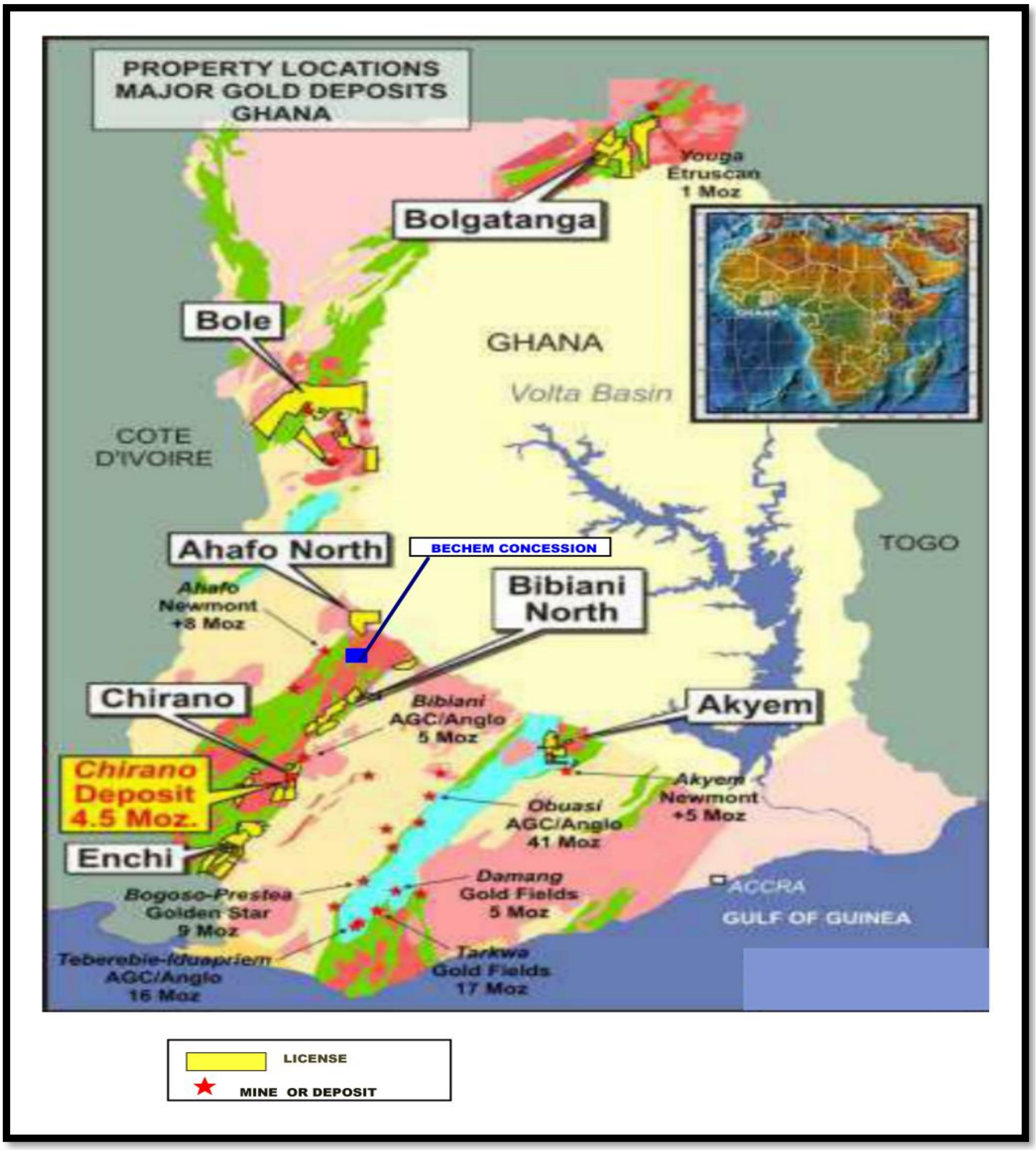


Figure 3.10 Major Gold Deposits of Ghana.

3.6 FUNDAMENTAL PRINCIPLES AND FLAWS OF ARTISANAL AND SMALL-SCALE MINING

- In Ghana, small-scale miners often use mercury in their mining activities, which poses a significant risk to human health and the environment. To address this issue, the government has implemented a Mercury-free Mining Initiative that promotes the use of alternative technologies.

- Small-scale miners often work in hazardous conditions with inadequate safety equipment. To address this issue, the government has implemented a Small-Scale Mining Development Policy that provides guidelines for improving safety and working conditions (Smith et al., 2016).

- · Small-scale miners often operate illegally, which makes it difficult to regulate their activities and protect the environment. To address this issue, the government has implemented a registration process for small-scale miners and increased enforcement of mining laws (Kambani, 2003).

- · Small-scale miners often work without proper land tenure or mining rights, which can lead to conflicts with other stakeholders. To address this issue, the government has implemented a land tenure reform program that aims to clarify and secure land rights for small-scale miners (Brottem and Ba, 2019).

- Small-scale mining sometimes is often associated with child labour and other forms of exploitation. To address this issue, the government has implemented a Child labor Monitoring System that aims to identify and address child labour in the mining sector.

- Most often some small-scale miners often use crude methods that damage the environment and cause health problems. To address this issue, the government has implemented a National Environmental Standards and Regulations Enforcement Agency that provides guidelines for responsible mining practices.

- Small-scale mining is often associated with conflict and violence. To address this issue, the government has implemented a Community Rights Law that gives communities greater control over their natural resources and the power to negotiate with mining companies.

- Small-scale mining is often associated with illegal trade in diamonds and other minerals.

• A study showed that small-scale mining is often associated with environmental degradation and loss of biodiversity. To address this issue, the government has implemented a National Biodiversity Strategy and Action Plan that aims to promote sustainable mining practices.

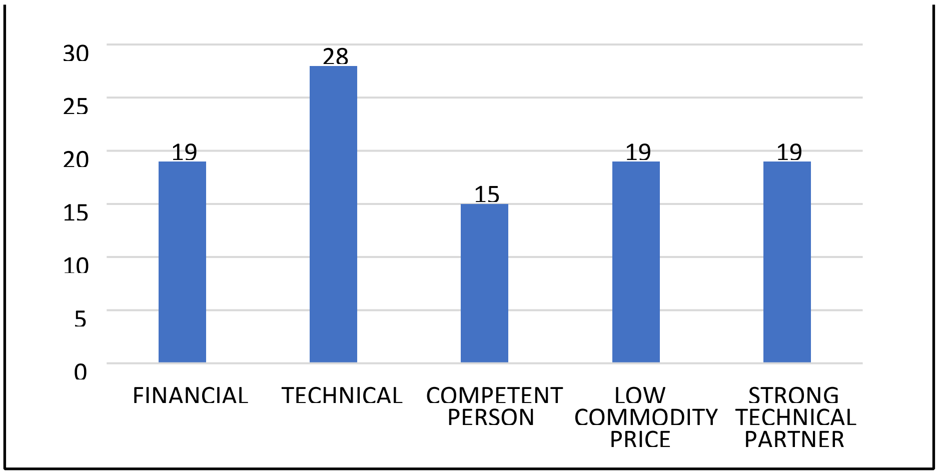


Figure 3.11: Some Components (%) Contributory to Failures in Alluvial Gold Operations in Ghana

Despite the stated challenges, the Artisanal and Small-Scale Gold Mining (ASGM) Industry is contributing significantly to the national gold production. See figure below.

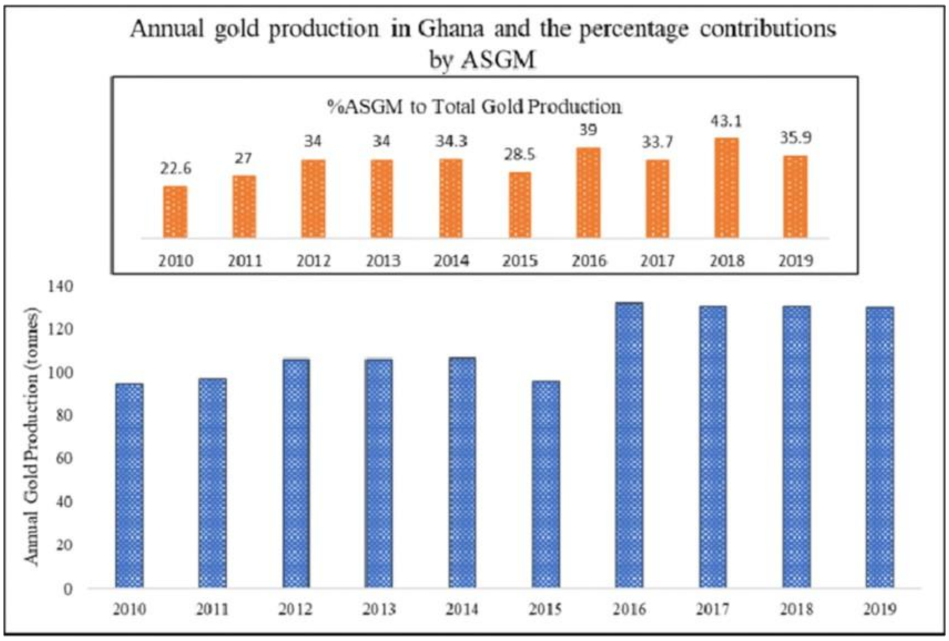


Figure 3.12: ASGM Contribution to Annual Gold Production in Ghana.

CHAPTER 4

4.1 ALLUVIAL GOLD EXPLORATION WORK ON THE OWORAM CONCESSION

4.1.1 Pitting and Sampling

Pitting exploration methods were applied to test the alluvial gold potential of the Oworam concession. Largely, the pitting method was used over the concession.

The planned exploration works were to determine the grade and size potential of the largely untested alluvial gold resource along the Ayensu River and its tributaries within the concession. The systematic alluvial gold exploration was aimed at identifying large volume areas with low to moderate grade values that could support a medium-to-large scale alluvial gold mining operation.

The strategy to effectively verify and evaluate the alluvial resource potential involved the following:

- Establishment of Local Grid System
- Pitting and Logging
- Sampling and Grade Determination

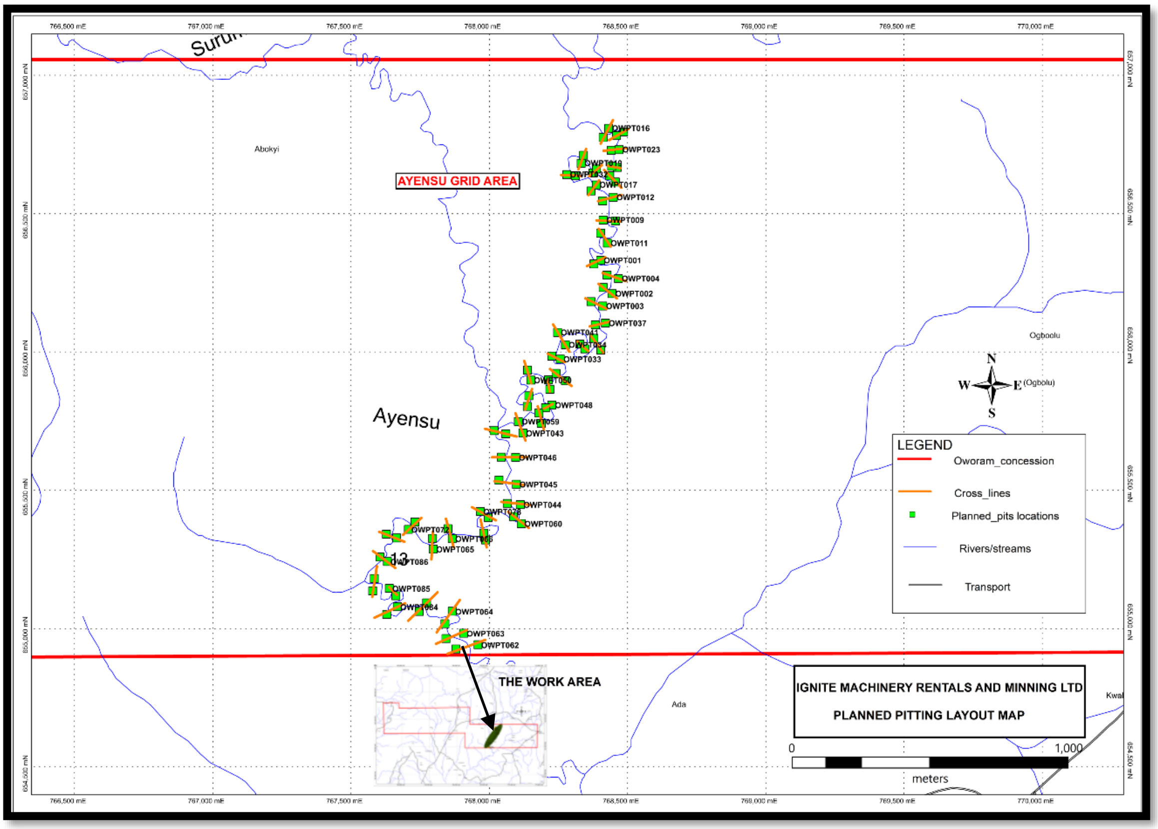


Figure 4.1: Planned exploration grid on the Eastern portion of Oworam concession.

4.1.2 Oworam Pitting and Sampling Processes

This pitting and sampling were carried to cover the eastern through the central portion of the concession from the south towards northern zones of the Oworam concession. One of the several streams of the concession area is the Ayensu River which flows from the south to the northern part through the eastern portion of the concession with its tributaries flowing to the east. The tributaries of the Ayensu and other streams which derive their source from the Atewa range flow through the concession. The prospecting covered 8.2 sqkm of the total concession of 32.57 sqkm. The pitting and sampling program were executed in a period of 3-months within the long Ayensu River area and accompanying streams. The program was executed to verify and outline the alluvial gold resources in the valley flats and terraces of Bosu and other streams for varying distances in the Oworam concession. The evaluation work also determined the grade and size potential of the alluvial gold reserve along the rivers and streams within the concession. A total of ninety-four (94) pits were planned along 100x 100m grid lines. The current pit area concentrated mainly along the Ayensu areas within the valleys of the Ayensu River and tributary flowing south. Pits were situated from the southern to northern portion towards the eastern boundary of the concession as well. The total number of pits sunk in all grids was 94 pits. Below are the bearing, baseline table and map of the planned pitting program.

Table 4.1. Bearing and baseline layout of the Alluvial Pitting Program.

Axis	Grid	Layout	
Axis Baseline	Grid Area	Bearing (mag)	Baseline Length
AYENSU	AYENSU	035°	3.2km

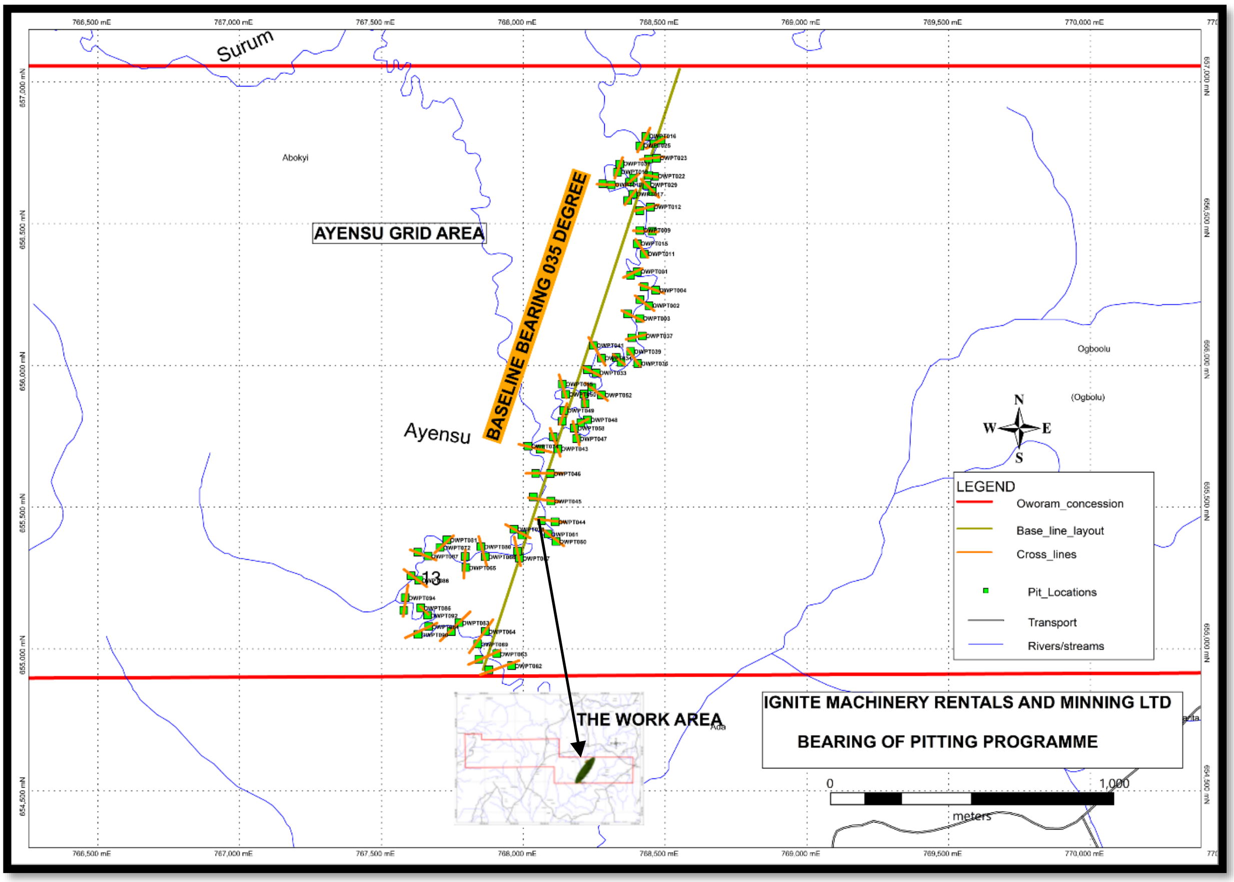


Figure 4.2 Map showing the bearing and baseline layout of the Pitting Program.

4.1.3 Pitting Techniques

Samples were recovered from the valley flat using 1-metre square pits sunk from the surface through the overburden and gravel layers and about 30 centimeters into the bedrock. The dimensions of the pits were carefully controlled to maintain the 1-metre square cross-section. A good number of pits required pumping to control water and cribbing to support the walls during excavation. All gravels recovered from the pits were labeled and later sluiced to obtain concentrates. The lithology and characteristics of the alluvium were also logged.

Timbering was used where the gravel layer was found to be water-saturated and where water inflow could cause caving or sample contamination. Frames are placed in the pit at 0.5m intervals with each frame tied to the previous one. When a point about 0.5m above the gravel layer has been reached, planks are sent down the sides of the pit between frames and the pit wall, to seal off the pit wall from the inside working area. The pit is then deepened using digging bars, removing the gravels by buckets on ropes. Gravel samples are then separated into 0.5m intervals as the pit is deepened. The planks are continually hammered down to the bottom of the pit to ensure no run-in of gravels.

Shallow water tables and high seepage rates necessitated the use of 2inch and 3inch self-priming water pumps to keep water out of the pits while excavation is in progress.

The soil profile in the Oworam area consists of a thin humus topsoil (dark clayey sand) that grades into an overburden of brown clayey soil devoid of pebbles. The gravel horizon that lies below the overburden is normally covered by a thin marker layer of greyish sand. The gravel layer is dominated by poorly sorted sub-angular to sub-rounded quartz pebbles and cobbles. The volume of gravel collected from each hole at 0.5 intervals from the bottom of the overburden to about 0.3m into the bedrock was thoroughly washed and processed using a sluice box that was astro- turfed to trap ore minerals like gold. The weathered bedrock is processed separately as it frequently contains significant gold and mining invariably involves digging into the top layer of the bedrock.

Figure 4.3: Pitting at Oworam

4.2 SAMPLING AND GRADE DETERMINATION

4.2.1 Sampling Technique

Systematic sampling of the gravel layer was carried out at intervals of 0.5 meters. Pit bulk samples were normally taken over 0.5m vertical intervals, with the exception of shorter sample intervals immediately above and into the weathered bedrock. Each $0.5m^3$ sample was placed on tarpaulin or plastic sheet beside the pit and labeled before removal for processing. Sampling in this manner was conducted over the entire gravel horizon exposed in the pit.

For the material excavated from the pits over each 0.5m interval, volumes were measured in a wooden box of 0.1m3 capacity (0.5x0.5x0.4m) and then stored on a plastic sheet.

In each pit excavated, the gravels were removed, head-carried by contract Labour and stockpiled on tarpaulins/plastic sheet at the treatment site.

4.2.2 Sample Processing and Treatment

The pit gravels were processed through a small portable alluvial wash-plant ("Proline Alluvial Prospector") to recover concentrates. The plant consists of small sluice box with a water pump machine and a 6mm screen deck. The plant employs the gravity method to hold concentrates of gold. High pressure water to wash the gravels which are fed by hand and partially scrubbed.

Each sample was fed onto the sluice box which is fitted with riffles and astro turf matting and puddled to remove clay and separated into plus 6mm and minus 6mm fractions. After washing, the entire minus 6mm fraction was reduced into a concentrate of gold and other heavy minerals by panning in a circular wooden bowl.

Each panned concentrate was physically dressed to recover all heavy minerals other than gold at the project camp site. Raw gold or free extracted gold by mercury amalgamation were then weighed with an electronic balance to the nearest milligram.

Figure4.4: Processing of pit samples at Oworam

4.2.3 Grade Determination

Grades in grams per cubic meter were calculated from the known gravel volumes and weighed gold. The grade was calculated by dividing measured sample volume expressed in cubic meters, by the weight of recovered raw gold in grams.

4.2.4 Chemical Characterization

The alluvial concentrate samples are mainly made up of precious gold minerals and weathered oxide minerals of iron, titanium, silicon and potassium.

4.2.5 Mineralogical Appraisal

The alluvial concentrate samples mainly contained the ore minerals gold, ilmenite, rutile garnet, magnetite, hematite and some amounts of quartz and feldspar as gangue minerals derived from their host Birimian volcanic and granitoid rocks.

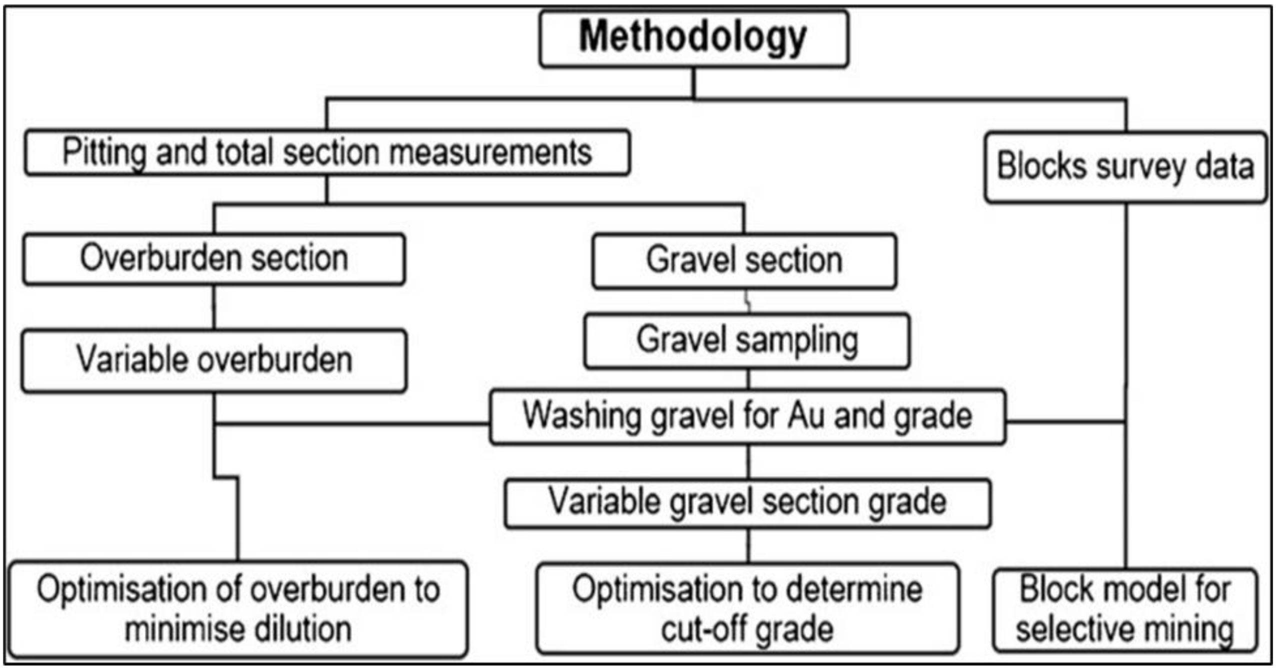


Figure 4.5: Characterization of Alluvial Gold Exploration Data to Improve Gold Recovery

Alluvial gold recovery methods involve extracting gold from riverbeds and other areas where it's deposited by water. Common methods include panning, sluicing, dredging, and more advanced techniques like gravity separation, flotation, and cyanide leaching. The choice of method depends on factors like the size of the deposit, the size of the gold particles, and the desired recovery rate.

Methods for Alluvial Gold Recovery:

- **Panning:**

  A manual method where a shallow pan is used to agitate sediment with water, allowing heavier gold to settle at the bottom.

- **Sluicing:**

  A sluice box uses running water and riffles (small barriers) to trap gold particles. Miners shovel material into the box, where gold is caught in riffles while lighter material flows out.

- **Dredging:**

  Dredging involves using suction or mechanical buckets to remove gold-bearing sediment from riverbeds, often used in large-scale operations.

- **Trommel Screens:**

  Rotational cylindrical screens used to wash and separate materials by size, often used in conjunction with sluices or shaking tables.

- **High bankers and Wash Plants:**

  Portable wash plants with highbankers, trommels, and gold concentrators for washing, classifying, and concentrating gold in one integrated system.

- **Gravity Separation:**

  A method using the density difference between gold and other minerals to separate them, commonly used with equipment like jigs, shaking tables, and spirals.

- **Flotation:**

  A method where gold is attached to bubbles and floated to the surface, separating it from other minerals.

- **Amalgamation:**

  A method that uses mercury to form an amalgam with gold, although it's discouraged due to environmental and health risks.

- **Cyanide Leaching:**

  A chemical process that dissolves gold in a cyanide solution, which is then recovered.

4.3 RESULTS AND INTERPRETATION

4.3.1 Nature of Alluvial Deposit

The alluvial profile mapped in the pits, was overburden, consisting of topsoils from 0.2 to 4.2m, clays and sandy clays, between 0.2m and 4.2m in thickness, overlying unconsolidated sand to pebble/cobble-sized alluvial deposits with average thickness between 1.5m and 0.3 to 2.6m.

The alluvial deposit is an example of a narrow high-energy gravel channel deposit in which the gold is generally medium to fine but with significant proportion of fine gold.

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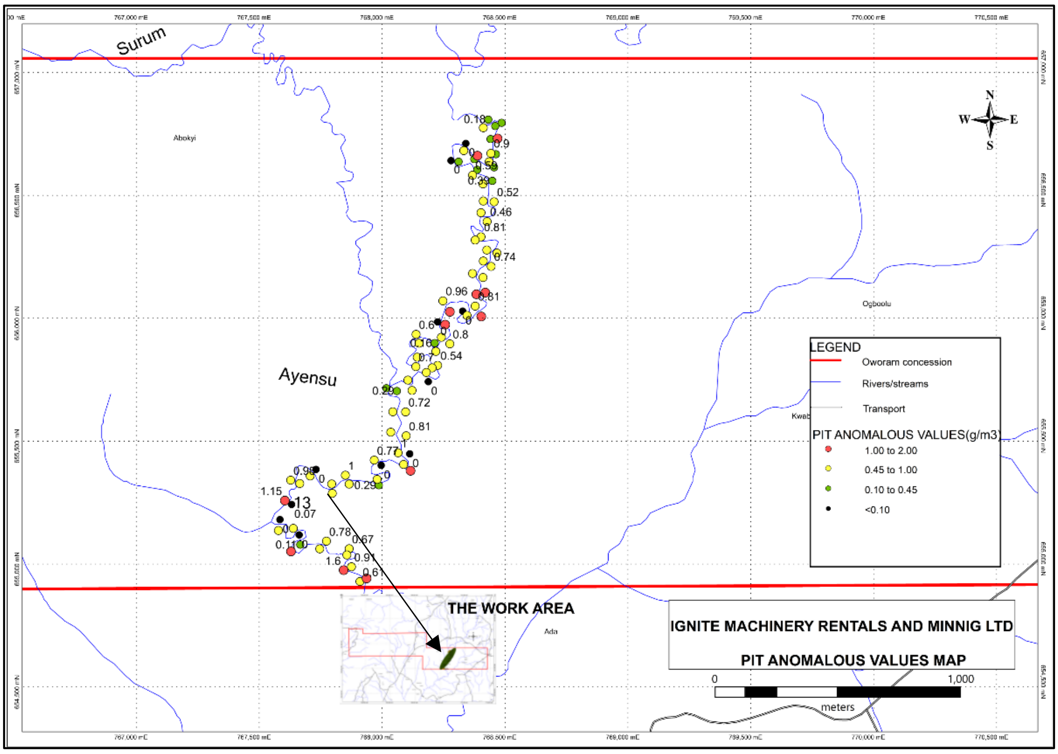

Figure 4.6: Assay plot of Gold Anomalies, AYENSU Grid-Oworam Concession.

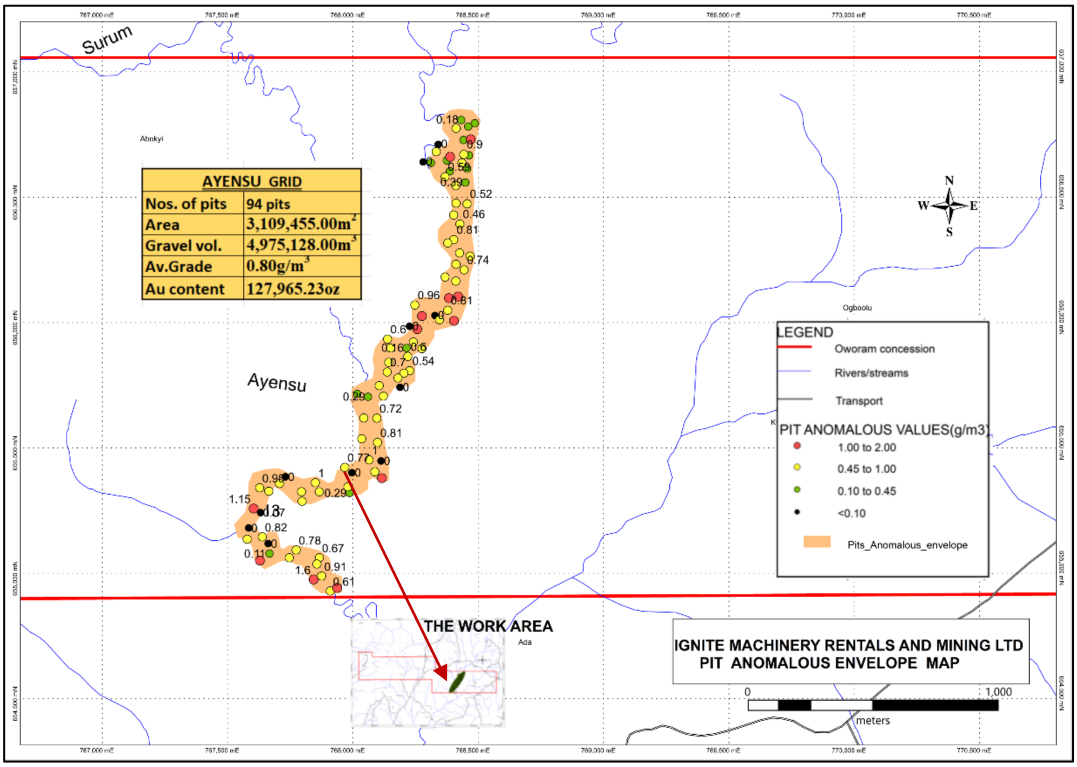

Figure 4.7: Model of Gold Anomaly Envelope, AYENSU Grid-Oworam Concession.

Exploration targets results summarized as:

- Average Gold Grade (Avg) 0.80g/m^3
- Average Ore Thickness 1.6m
- Average Overburden thickness 1.2m

Table 4.2: Pit Results-Gold Grades in Grid Areas

Areas	Range of Values (g/m^3)	Av. Grade (g/m^3}
AYENSU	0.07-1.60	0.80

Gold grades recorded in the Ayensu grid area are in the range 0.07-1.60 g/m^3 and averages grade is 0.80g/m^3 Gold is found throughout the gravel section although higher values and coarser flakes are usually concentrated in the bottom meter and 0.1 meters into weathered bedrock. Since the alluvial deposit is an example of a narrow, high-energy gravel channel deposit in which much of the gold recovered is fine to flour gold, much larger samples (mechanically excavated samples) would have yielded much more reliable and satisfactory results.

4.3.2 Overburden

The overburden exposed and examined is usually massive and comprised of soft to moderately soft mottled yellowish-brown clays, pale brownish clay and sandy clays. In some areas close to the terraces, thin horizons of ferruginous or lateralized layers have been encountered. The overburden thickness in all the gridded areas investigated ranges from 0.20m to 2.40 m in the Ayensu area and averaging 1.20m in the Ayensu Grid area and these are summarized in Table below:

Table 4.3: Pit Results-Overburden Thickness in Gridded Areas

Areas	Range of Thickness (m)	Av. Thickness (m)
AYENSU	0.20-2.40	1.20

The table above indicates the thickness of the Ayensu grid area. The thickness ranges from 0.20m to 2.40m and averages 1.20m at the AYENSU grid.

4.3.3 Gold-bearing Gravel

The beds underlying the overburden, collectively referred to as gravels, consist of clasts, cobbles and pebbles with a matrix of sand, silt and clay. The dominant rock type in the coarse clasts is white and brown stained quartz. Most of the coarse clasts are sub-angular to sub-round in shape. The matrix is quartz sand, silt and clay with minor amounts of heavy minerals. Within the gravel sections, the larger clasts occur close to the bedrock contact. Post depositional consolidation of gravel is moderate.

Table 4.4: Pit Results-Thickness of Gravel Beds in Gridded Areas

Areas	Range of Thickness(m)	Av. Thickness (m)
AYENSU	0.50-2.60	1.60

The table above indicates that the range of thickness of gravel layer range from a thickness of 0.50m to 2.60m with average of 1.60m at the Ayensu grid within the Oworam concession.

4.3.4 Bedrock

Bedrock, where identified in the pits, is typically basaltic flow, granodiorite and hornblende-biotite diorite that are metamorphosed. Mostly the pitting ended upon the observation of the bedrock. Where possible, a minimum of 30cm of bedrock was excavated at the bottom and this is aimed at checking for gold that may have migrated downwards and possibly allow for dilution during the mining operation.

4.3.5 Engineering Data

Various checks were performed on the gravels to determine the swell factor and particle size of the gravels. Gold particle size tests were also performed on the gold recovered from the sampling program and summarized in Tables

4.3.6 Swell Factor

For the material excavated from the pits over each 0.5m interval, volumes were measured in a wooden box of 0.1m^3 capacity (0.5x0.5x0.4m) and then stored on a plastic sheet. Material from the water table showed a fairly consistent swell factor of 22%, compared with a range of 30% to 70% in materials above the water table.

4.3.7 Gravel Particle Size Distribution

Samples of the gravels from sections of the river systems in the baseline areas were taken, dried and soured by hand and screened to establish approximate particle sizes of the gravels. The results are tabulated in the table below:

Table 4.5: Gravel Particle Size Distribution

Screen Size (mm)	% Fraction
+100	14
+50	22
+6	26
-6	14
Clay/Silt	24

From the analysis, it can be noted that 62% of the gravel is expected to fall in the +6mm range and above.

4.3.8 Gold Particle Size Distribution

Gold particle size test was performed on the gold recovered from the test pits at the Ghana Geological Survey Authority in Accra. Gold is generally medium to fine grained. Results from the test are summarized in Table below.

Table 4.6. Distribution of Gold Particle Sizes

Appearance	Sieve Mesh Size	Gold Particles (%)
Coarse	+20	4
Medium	+30	34
Fine	+60	43
Very Fine	+150	8
Flour Gold	-150	11

The breakdown shows a distribution of 77% +60 mesh and +30% mesh sizes together.

4.4 MINERAL RESOURCE

4.4.1 Estimate Resource Potential

The essential parameters used for the resource estimation on the Oworam concession was design to test and confirm gold grades and vital geological statistics needed to produce a resource estimate for investment analysis and budgeting. Among the key parameters are:

- ❖ Average gold grade
- ❖ Gold size distribution
- ❖ Gravel thickness
- ❖ Overburden thickness
- ❖ Gravel characteristics

The polygonal method of reserve estimation was used in calculating the total mineral resource of the eastern portion of the Oworam Concession. The method is a more traditional approach to reserve evaluation in order to arrive at a compliant mineral reserve estimate. Further work involving pitting and bulk sampling will be required if the current pit data is to be used in reserve estimation by the geostatistical approach. The Oworam alluvial deposit is an example of a narrow, high energy gravel channel deposit which requires much closer sample spacing.

4.4.2. Evaluation Procedure

The total pit section grade was calculated as the weight of gold (in grams) obtained from the gravel horizon of a pit divided by the in-situ volume of the entire pit. The average of the depths of all the gravel horizons within the pay zone was then computed. The pay-zones for this estimation were grades greater than or equal to $0.5g/m^3$, however other areas with grades above 0.45g/m3 may be mined in future when the initial reserve is exhausted. The volume of gravel within the pay zone was calculated as the product of the average gravel horizon thickness computed and the surface area of the delineated pay zone. The average gravel grade of the pay deposit was estimated as simple arithmetic average of all the pit gravel section grades within the pay zone. The quantity of gold in grams was then computed as the product of the average gravel grade of the pay zone and the volume of gravel estimated within the pay zone.

Table 4.7: Resource table for Oworam Boundary

PARAMETERS	QUANTITY	UNITS
Average Gold Grade (Avg)	**0.80**	**g/m^3**
Ore Thickness (gravel)	**1.60**	**m**
Average Overburden thickness	**1.20**	**m**
Total area of deposition	**3,109,455.00**	**m^2**
Total volume of ore (TVg)	**4,975,128.00**	**m^3**
Gold content (Potential Reserve)	**3,980,102.40**	**g/m^3**
Potential Reserve(oz)	**127,965.23**	**oz**
Gross Present value of ore (Base rate $3,000)	**383,895,675.66**	**USD**

The estimation of the Oworam resource is based on a 100x100m spaced pitting program. Since much of the gold is fine, much larger samples are required to yield satisfactory results. These two factors favor mechanical pitting (excavator pits), which yields much larger samples. Based on the information and data obtained during the pitting program, the volume estimate of the potential reserves of gold- bearing gravels along the channel and valley flats of the AYENSU grid, approximately **4,975,128.00 m^3**. This was estimated while considering a lower cut-off of $0.5 g/m^3$. The total gold content is **3,980,102.40 g/m^3** and the total gold content in ounces is **127,965.23oz**. The cut-off gives room to expand the area of interest if a much lower cut-off is considered as shown in the table below.

Table 4.8 Resource Table for Gravel and Gold Content at The Oworam Alluvial Prospecting Area

Grid	Average Gravel Thickness (m)	Block Area in (Sqm)	Block Gravel Volume(m^3)	Average Grade ((g/m^3))	Gold Content (g)	Gold Content (oz)
AYENSU	**1.60**	**3,109,455.00**	**4,975,128.00**	**0.80**	**3,980,102.40**	**127,965.23**

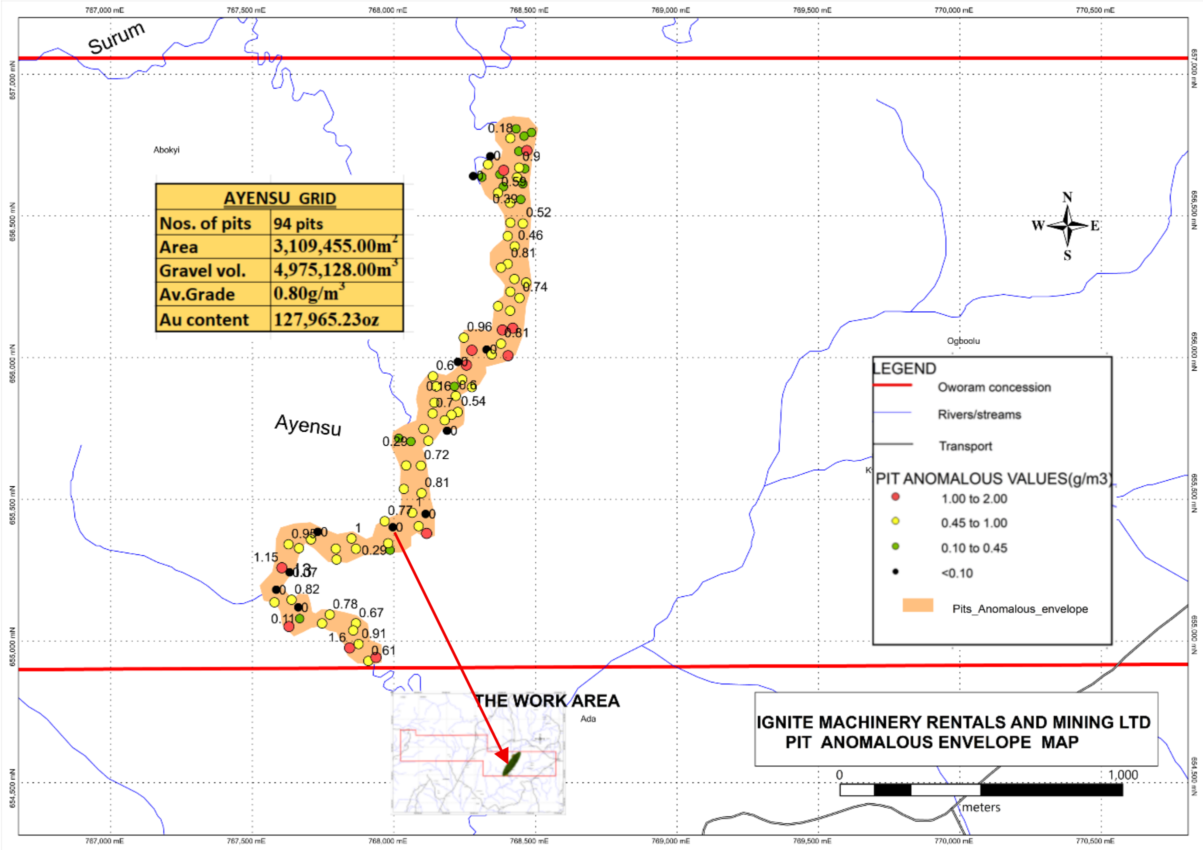

Figure 4.8: Model of Gold Anomaly Envelope- AYENSU Grid

The alluvial gold evaluation for the entire project area, established the existence of gold- bearing gravels with an average gold grade of 0.80g/m^3 and an average gravel thickness estimated at 1.60 m in pay-zone areas with a total area of 3,109,455.00m^2. Based on these results, the potential gold-bearing gravel volume in the valley flats and channels of the Concession are estimated to be in excess of 4,975,128.00 m^3. The in-situ gold content in the gravel block is estimated to be in excess of 3,980,102.40g thus estimated in ounces as 127,965.23 oz. The estimation was based on field data pitting projects carried out on the Oworam concession.

4.5 ESTIMATION OF MINE LIFE

Total Volume of gold bearing gravel estimated in the concession = 4,975,128.00 m^3

Capacity of Trommel = 160m^3/hr.

For 8 hours shift for a day, the total volume of gravel = 160m^3/hr. x 8hrs = 1280m^3

For a period of six working days per week, 1280m^3 x 6 days = 7,680m^3

For a period of four weeks per month, the total volume of gravel, 7,680 x 4wks = 30,720m^3

Total volume of gravel for one year (11 months), 30,720m^3 x 11 = 337,920.0m^3

NB: Management of IGNITE MACHINERY RENTALS AND MINING LTD intends to operate a day and night shift

The total volume of ore processed for two shifts (16 hours per day) in a year is

337,720 x 2 = 675,840 m^3

Hence the mine life of the main plant unit of Oworam Mine is

4,975,128.00 m^3 ÷ 675,840.00m^3 = 7.36 years

Therefore, the mine life for Ignite Machinery Rentals and Mining Ltd is approximately 7years.

Table 4.9: Summary of Mineral Reserved for Ignite Machinery Rentals and Mining Ltd

Grid	Average Gravel Thickness (m)	Block Area in (Sqm)	Block Gravel Volume(m^3)	Average Grade ((g/m^3))	Gold Content (g)	Gold Content (oz)	Gross present value of ore ($)3,000/oz
Total	1.60	3,109,455.00	4,975,128.00	0.80	3,980,102.40	127,965.23	383,895,675.66

Table 4.10: Summary of Mineral Reserved, Plant Equipment Setup and Mine Life for Ignite Machinery Rentals and Mining Ltd Oworam Concession

SUMMARY OF MINERAL RESERVED, PLANT EQUIPMENT SETUP AND MINE LIFE		
(PARAMETERS)	(QUANTITY)	UNITS
Total Volume of Gravel (m^3)	4,975,128.00	m^3
Plant Capacity	160.00	m^3 /H
Number of Plants	1.00	
Total Volume (M^3) Of Gravel Washed/Shift-Day (8hrs)	1,280.00	m^3
Total Volume (m^3) Of Gravel Washed Per Week (6 Working Days)	7,680.00	m^3
Total Volume (m^3) Of Gravel Washed Per Month	30,720.00	m^3
Total Gravel Washed Per Year (Single Shift-11 months)	337,920.00	m^3
Total Gravel Washed Per Year (Day & Night Shift)	675,840.00	m^3
Mine Life (total volume/gravel washed per year)	7.36	years
Mine Life (Years)-(LOM)	Approx. 11 years	

4.6 RESOURCE/RESERVE

The Potential Mineral Reserve of the entire Oworam concession is estimated to be 127.965.23 oz.

The estimate was defined based on exploration results obtained from the alluvial pitting and sampling programme on the entire concession. The assessment involved the calculation of average thicknesses of gold-bearing gravels for the resource areas in order to establish the volume of gold-bearing gravels in Oworam concession.

The alluvial gold pitting and sampling program in the project area, established the existence of gold-bearing gravels with an average gold grade of 0.80g/m^3. Based on these results, the potential gold bearing gravel volume in the entire concession is estimated to be in excess of 4,8975,128.00 m^3.

CHAPTER 5

5.0 MINING PROJECT

The Oworam Project will adopt the most practical method of exploiting the measured reserves, which is strip-mining, utilizing excavators for both removal of overburden and the excavation of the gold bearing gravels. The gold resources in the Oworam concessions are primarily concentrated in the gravel horizon, with very little occurring in the saprolite of the bedrock. Before removing the waste, the management of Ignite Machinery Rentals and Mining Ltd intends to conduct rigorous grade control and sampling within the concession boundaries of the delineated ore zones during the mining operation.

An excavator will remove the lateritic clay overburden up to the top gravel contact, which is often indicated by a horizon of white marker sand. The removed overburden material will be stockpiled in anticipation of its eventual use for reclamation purposes. The primary mining machinery to be used are the bulldozer, excavator, and dump trucks.

In selecting a mining technique for the project, the following have been considered:

- The mining method should be technically feasible and economical.
- The mining method should maintain a high standard of safety for the workers and inhabitants.
- The mining should have minimal impact on the environment.

The grade of the deposit necessitates a high-capacity, low-cost mining method. All the above requirements are met by the mining method chosen. The Oworam Project will employ the most practical method of exploiting the proven (measured) reserves, which is strip-mining with excavators for both overburden removal and excavation of the gold bearing gravels. The conditions for dry stripping techniques are ideal, allowing for a quick and cost-effective extraction rate.

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5.1 STRIP MINING CONCEPT

The Strip-mining approach is grounded in a tried-and-tested technique used for mining placer deposits that share similarities with the Oworam deposit. This method has demonstrated success in various locations in Ghana and other regions of West Africa over the years. The proposed strategy for extracting the confirmed (measured) reserves at the Oworam concession involves a phased approach, including:

- Surface Preparation
- Overburden Removal
- Excavation and loading of the ore
- Transportation of the ore to the processing plant
- Processing of the ore
- Restoration

5.1.1 Surface Preparation

In anticipation of subsequent mining activities, the initial phase involves the meticulous preparation of the mining site. To clear the area, a chain saw machine will be employed to fell trees. During this process, valuable timber species will be selectively collected for utilization in civil works within the mining area. The remaining timber will be cut into manageable pieces and dried, providing a sustainable source of firewood for local residents.

Furthermore, the surface preparation will entail the systematic removal and stockpiling of topsoil. This nutrient-rich layer, which serves as the foundation for the existing vegetation, will be strategically preserved for future re-vegetation efforts. The use of a Caterpillar D6 or D7, is deemed suitable for the efficient execution of this task. This method ensures a careful and environmentally conscious approach to site preparation, emphasizing the importance of sustainability in the mining process.

5.1.2 Overburden Removal

Overburden, the non-ore material overlaying the valuable ore deposits, must be systematically cleared to facilitate access to the desired resources. In our mining approach, the overburden will not be transported; instead, it will be strategically piled alongside the excavation for future utilization during the reclamation process.

The backhoe excavator is considered for the efficient removal of the overburden. This decision is based on a cost-effective approach. By opting for a backhoe and bulldozer combination, we ensure a balanced blend of efficiency and cost-effectiveness in managing the overburden removal process. This strategic choice aligns with our commitment to optimizing resource allocation while maintaining environmental stewardship throughout the mining operation.

5.1.3 Excavation and Loading of Ore

Following the successful removal of overburden, the ore is now exposed, marking the beginning of the excavation, and loading phase. Notably, some ore deposits may exhibit compact characteristics, demanding a substantial amount of force for effective excavation. Considering this, the selection of appropriate equipment becomes paramount.

Given the need for robust force in ore excavation and considering pit conditions, the backhoe excavator emerges as the preferred choice. This decision aligns with our commitment to operational efficiency and cost-effectiveness. The backhoe excavator's versatility and higher breakout force make it well-suited for the challenges posed by compact ore deposits, ensuring a seamless and effective excavation and loading process.

5.1.4 Transportation of Ore

Transporting the gold-bearing gravel from the mining site to the processing plant is a crucial aspect of our operational plan. The haul distances may vary, spanning from a few hundred meters to several kilometers. To efficiently carry out this task, heavy-duty dump trucks have been strategically chosen for their robust capabilities and load-bearing capacity.

The selected heavy-duty dump trucks are designed to navigate diverse terrains, ensuring the secure and reliable transportation of the ore material. Their suitability for varying distances within our operational range positions them as the optimal choice for this crucial phase. This decision underscores our commitment to a well-planned logistics strategy, enabling the seamless and timely delivery of the gold-bearing gravel to the processing plant.

5.1.5 Processing of Ore

The processing of the extracted ore will be carried out through a systematic series of operations within a dedicated plant. This facility is equipped with essential components to ensure efficient recovery and extraction. The key elements of the processing plant include; Vibrating Grizzly, Scrubber cum Screen Trommel, Knelson Concentrators and a Gemini Shaking Table. This integrated approach to ore processing underscores our commitment to employing advanced technologies for optimal recovery and resource utilization.

5.1.6 Restoration

In an integrated approach to resource management, the processing plant tailings, composed of washed boulders, cobbles, pebbles, and sand, will serve a dual purpose. These tailings will be systematically conveyed to the mined-out areas, contributing to backfilling operations. Importantly, this backfilling process will run concurrently with the mining operations, ensuring a comprehensive and sustainable utilization of extracted materials. The logistical efficiency of this process is maximized by utilizing the same dump trucks responsible for transporting ore to the processing plant. On their return journey to the mining face, these trucks will be loaded with washed gravel or tailings, minimizing transportation-related overhead.

To facilitate this multifaceted operation, specialized equipment will be strategically positioned at the tailings site. This machinery not only loads dump trucks efficiently but also plays a crucial role in de-silting the sandpit. Considering the diverse responsibilities, an excavator with an extended boom and a light to medium weight profile emerges as the ideal choice for this task. This tailored approach aligns with our commitment to optimizing operational efficiency and environmental stewardship.

5.2 EQUIPMENT AND MATERIAL REQUIREMENTS

5.2.1 Mining Excavators

As previously outlined, a single backhoe excavator, complemented by a bulldozer, will play a pivotal role in both overburden removal and ore excavation and loading. The choice of excavator size is directly correlated with our planned production rate, a crucial factor in ensuring operational efficiency.

The treatment plant designated for use boasts a capacity of 160m^3/h. Consequently, the selected excavator must possess the capability to strip overburden and load more than 100m^3 of gravel within a one-hour time frame. Given a stripping ratio of 1:2, where stripping precedes mining to expose ore for extraction, an additional excavator, supported by a dozer, will be employed specifically for the stripping process ahead of mining operations.

To meet our production goals and create a sufficient ore stockpile for the treatment plant, the ore mining crew aims for a production rate of 100m^3/h. The choice of the ore mining excavator will be made to achieve this targeted mining rate. The productivity of backhoe excavators, particularly under medium digging conditions, is outlined in Table 5.1, guiding our selection process for optimal performance and output. This meticulous approach to excavator selection aligns with our commitment to precision and efficiency in meeting production targets for a seamless mining operation.

Table 5.1 Productivity of Backhoe Excavators

Bucket Size (m^3)	Cycle Time T(sec)	Production Rate Q(m^3/h)
0.9	17	79.9
1.0	17	87.8
1.2	18	98.7
1.4	19	107.7
1.6	19	121.8
1.7	20	129.9
2.3	21	162.8
2.8	22	186.7
3.7	24	227.2
4.6	26	262.5
9.1	27	205.5

For the required production rate, an excavator with a bucket size of 1.4 - 4.6 m^3 will suffice. In this case, a CAT 345 excavator with 1.5 - 4.6 m^3 bucket capacity or equivalent will be used.

5.2.2 Feeding Excavator

To efficiently feed the processing plant with a capacity of 100m^3/h, an excavator with a minimum capacity of 2.5m^3/min is required. Therefore, it is recommended to employ a CAT 345 excavator or an equivalent model for this essential function.
The selection of this specific excavator ensures not only the necessary capacity for plant feeding but also takes into consideration the operational efficiency and reliability associated with the CAT 345 or its equivalent. This strategic choice aligns with our commitment to using equipment that meets the demands of the processing plant while maintaining high standards of performance.

5.2.3 Tailings Handling Excavator

In adherence to our design considerations, a single excavator will be assigned the crucial tasks of de-silting the sandpit and loading all tailings material onto trucks. Effective desilting requires the excavator to have a relatively long boom for optimal reach and performance.

However, it is essential to strike a balance between functionality and weight to avoid sinking while cleaning the sandpit. In consideration of these requirements, both the CAT 320 and Daewoo 220 excavators are deemed suitable for this dual-purpose operation. Their design features, including a long boom for effective desilting, coupled with an optimal weight profile, make them well-suited for the demands of the task at hand.

This strategic choice ensures that our equipment is not only capable of efficient sandpit maintenance but also minimizes the risk of environmental impact, aligning with our commitment to responsible and sustainable mining practices.

5.3 TRUCKS

The choice of trucks for specific tasks is directly influenced by the size of the loader used in the excavation process. In this project, the proposed mining excavator is designed with a 2.5m^3 bucket size. The volume estimation for a 2.5m^3 excavator operating under optimum conditions is detailed in Table

Table 5.2 Volume Estimation of Truck Loaders

No. of Passes	Volume (m^3)
6	15
7	17.5
8	20
9	22.5

From Table above, it is determined that 22.5 m^3 CAT articulated trucks, requiring nine passes of the excavator per truckload, are a suitable choice for efficient material transportation.

The trucks will carry ore to the treatment plant and carry tailings on their return trip. Cycle time studies of similar trucks working under similar conditions in a similar mine in the country have an average cycle time of about 18 minutes. An average cycle time of 18min is therefore used to determine the number of trucks required to meet the production target.

Truck output formula is given as:

$$P = 60 * C * \frac{E}{T}$$

Where P = Truck Output (m^3/h)

C= Truck Capacity (22.5 m^3)

E= Truck Efficiency

T = Truck Total Cycle Time (18mins)

Assuming an Efficiency of 85%

$$P = 60 * 22.5 * \frac{0.85}{18}$$

$$P = 80m^3/h$$

$$Number\ of\ Trucks = \frac{\text{required production}}{\text{truck output}} = \frac{160m^3/h}{80m^3/h} = 2.0 \approx 2.0$$

Therefore, two trucks will be sufficient. However, to take care of maintenance schedules an additional truck will be required. Two trucks will therefore be needed for the project. Table shows the machinery requirements for the operation. Table shows the machinery requirements for the operation.

Table 5.3 Summary of Selected Equipment

Project Item	Quantity
CAT 14G Grader	1
CAT 345 Excavator	1
CAT 966F Front End Loader	1
CAT D6 Bulldozer	1
Dewatering Pump	2
Dump Trucks	2
Water Tanker	1

Generator	4
Gold Trommel Wash Plant	1
Nissan Bus	2
Office Equipment	1
Office furniture & fittings	1
Pick Up (4 X 4)	4
Station Wagon	4
Tractor Loader Backhoe	1
Utility Truck	2

5.4 MINE SAFETY

5.4.1 Geotechnical Considerations

The alluvial gold is embedded within loose gravel, predominantly composed of quartz pebbles and cobbles cemented in a clayey sand matrix. The extraction of these unconsolidated gravels for treatment in the washing plant does not involve hard rock drilling or blasting with dynamite. To uphold the highest standards in geotechnical aspects, the company has implemented comprehensive procedures and protocols in collaboration with a private geotechnical consultant. This ensures the effective management of alluvial surface mining operations.

Monitoring and Deformation: Rigorous geotechnical monitoring is conducted on pit wall faces, upper slopes, and accessible benches to assess their rates of deformation. This proactive approach helps in identifying and addressing potential stability issues promptly.

Settling Pond Construction: The sites chosen for the construction of settling ponds are underlain by weathered sedimentary rocks covered by overburden. To guarantee the stability and longevity of the settling ponds, a thorough foundation investigation will be carried out. This investigation aims to identify and analyze key geotechnical factors that need consideration during the construction phase.

Key Considerations:

1. **Slope Stability:** Ensuring the stability of the side walls of the ponds is of paramount importance. Rigorous analysis and reinforcement measures will be implemented to mitigate the risk of slope instability.

2. **Water Seepage Prevention:** Preventing potential water seepage through buffer zones during heavy rainfall is a critical factor. Robust engineering solutions will be employed to safeguard against any adverse effects of water seepage.
3. By integrating these geotechnical considerations into the mining and construction processes, the project aims to ensure the safety, environmental sustainability, and long-term viability of the project.

5.4.2 Dewatering and Surface Run-off Management

Processed water will be recycled as required to meet discharge permit regulations and support mining operations during low precipitation periods, ensuring a zero-discharge operation at the project site. Collected water includes precipitation pumped from the mine, direct precipitation on the dust control pond, Bulk Storage Reservoir, off-channel storage reservoir, and dewatering well yields.

To manage inflows to the Tailings Storage Facility (TSF), tailings water and runoff will be directed to the Bulk Storage Reservoir, minimizing water storage in the TSF. Losses include water retained in tailings, evaporation, and seepage. The remaining water is recirculated to the processing plant via the Bulk Storage Reservoir, with recirculation from the tailing's thickener. Because the tailings will be non-segregating thickened tailings, the volume of water available from tailings storage for return to the process plant is significantly reduced in the later years of production.

A portion of this flow will meet processing plant freshwater needs, and excess water from wells can be used for process water or dust control. Permission from the Water Resources Commission (WRC) will be sought by the company if drawing from water bodies is necessary to maintain operational flow balance.

Mine-pumped precipitation is stored in the dust control pond. Any deficit for dust control is compensated with water from dewatering wells or the off-channel reservoir. In a wet year, excess runoff can be held in pit sumps, and unused water can be directed to the process plant.

The main sources of water inflow into the pit will be ground water filtration and surface run-off. Oversize tailings from the plant as well as overburden from the next cut will be dumped into the previous cut. This will be done immediately after mining to ensure minimum ground water seepage into the pit. However, during heavy downpours or mining at areas with high seepage of water, all water will be directed in the lowest point created by deepening the first bite from which the excess water will be pumped into the settling ponds. Surface run-off will be prevented from entering the pit by constructing small berms using overburden material around the perimeter of the pit to prevent excessive water entering the pits during rainfalls.

Alternatively, the preferred and least expensive option of dewatering a pit is to divert the water off the pit to a safe place downstream. Management of the company considers the construction of temporary drainage system 5-15 m behind the high wall to intersect storm water from hills and groundwater shedding from swamps into the pit. The water is then collected by gravity into a sump and discharged safely to a downstream water source. The company has conducted an estimate for sump dewatering based on the pump capacities and the effective pumping hours

5.5 PRE-STRIPPING

Topsoil, a precious resource considering its scarcity, should be extracted separately, stockpiled, and subsequently replaced as a surface covering during reclamation and restoration efforts whenever feasible. Placing the stockpile as close to the mining block as possible, without impeding subsequent operations, is crucial. It should remain in its designated location until the land is fully reclaimed. To preserve the quality and fertility of the soil source, minimizing topsoil stockpiling is essential to prevent compaction and reduced fertility. Soil handling, removal, and re-spreading should only be carried out in dry conditions.

Caution must be exercised during gravel excavation to minimize dilution. Rigorous monitoring and continuous sampling are imperative to keep under/over mining to a minimum, especially in regions with undulating bedrock. Expect higher gold values within the base gravel horizon, making it necessary to include a minimal amount of bedrock with the bottom gravels.

5.6 MINING (OVERBURDEN REMOVAL AND GRAVEL EXTRACTION)

The excavator is preferred due to its adaptability and resilience in handling challenging ground conditions. Its short cycle time and versatility extend its utility for general tasks across the mining site. The selected CAT 345, equipped with a 4.6m³ bucket, aligns perfectly with these operational needs. For the efficient transportation of gravel to the processing plant, the recommendation is for all-wheel drive articulated trucks. It's envisioned that these trucks will double their utility by transporting tailings back to the mined-out areas on their return trip. To maintain continual operations, four trucks are initially proposed, with a prudent addition of an extra unit to account for breakdowns and maintenance considerations.

The bulldozer takes on a pivotal role, primarily involved in clearing, grubbing, constructing access roads, and waste management. Anticipating wet conditions, the recommendation is for a Caterpillar D6 or its equivalent, featuring low ground pressure (LGP) undercarriage and extensive width tracks. This dozer also plays a complementary role, working alongside the excavator in the crucial task of stripping ahead of mining operations.

The machinery selection, including the excavator, trucks, and bulldozer, is meticulously tailored to ensure operational efficiency and adaptability to varying mining conditions. The inclusion of an extra truck as a backup underscores the commitment to uninterrupted operations, accounting for potential breakdowns and necessary maintenance.

5.7 LIFE OF MINE

Operational Parameters

Guided by process test work, the gravity separation processing plant will operate at a feed rate of $100m^3/h$ to ensure a buffer against equipment breakdowns and weather uncertainties.

Total Volume of gold bearing gravel estimated in the concession = $4,975,128.0m^3$

Capacity of Trommel = $160m^3/hr$.

For 8 hours shift for a day, the total volume of gravel = $160m^3/hr.$ x 8hrs = $1,280m^3$

For a period of six working days per week, $1,280m^3$ x 6 days = $7,680m^3$

For a period of four weeks per month, the total volume of gravel, 7,680 x 4wks = $30,720m^3$

Total volume of gravel for one year (11 months), $30,720m^3$ x 11 = $337,920m^3$

NB: Management of **Ignite Machinery Rentals and Mining Ltd** intends to operate a day and night shift.

The total volume of ore processed for two shifts (16 hours per day) in a year is

337,920 x 2 = $675,840\ m^3$

Hence the mine life of the main plant unit of Oworam Mine is

$4,975,128.00\ m^3 \div 675,840.00m^3 = 7.36$ years

Therefore, the mine life for Ignite Machinery Rentals and Mining Ltd is approximately 7years.

Life of Mine Calculation:

$$\text{Life of Mine(LOM)} = \frac{\text{Reserve}}{\text{Plant Annual Throughput}} = \frac{4,975,128.00\text{m3}}{675,840\text{ m3}} \approx 7 \text{ years}$$

As common industry practice, ongoing intensified exploration of the property is anticipated. Any successes in exploration efforts have the potential to extend the Life of Mine beyond the initial estimate. **Ignite Machinery Rentals and Mining Ltd** strategic planning considers operational parameters, maintenance schedules, and the potential impact of ongoing exploration activities, demonstrating a proactive approach to project longevity and adaptability to industry trends.

CHAPTER 6

6.0 MINERAL PROCESSING

6.1 GRAVEL PARTICLE SIZE DISTRIBUTION

Samples of gravel sourced from specific sections of the river systems in the baseline areas underwent a meticulous process. The process included drying, manual sorting, and screening to establish approximate particle sizes of the gravels. The results are presented in Table below.

For the material extracted from the pits at each 0.5m interval, volumes were measured using a wooden box with a capacity of $0.1m^3$ (0.5x0.5x0.4m). Subsequently, the measured material was carefully stored on a plastic sheet. Materials excavated from the water table consistently exhibited a swell factor of approximately 20%. In contrast, materials above the water table demonstrated a broader range, spanning from 30% to 70%.

These findings provide valuable insights into the characteristics of the extracted gravels, crucial for effective mineral processing and subsequent gold recovery efforts.

Table 6.1 Distribution of Gold Bearing Gravel

Screen Size (mm)	% Fraction
+100	14
+50	22
+6	26
-6	14
Clay/Silt	24

From the analysis, it can be noted that 62% of the gravel is expected to fall in the +6mm range and above.

6.1.1 Gold Particle Size Distribution

Gold particle size test was performed on the gold recovered from the test pits at the Ghana Geological Survey Authority in Accra. The gold is generally medium to fine grained. Results from the test are summarized in Table below.

Table 6.2 Distribution of Gold Particle Sizes

Appearance	Sieve Mesh Size	Gold Particles (%)
Coarse	+20	4
Medium	+30	34
Fine	+60	43
Very Fine	+150	8
Flour Gold	-150	11

The breakdown shows a distribution of 77% +60 mesh and +30% mesh sizes together.

6.1.2 Gold Fineness Determination

The gold specks were digested with aqua-regia and the subsequent determination of the grade was achieved through Atomic Absorption Spectroscopy (AAS). The outcome revealed a gold purity of 90%, equivalent to 21.59 carats. Notably, silver emerged as the predominant impurity, constituting 4% of the composition. It's imperative to acknowledge that without conducting broader and more comprehensive testing, the exact percentage of minerals present in the saleable concentrate remains unknown, underscoring the importance of future analyses for a more detailed understanding.

6.1.3 Gold Liberation Characteristics

Within the confines of the Oworam concession, the gold particles stand as liberated entities, a state often referred to as free gold. Notably, the predominant presence of small quartz inclusions nestled within larger nuggets serves as the primary source of locked gangue particles. Encouragingly, the gangue encapsulated within these larger nugget particles can be readily separated and removed during the subsequent smelting process. This liberation characteristic bodes well for the efficient extraction and processing of gold from the geological matrix.

6.1.4 Pre-Concentration Amenability of Ore

The gravel within the Oworam concession exhibits favorable characteristics for pre-concentration. A comprehensive gravel particle size analysis reveals that 54% of the gravel falls within the +6mm range. The pre-concentration process will be seamlessly integrated into the washing and screening operations. This strategic approach involves the removal of all particles larger than the anticipated gold size, effectively enhancing the grade of the remaining material. The incorporation of pre-concentration techniques is a pivotal step, ensuring the optimization of ore processing and the extraction of valuable gold resources.

6.1.5 Gravel Washing Characteristics (Gravel Washability)

The washability of gravel serves as a critical indicator of its susceptibility to washing. Generally, the higher the clay content in the gravel, the more challenging it becomes to wash. On the Oworam concession, the presence of a sandy clay matrix with minimal clay content renders it free washing.

Recognizing the tendency of clay to entrap gold particles, a proactive approach is taken to prevent such entrapment. A clay blaster, an integral component of the washing plant, will be employed. This specialized equipment ensures thorough washing and subsequent separation of the ore into its constituent minerals. To enhance the washing process, lifter bars will be incorporated into the Trommel/Scrubber, generating agitation and a cascading effect.

Efficient washing is further guaranteed by the application of a high volume of spray water in a counterflow manner to the screen surface. The residence time within the Trommel ensures the complete liberation of particles, setting the stage for the ensuing gravity separation process. This multifaceted approach underscores the commitment to effective gravel washing, a pivotal step in optimizing gold recovery.

6.2 PLANT DESIGN

The design of alluvial gold treatment plant must take many factors into consideration in order to achieve an optimum recovery of gold. Among the factors taken into consideration are:

- Characteristics of ore to be processed.
- Size distribution of ore and gold particles. Fractions of the gravel reporting to the various sections of the plant.

- Volume of ore/gravel to be processed per day to recover gold to maintain economic viability for the project.
- The effective maximum and minimum sizes of the liberated gold particles and their distribution in the ore.
- Optimum operating conditions of the gravity separating equipment.
- Infrastructure requirement
- Robustness of the final design regarding operating conditions.

6.2.1 Selection of Processing Plant

The gravel and gold within the Oworam concession exhibit distinct characteristics, including the presence of free washing and normal washing gravel, as well as a gold particle size spectrum ranging from fine to coarse grains. Based on these specific attributes, the chosen method for treatment involves the utilization of sluice boxes, complemented by further concentration through a 7.5" Knelson Concentrator.

The treatment process begins with the initial washing and screening of the gravel before directing it to the sluice boxes for gravity concentration. Sluice boxes stand out as a prevalent gravity separation device in the industry, valued for their simplicity and effectiveness in heavy mineral recovery. They prove efficient in recovering gold particles ranging from 25mm to 100 microns. However, the efficacy of gold recovery varies based on operational practices, with well-operated sluices achieving recovery rates between 80 to 100%.

The chosen PROLINE sluice box design has been successfully employed in similar projects with comparable gravel compositions within the country. Noteworthy advantages of the PROLINE design include:

- Fast Waste Flush: Efficient removal of sands and gravels.
- High Retention: Excellent retention of all heavier minerals.
- Variable Flow Adjustment: Allows for adjustment between fast and slow waterways.
- Water Stop Feature: Facilitates clean-up or quick assessment of current gold levels in the sluice.
- Selective Capture Zones: Captures coarse gold in lower flow zones, while fine gold recovery is enhanced by adjusting the flow rate.

The PROLINE design operates by creating a continuously lowering water pressure zone, enabling the sequential settling of gold particles of varying sizes and weights at different intervals. This approach ensures a constant capture system throughout the entire length of the sluice box, optimizing the recovery of gold from the diverse particle sizes present in the gravel.

6.2.2 Processing Plant Flow Sheet

The flow sheet for the Oworam Project has been designed to ensure the optimum recovery of gold from the treatment plant. This comprehensive design incorporates insights gained from on-site gravel assessments, logistical considerations in the operational area, and valuable input from seasoned professionals with extensive practical experience in plant design. The culmination of these factors ensures a robust flow sheet aligned with the specific characteristics of the ore.

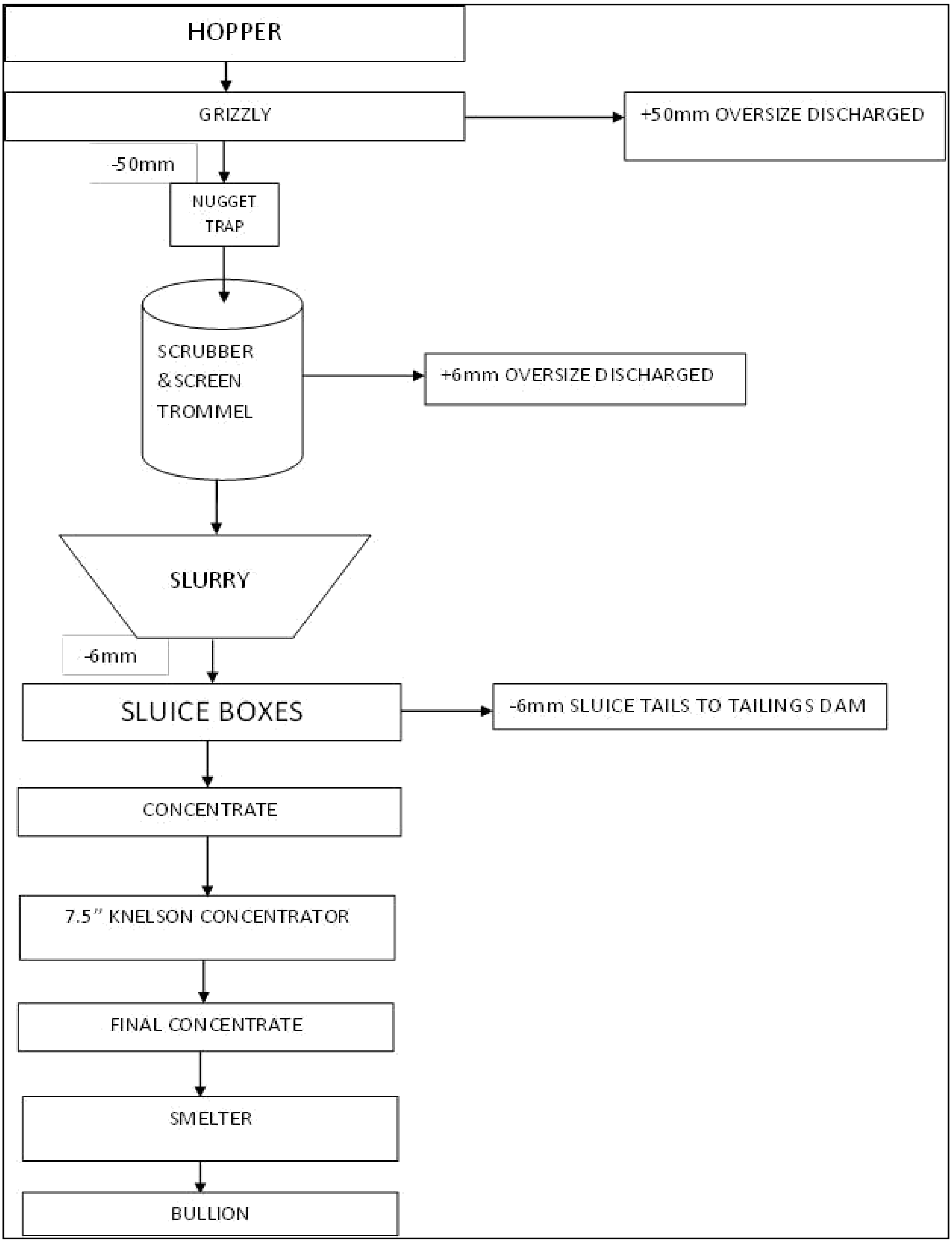

Figure 6.1: Proposed Plant Flow Sheet

6.2.3 Processing Plant Flow Sheet Description

The intricately designed flow sheet for the Oworam Project Processing Plant, illustrated in the Figure above, outlines a systematic approach for optimal gold recovery. The plant's ore intake involves direct dumping from 22.5m³ trucks at the mining face into the feed hopper or, alternatively, feeding from a stockpile using a Front-End-Loader. To minimize truck idle time, an ore stockpile located close to the feed hopper ensures a seamless operational flow.

To regulate the ore flow onto the grizzly and prevent uncontrolled spillage, chains are integrated into the hopper. An 8” giant water monitor facilitates an even feed rate of 120m^3/hr. onto the grizzly, with water from the monitor serving as added moisture at a rate of 750 gal/min. Positioned at the grizzly's edge, a clay blaster tackles clayey ore. The +50mm fraction is conveyed to an oversize stockpile, while the undersize -50mm progresses through the scrubbing trommel and into the screen trommel.

A nugget trap strategically placed at the trommel entrance captures gold nuggets. The +6mm fraction from the screen trommel is directed to a conveyor and subsequently discharged to the oversize stockpile. The +50mm and +6mm oversize material are stored for transport back to the mined-out pits on return trips of trucks. The -6mm fraction (undersize of the screen trommel) gravitates into a slurry tank, flowing through 18” pipes to three sluice box units.

From the 18” pipe, the material is divided into three 14” pipes, each leading to a sluice box. Control valves on the 14” pipes enable selective shut-off during sluice box cleaning intervals (twice per shift, i.e., every 4 hours). Sluice tails follow gravity to a three-stage settling pond. The concentrate extracted from the sluice boxes undergoes an upgrade process through a 7.5” Falcon Concentrator. Subsequently, the concentrated material is smelted to produce gold bars. Power, lighting, and other energy requirements for the operations are supplied by a 550 KVA diesel generator.

6.2.4 Process Water

The treatment plant's process water requirements will be sourced from the four-stage tailings settling ponds post-decanting, ensuring an eco-friendly approach with minimal discharge into the environment.

The initial supply of process water will be facilitated by an 8” diesel pump drawing from an off-channel reservoir, strategically constructed from one of the tributaries within the nearby Pamu river drainage system, secured with the necessary permits from the Water Resources Commission. Subsequent process water needs will predominantly be met by the settling pond. Water loss is anticipated due to evaporation and spillage at the plant. Supplemental makeup water will be drawn from the nearby river tributary/streams as required.

The tailings and recycle ponds will be ingeniously constructed from the initial excavation of mining pits, serving as reservoirs for slurry tailings, pit water, and runoff during mining operations. This sustainable water management approach aligns with the best environmental practices, ensuring responsible use and conservation of water resources.

6.2.5 Tailings Management System

The processing plant at the Oworam Project will generate two types of waste: oversize material (derived from the grizzly and screen trommel) and tailings from the sluice boxes. Oversize material will be conveyed and stacked for reclaiming mined-out pits and constructing haul roads. The tailings from the sluice boxes will undergo gravity discharge into a three-stage settling pond system, a design integral to the initial excavation of mining pits. The pond walls will be constructed with sufficient width to serve as functional haul roads.

The three-stage settling pond arrangement is meticulously designed to facilitate the settling of coarse particles in the first pond, where coarse sand pebbles will settle near the sluice discharge area. This material will be periodically excavated for reclamation purposes. The second pond, connected via a weir, allows for continued settling of suspended particles, ensuring fines settle during the gravity flow through the arrangement. Clean water, separated from suspended particles, will flow to the third pond through another weir.

The third pond will be interconnected with the off-channel reservoir. Utilizing gravity, clean water from the third pond will be directed to the reservoir. Subsequently, this water will be pumped back to the treatment plant for efficient reuse, emphasizing a sustainable approach to water resource management within the project.

6.3 INFRASTRUCTURE

The Oworam township has not seen much infrastructure in terms of schools, hospitals, tarred roads, police station etc. but some fundamental facilities including potable drinking water, electricity, and good mobile communication and internet services are available. During the construction phase of the mining site, essential physical structures pivotal for the seamless implementation of the project will be established. The key facilities to be erected on-site include: Operating plant (Washing plant site with equipment)

1. Operating Plant (Washing Plant Site with Equipment)
2. Maintenance Workshop and Warehouse
3. Fuel Storage Facilities
4. Water Storage Facility
5. Office Accommodation
6. Washroom and Toilet Facilities
7. Changing Room
8. Canteen
9. Site Clinic

6.3.1 Plant Area Buildings

The warehouse will be built of blocks and used for the storage of spare parts, lubricants, tagged instruments, paints and general supplies that will be needed for welding works, mechanical, electrical works, civil works etc. It will also store supply equipment such as generators, water pumps, compressors, welding machines, conveyor parts etc. Flammable materials such as lubricants, thinner and paints will be stored in accordance with their environmental and Material Safety Data Sheet (MSDS) information. As a safety measure, no hazardous material may come to the project site without the Company securing MSDS information and ensuring that the required storage and handling techniques are adhered to.

The facility will have an office complex which consists of offices for administrative work, a washroom, workers' canteen and first aid facilities for workers.

6.3.2 Energy Supply

a. Power Supply Requirements: The power supply for the mining operations will be obtained from the National Grid for which the necessary arrangements for the connection will be made with the Electricity Company of Ghana. The main consumer of electric power at the mine site will be the gravel treatment plant. Other power outlets will be offices, stores, security lights at strategic places like the open pits, fuel and lubricant storage depots and the main gate to the site.

b. Power Supply Option: Apart from the high- tension power supply to the mine site, stand-by generator sets (2) will be procured and installed at the site for use during external power failures. Power consumption specifications are as follows:
Main High-Tension Power = 340 KW
Maximum Operating Power = 280 KW
Average operating Power = 240KW

c. Slimes Disposal: A power generating set will be provided for the pumping of slimes for safe disposal at constructed dam sites away from water bodies. The generating set will be rated at 200 KVA.
Connected power supply = 130 KW
Average Operating Power = 110 KW

d. Lighting of Mine Area: Generator rating = 7.9 KVA
e.
f. Dewatering of Mine Pits: Pump engine rating = 11 KW

6.3.3 Fuel Storage Facilities

Diesel fuel will be the main fuel to be consumed on site for fueling the standby generator and other trucks. Diesel storage tank will be provided on site. The fuel storage area will be constructed with concrete base to prevent leakage into the ground in the event of a spill during fueling.

6.3.4 Water Supply

a. Operational Water Supply: Water supply to the treatment plant will be obtained from an intake point located on the banks of the streams upon attaining water use permit from the Water Resources Commission in accordance with the laws of Ghana. A mobile generator will be placed at the intake point for the pumping of water into the settling ponds from where other pumps will work to deliver water to the treatment plant for gravel washing. Dust control on accesses, haulage roads and aggregate areas will be catered for by pumping water from the streams into tankers for suppression purposes. The pumps are rated at 200KVA.
Connected power supply = 130 KW
Average Operating Power = 110 KW

b. Raw Water Supply: The streams and rivers within the concession will be the main source of raw water supply for the gravel washing plant of the proposed Ignite Machinery Rentals and Mining Ltd project.

c. Potable Water: Generally, Ignite Machinery Rentals and Mining Ltd will source potable water from Ghana water company into polytanks for use at the site. Additional boreholes will be constructed on the mining site as alternative source of water to the site office buildings for drinking and cooking in the canteen.

d. Sewerage System: Plumbing works will be done to ensure efficient discharge of sewerage from the washrooms to the manholes to be constructed at the site. Waste water from the workers' canteen and dirty oil and lubricants will appropriately be discharged through drains into a special container for collection and disposal in an environmentally safe manner. All these measures will be taken to prevent the pollution of water bodies in the area.

6.3.5 Tailings Storage Facility (TSF)

Provision has been made for the construction of a Tailings Storage Facility at the site. Waste gravel with different sized material will be stored for future uses in back-filling the mined-out pits. Fine tailings will also be stockpiled separately to be used in the future land reclamation programme. The TSF type will be shallow open pits to be constructed at the mine site. The tailings collected from the processing plant will be stockpiled here for use.

6.3.6 Communications

The company will make available a two-way radio system for the mining, plant and security departments. Other essential departments will also be provided with the two-way radios. General communication will be via cellular telephone to all staff. Vodafone, MTN, Airtel-Tigo are available in the area. Credits for cellular phones will be provided for management staff, supervisors and other essential staff. Modems of the various mobile phone companies will also be used for internet connections.

6.3.7 Accommodation

The Company plans to construct modest houses to provide residential accommodation for and local senior staff of the mining project. Since the Company will largely employ its labour force from the catchment area, it will assist the junior workers to rent accommodation from the nearby towns and villages to save costs. Transportation arrangement will be made in such instances.

6.3.8 Security

The security of the mining operations will be ensured through putting in place adequate safety and protective measures. A security company with trained personnel will be engaged to oversee the security arrangements of the mining activities. A security post will be constructed at the main entrance to the mine site and manned for 24 hours on rotational basis. The washing plant and office block areas will be protected with a wire mesh fence. The key places at the site that will be kept under security surveillance always include: the gold room and the open pits.

CHAPTER 7

7.0 HUMAN RESOURCE REQUIREMENTS

The human requirement for this project is as follows in Table below. Most of the employees, apart from the key technical personnel, will be sourced from the local communities. If there are technical personnel from local communities who meet the competence requirements, they will be given priority during employment.

Table 7.1: Human Resource Requirements

Resource Item	Number-Ghanaian	Number-Other National
Managing Director	1	-
Administration and Finance Manager	1	-
Secretary	2	-
Mine manager	1	-
Project Geologist	1	-
Accounts Officer	2	-
Pit Foreman	2	-
Administration Officer	1	-
Metallurgist	1	-
Metallurgist Assistant	1	-
Mechanical Engineer		
Environment, Health & Safety Officer	1	-
Chief Security	6	-
Mine surveyor	1	-
Transport Officer	1	-
Electrical Officer	1	-
Plant Labour	2	-
Operators	6	-
Auto Technician	2	-
Cook	2	-
Mining Labour	20	-
TOTAL	55	

ORGANIZATIONAL STRUCTURE FOR IGNITE MACHINERY RENTALS AND MINING LTD

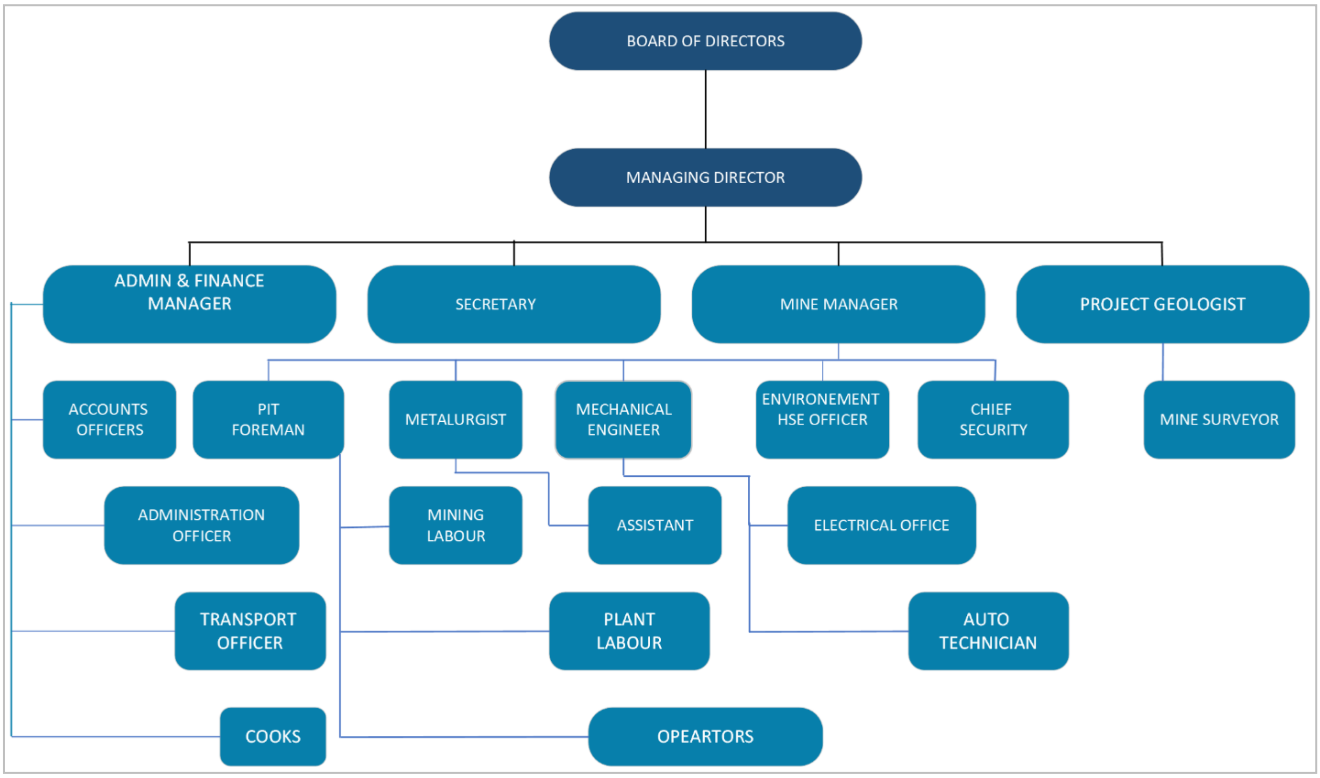

Figure 7.1: Organizational Structure

CHAPTER 8

8.0 ENVIRONMENT

An Environmental Impact Assessment (EIA) for developments, projects, or undertakings has been a requirement in Ghana since 1989. In June 1995, the Ghana Environmental Protection Agency (EPA) established new procedures for EIAs involving gradual phases depending upon the nature, complexity, and location of the undertaking (Ghana Environmental Impact Assessment Procedures, 1995). Between 1995 and 1999, the EPA reviewed and revised these procedures. In June 1999, the revised procedures were adopted and passed by Parliament as Legislative Instrument 1652 Environmental Assessment Regulations (L.I. 1652). These procedures require that an Environmental Impact Statement (EIS) be submitted to the EPA for review and be approved to obtain an Environmental Permit, which allows a project to proceed on environmental grounds.

8.1 Summary of the Baseline Information

The topography of the district is characterized by scattered rock-outcrops and upland slopes with relatively undulating lowlands with gentle slopes ranging from 1° to 5° gradient. The concession area is generally low lying characterized as a flood plain.

The project area is drained by the Red Volta and its tributaries.

8.2 Air Quality

The Project area is entirely rural and ambient air quality is good. Particulate pollution from the seasonal dust laden Harmattan winds occurs principally during the three months of the dry season, from December to February. During this period, it is not exceptional to record a background level of total inhalable dust exceeding the maximum 24 hours peak of 500 mg/m³ quoted by the World Health Organization (WHO).

The project area is entirely rural with relatively good background air quality. Impacts prior to mining was primarily related to smoke from slash/burn agricultural practices, charcoal manufacturing, and fires set naturally or by other human-related activities such as hunting. Background dust impacts were primarily related to vehicular traffic along the mostly unpaved road network within the project area and the dust laden harmattan winds occurring annually from December through February.

8.3 Surface Water

Generally, surface water will be diverted around areas of disturbance to avoid impacting water quality. Surface water flows into the Red Volta basin running through the center of the concession, proximal sub-basins are impacted due to the presence of the tailing's storage facility and water storage facility.

8.4 Flora and Fauna

The vegetation is Sudan Savanna consisting of short drought and fire-resistant deciduous trees interspersed with open savanna grassland. Grass is very sparse and in most areas the land is bare.

Mining activities throughout the life of mine, from exploration and development to post-closure, impact social and environmental systems, including ecosystem services (ES) (Neves et al., 2016) which contribute to human wellbeing (Crossman et al., 2013; MEA, 2005).

As a result, the company will assess, analyse and manage threats to ecosystem included in mining industry performance standards (ICMM, 2020).

8.5 Dust Generation

Vehicular movement on gravel roads may lead to generation of dust within the working communities. Frequent exposure to dust may result in lung related disease. The company will ensure that its vehicles are driven within speed limits in the host communities. During drilling, dust suppression equipment will be used on rigs to minimize exposure of workers to dust. This will be augmented with respiratory protection to be worn as Personal Protective Equipment (PPE).

8.6 Noise

Drilling sites are noisy places, as the use of rigs and heavy machinery generate a lot of noise with potential for hearing damage. To protect workers against noise, the company will evaluate working conditions and noise exposure through risk assessments. Avoiding and reducing exposure can be achieved by applying engineering controls at the noise source or along the noise path to reduce exposures. Regular maintenance of vehicles and machines is also essential to reducing noise. The company will ensure proper use of personal hearing protection amongst noise-exposed employees, while providing necessary health and safety training and maintaining up-to-date health surveillance records.

8.7 UV Exposure

Exploration is often carried out under harsh weather conditions. Exploration workers often spend whole days out in the baking hot sun. Thus, understanding the risk of over-exposure to UV (ultraviolet) radiation in sunlight is essential as it can expose workers to risk of skin cancer or cause serious damage to the eyes if protective glasses are not worn. In the short term, overexposure to the sun can cause dehydration, headaches and nausea. Workers will be required to use the right PPEs in order to minimize exposure to sunlight.

8.8 Stakeholder Consultations

The Company will engage with community leaders and sometimes wider communities on the scale and anticipated impacts of planned mining and related activities in the communities in which we operate. We commit to design projects that take into consideration the concerns and aspirations of community members; and to explore opportunities and strategies for maximizing benefits of our projects. Our engagement strategies will include grievance mechanisms to allow for peaceful resolve of community grievances before they become risks and escalated conflicts.

8.9 Impact Identification

Project-related air quality impacts include non-point source impacts such as dust generation attributable to mining activities (mainly hauling) and emissions from the mining equipment fleet. Point source emissions include emissions from the trash/medical waste incinerator and emissions from the gold room processing facility. Management of air quality impacts are primarily focused on dust control. Measures including extensive road watering, applying rock running courses to road surfaces, and speed limit controls are utilized to minimize dust. An air quality monitoring program is in place to evaluate respirable dust (PM10) and dust accumulation. Source testing of the point source emissions will be conducted periodically.

8.10 Corporate Social Responsibility Programme

In addition to doing its core business of mining, the company is committed to demonstrating social responsibility towards the communities that host our projects. The scope of our citizenship moves beyond traditional philanthropy to include engagements and critical social interventions that will create shared value for us and the local communities. These will include provision of educational and health facilities, potable water supply, etc.

Consultation with stakeholder communities, traditional and elected leaders, and regulators forms an integral component of the project environmental and socioeconomic impact assessment process.

Stakeholder consultation follows a typical pattern reflecting the stages of evolution of the project knowledge and permitting. Initial consultations will occur to sensitize the community on the project, and gain relevant information for the documentation for the Project Environmental Scoping Report.

In addition to compliance to regulatory requirements, the company will design and commit to a robust Corporate Social Responsibility (CSR) program which cascades across the project life up to mine closure. At the core of the program is our belief to be a responsible mining operator in our host communities by innovating and demonstrating responsibility for enhancing social well-being, protecting the environment, whilst maintaining financial viability of our operations. The CSR program comprises of four key activities, namely; Stakeholder engagement; Employee Engagement; Community development; and Environmental stewardship. Each of these activities are mutually-reinforcing and will be planned and implemented in a way that they extend credible development opportunities for local communities.

8.11 Community Development

The company believes in creating shared value with local communities through successful mine operations. As part of this, the company will work with the District Assembly to prioritize and invest in areas of Education and Skills development, Health infrastructure, Local economic development, as well as Water and Sanitation facilities. In particular, the skills development module will be aligned with the labor requirements of the project to ensure that youth are prioritized for employment along the project life. Added to the operational requirements, the initiatives under the each of the prioritized modules above must align with the medium-term development plan of the District Assembly as a condition of company’s support. Ultimately, our investment will help the capacity of the district assembly to deliver on its development mandate. The community development activities that will be considered include the following:

Table 8.1: Priority Areas for Ignite Machinery Rentals and Mining Ltd in CSR Activities

Priority sector	Potential activities
Education and skills development	Scholarships and bursaries for tertiary and technical education. Support for technical education will target youth interested in developing their skills in craft and artisanship e.g. dump truck operators, auto-mechanics, welders, carpenters, hospitality stewards, Integrate skills training with company's operational requirements
Health infrastructure and services	Construction of Wards at Regional/district hospitals Clinics/CHPs Compounds in local communities Donation of medical equipment Short-term staff development training in China for health professionals
Local economic development	Sponsorship for agriculture-based initiatives e.g. National farmers' day Training and technical support for women in agro-processing initiatives Crop-based livelihoods activities linked with livelihood restoration programs for Project-affected persons Local Employment Strategy – 100 percent recruitment of unskilled labour from affected communities
Water and Sanitation	Provision of boreholes and small-town water systems Provision of household latrines
Others	Cash donations for community driven projects and initiatives including

CHAPTER 9

9.0 PROJECT COST ESTIMATION

The cost analysis of mining projects base on gold price trends is a key factor in making a decisive decision of the return on investment of the project. Mining is a very capital-intensive venture due to advance methodology for extraction and its environmental consequences. The proposed Oworam commercial mining project is therefore expected to have cost elements. This includes Capital Expenditure and Operating Costs. At this time, the company will be arranging equity and debt funding for the proposed mine. All financial projections were done using an exchange rate of $1 = GH₵11.00 as of January,2026 an inflation rate of 10%, 5% royalty and a corporate tax of 35%. The following assumptions were also made in the financial inclusion of the project.

- In estimating the capital costs, it is assumed that basic mining equipment is exempted from import duties as outlined in the minerals code of Ghana.
- In estimating the operating costs, it is assumed that both skilled and unskilled labour is available in Ghana. Therefore, local labour will be used for the project and where training is required the company will ensure that it is given to suitable candidates.
- The remuneration for employees has been determined based on consultation with other mining companies currently operating in Ghana.

9.1 Projected Cost Estimation

The Capital Expenditure covers the machinery, heavy duty trucks and many other items that the company plans to acquire for the implementation of the mining project. The Capital Expenditure is estimated at $51,243,900.00. The estimated costs of the various items are based on price quotations on the Pro forma Invoices obtained from the relevant suppliers and manufacturers. The total Operating Costs of the proposed Oworam project is estimated over the 7-year period of the mine.

Other operational expenses include:

Table 9.1 Additional operating cost

Item	Amount (US $)
Land Compensation (EPA Bond)	10,000,0000.00
Licensing and permits	1,000,000.00
TOTAL	11,000,000.00

The Summary of the Financial Projections of Oworam proposed mining project is as follows:

1. Capital Costs - **US$ 51,243,900.0**
2. Operating Cost for Year 1 - **US$ 9,002,345.00**
3. Additional Operating Cost - **US$11,000,000.00**
4. Mine Life Operating Cost (7 years) is **US$ 67,915,417.00**

The price quotations on which the capital costs are based are likely to be increased at the time that the items are purchased in case there occurs inflation and change in exchange rates.

9.2 Project Revenue Estimation from Gold Sales

Average gold price of gold as of January,2026 = US$ 4,216/oz. conservatively = US$3,000/oz

However, a base of $3,000/oz is used for Ignite Machinery Rentals and Mining Ltd proposed project.

Total ounces of gold in Oworam Concession: 127,965.23oz.

Total Gold sales (Gross Revenue) = 127,965.23 oz. x US$3,000.00 = US$383,895.676.00

Hence expected revenue per year

= $ 383,895,676.00÷ 7yrs

= $ 54,842,239.38 per year

Table 9.2: Income statement for Ignite Machinery Rentals and Mining Ltd

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	127,965.23
Gold Price	US$/oz	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	
Revenue	US$		54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	383,895,690.00
Operating Cost			9,002,345.00	9,003,455.00	9,126,691.00	9,247,722.00	10,100,455.00	10,255,809.00	11,178,940.00	67,915,417.00
Gross Profit			45,839,896.43	45,838,786.43	45,715,550.43	45,594,519.43	44,741,786.43	44,586,432.43	43,663,301.43	315,980,273.00
Royalty (1)	0.05		2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	19,194,784.50
Pre-Tax Profit			43,097,784.36	43,096,674.36	42,973,438.36	42,852,407.36	41,999,674.36	41,844,320.36	40,921,189.36	296,785,488.50
Corporate Tax	0.35		15,084,224.53	15,083,836.03	15,040,703.43	14,998,342.58	14,699,886.03	14,645,512.13	14,322,416.28	103,874,920.98
Net Profit			28,013,559.83	28,012,838.33	27,932,734.93	27,854,064.78	27,299,788.33	27,198,808.23	26,598,773.08	192,910,567.53
Cumulative Earnings			28,013,559.83	56,026,398.16	83,959,133.10	111,813,197.88	139,112,986.21	166,311,794.44	192,910,567.53	

Table 9.3: Cash flow of proposed Ignite Machinery Rentals and Mining Ltd

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	127,965.23
Gold Price	US$/oz	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	
Revenue	US$		54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	383,895,690.00
Operating Cost			9,002,345.00	9,003,455.00	9,126,691.00	9,247,722.00	10,100,455.00	10,255,809.00	11,178,940.00	67,915,417.00
Gross Profit			45,839,896.43	45,838,786.43	45,715,550.43	45,594,519.43	44,741,786.43	44,586,432.43	43,663,301.43	315,980,273.00
Royalty	5%		2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	2,742,112.07	19,194,784.50
Pre-Tax Profit			43,097,784.36	43,096,674.36	42,973,438.36	42,852,407.36	41,999,674.36	41,844,320.36	40,921,189.36	296,785,488.50
Corporate Tax	35%		15,084,224.53	15,083,836.03	15,040,703.43	14,998,342.58	14,699,886.03	14,645,512.13	14,322,416.28	103,874,920.98
Net Profit			28,013,559.83	28,012,838.33	27,932,734.93	27,854,064.78	27,299,788.33	27,198,808.23	26,598,773.08	192,910,567.53
Cumulative Earnings			28,013,559.83	56,026,398.16	83,959,133.10	111,813,197.88	139,112,986.21	166,311,794.44	192,910,567.53	778,147,637.15
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	27,341,596.00
Capex (Project Cost)		- 51,243,900.00								-
Net Cash Flow		- 51,243,900.00	28,322,957.83	28,399,936.33	28,519,527.93	28,652,409.78	28,193,572.33	28,102,677.23	27,901,442.08	146,848,623.53
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 51,243,900.00	25,748,400.97	23,469,707.38	21,426,721.34	19,569,595.88	17,505,389.06	15,863,961.30	14,319,020.08	86,658,896.00
IRR		0.52								
NPV		86,658,896.00								

Table 9.4: Sensitivity Analysis (OPEX > 10%)

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	127,965.23
Gold Price	US$/oz	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	
Revenue	US$		54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	54,842,241.43	383,895,690.00
Operating Cost			9,902,579.50	9,903,800.50	10,039,360.10	10,172,494.20	11,110,500.50	11,281,389.90	12,296,834.00	74,706,958.70
Gross Profit			44,939,661.93	44,938,440.93	44,802,881.33	44,669,747.23	43,731,740.93	43,560,851.53	42,545,407.43	309,188,731.30
Royalty	5%		2,246,983.10	2,246,922.05	2,240,144.07	2,233,487.36	2,186,587.05	2,178,042.58	2,127,270.37	15,459,436.57
Pre-Tax Profit			42,692,678.83	42,691,518.88	42,562,737.26	42,436,259.87	41,545,153.88	41,382,808.95	40,418,137.06	293,729,294.74
Corporate Tax	35%		14,942,437.59	14,942,031.61	14,896,958.04	14,852,690.95	14,540,803.86	14,483,983.13	14,146,347.97	102,805,253.16
Net Profit			27,750,241.24	27,749,487.27	27,665,779.22	27,583,568.91	27,004,350.02	26,898,825.82	26,271,789.09	190,924,041.58
Cumulative Earnings			27,750,241.24	55,499,728.51	83,165,507.73	110,749,076.65	137,753,426.67	164,652,252.49	190,924,041.58	770,494,274.88
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	5,181,956.00
Capex (Project Cost)		- 51,243,900.00								-
Net Cash Flow		- 51,243,900.00	28,059,639.24	28,136,585.27	28,252,572.22	28,381,913.91	27,898,134.02	27,802,694.82	27,574,458.09	144,862,097.58
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 51,243,900.00	25,509,018.03	23,252,074.07	21,226,157.51	19,384,847.20	17,321,951.42	15,694,621.23	14,151,211.89	85,295,981.35
IRR		0.52								
NPV		85,295,981.35								

Table 9.5: Sensitivity Analysis (Revenue < 10%)

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	18,280.75	127,965.23
Gold Price	US$/oz	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	
Revenue	US$		49,358,017.29	49,358,017.29	49,358,017.29	49,358,017.29	49,358,017.29	49,358,017.29	49,358,017.29	345,506,121.00
Operating Cost			9,002,345.00	9,003,455.00	9,126,691.00	9,247,722.00	10,100,455.00	10,255,809.00	11,178,940.00	67,915,417.00
Gross Profit			40,355,672.29	40,354,562.29	40,231,326.29	40,110,295.29	39,257,562.29	39,102,208.29	38,179,077.29	277,590,704.00
Royalty	5%		2,017,783.61	2,017,728.11	2,011,566.31	2,005,514.76	1,962,878.11	1,955,110.41	1,908,953.86	13,879,535.20
Pre-Tax Profit			38,337,888.67	38,336,834.17	38,219,759.97	38,104,780.52	37,294,684.17	37,147,097.87	36,270,123.42	263,711,168.80
Corporate Tax	35%		13,418,261.04	13,417,891.96	13,376,915.99	13,336,673.18	13,053,139.46	13,001,484.26	12,694,543.20	92,298,909.08
Net Profit			24,919,627.64	24,918,942.21	24,842,843.98	24,768,107.34	24,241,544.71	24,145,613.62	23,575,580.22	171,412,259.72
Cumulative Earnings			24,919,627.64	49,838,569.85	74,681,413.83	99,449,521.17	123,691,065.88	147,836,679.50	171,412,259.72	691,829,137.58
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	27,341,596.00
Capex (Project Cost)		- 51,243,900.00								-
Net Cash Flow		- 51,243,900.00	25,229,025.64	25,306,040.21	25,429,636.98	25,566,452.34	25,135,328.71	25,049,482.62	24,878,249.22	125,350,315.72
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 51,243,900.00	22,935,707.21	20,912,911.63	19,105,286.26	17,461,886.95	15,606,525.60	14,140,432.94	12,767,517.50	71,686,368.08
IRR		0.46								
NPV		71,686,368.08								

Table 9.6: Sensitivity Analysis Ignite Machinery Rentals and Mining Ltd

Scenario	NPV $'M'	IRR
Base Case	86,658,896	52%
Operational Cost Increase 10%	85,295,981	52%
Revenue Decrease 10%	71,686,368	46%

CHAPTER 10

10.0 MINE CLOSURE AND REHABILITATION

10.1 Introduction

The Company is required by legislation to implement a land rehabilitation program and prepare a decommissioning plan as part of its overall environmental management strategy. Mine closure and rehabilitation will be done with strict adherence with all laws and regulatory provisions provided by the institutions that are mandated to oversee its successful implementation.

The EPA requires a Reclamation Bond to be posted as part of any mine permitting process. The bond is required to provide financial security against non-compliance under the approved Closure and Reclamation Plan and is required within six months after the start of operations. Projected reclamation liability for the Oworam Project is approximately US$ 2.25 million.

This compacted amount will be lodged into an escrow account with both the Company and EPA as co-signatories to ensure that the desired reclamation program is achieved.

10.2 General Provisions for Mine Site Rehabilitation and Closure

As has been stated by the company, mining and reclamation of mined out pits shall be done concurrently. The company will liaise with the inspectorate division on monthly update of mining and reclamation progress. The inspectorate due to scheduled visits has the authority to crosscheck and satisfy all required provisions required to be fulfilled with regards to mine closure Ignite Machinery Rentals and Mining Ltd will submit a mine closure plan to the Inspectorate Division of EPA for approval before closure initiation kickstart. This plan will include but not limited to surveyed mine layout plan that shows all final workings as prescribed by L.I. 2182, regulation 67.

Furthermore, the Company is of the view that a successful closure could be achieved through progressive closure commitment and planning and in constant consultation with the inspectorate division to be in a position to satisfy the Chief Inspector of Mines that each source of potential pollution and component of the mining project that is to be closed are designed to be stable in the long term.

Mine closure processes consist of several steps including the following:

1. Decommissioning and rehabilitation
2. Relinquishment (issuing of a mine closure certificate)
3. Management of residual and latent risks
4. Mine site reclamation
5. Continuous closure
6. Operating the mine for closure
7. Post closure monitoring and maintenance
8. Engaging stakeholders in the mine closure process

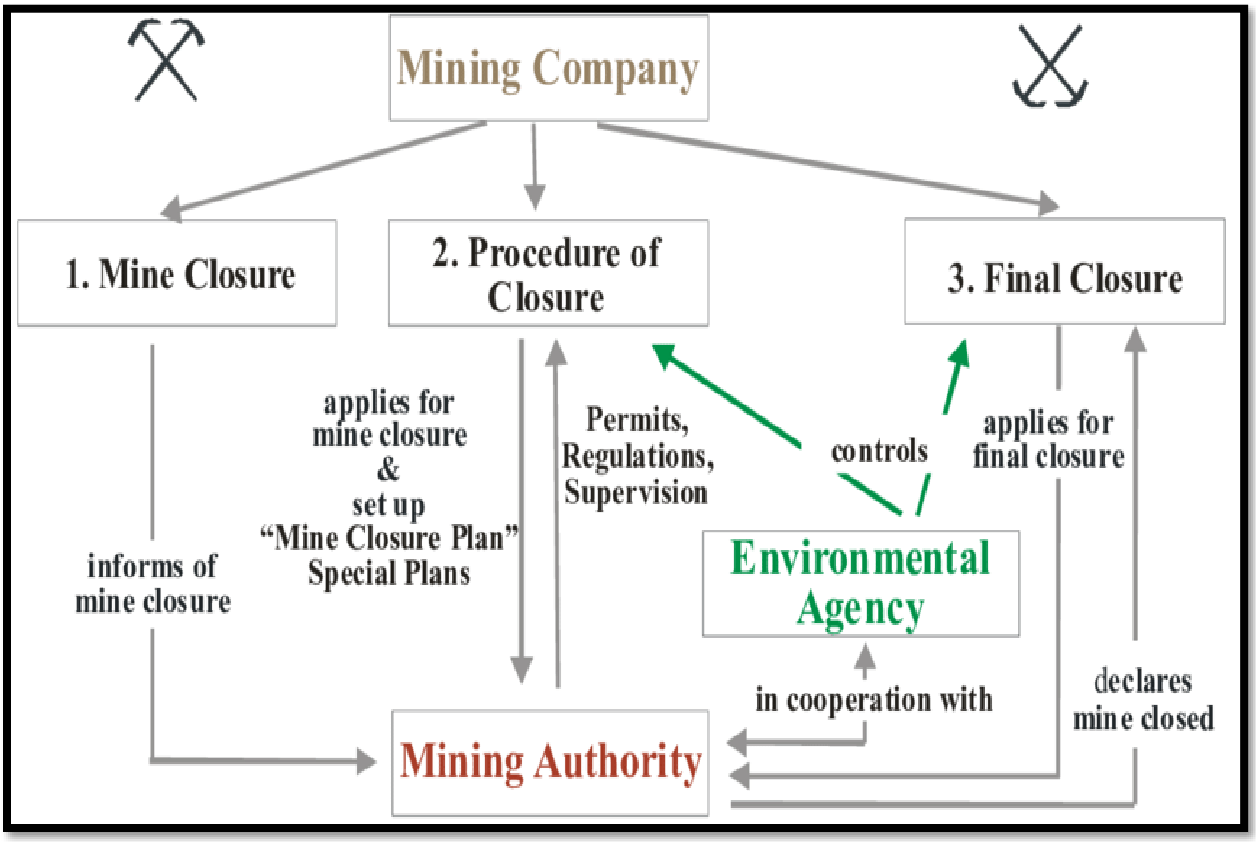

Figure 10.1: Mine Closure Processes and Responsibilities.

10.3 Reclamation Processes

As already stated, the company is committed to rehabilitation and Mine Closure to ensure that the environment is restored to its primary state. The waterbodies, the air, the trees, cash crops, infrastructure sited on arable lands, dams, tailings dams, ponds and sundries shall all be restored, and the report submitted to the Mine Inspectorate division not later than sixty (60) days before abandonment. Maps for areas that have been mined, partially backfilled, reclaimed, and revegetated shall be prepared monthly to know the deficit and the areas to zero-in on during the mine closure and folding periods. The company is going to ensure that total mine-out figures and maps are well documented, and budgets prepared for the environmental department to embark on seamless backfilling and revegetation during and after the mine life. Consultation with landowners will help to define a map out plan as to how their respective lands shall be put to use. This will inform the environmental unit as the end-use of their land before commencing the restoration.

The site infrastructure including dams, ponds, offices, routes, bridges, culverts, workshops, processing plant; shall be restored and some be marked out as assets for the community with the agreed compensation given the landowners for the loss of their lands.

Ignite Machinery Rentals and Mining Ltd shall comply with all the Commissions regulations concerning closure of processing plants, removal of hazardous plants and closure of ancillary facilities if we are given the Lease. It is, however, worth noting that no chemical or chemical precipitates would be used in the gold recovery process.

10.4 Reclamation and Restoration Plan

The Company has drawn up a program to reclaim all areas that will be affected by the mining project, in accordance with Ghana's mining and environmental regulations. The reclamation program is made up of the following components:

- Dewatering
- Backfilling: which involves the use of waste gravel and overburden to re-fill mined out areas
- Top-soiling: Spread of topsoil in preparation for re-vegetation.
- Re-vegetation: The planting of selected tree species over areas that have been back-filled and top soiled.
- Monitoring

The reclamation process will be done in constant consultation of authorizing institutions such as the Minerals Commission, Mines Department, Environmental Protection Agency (EPA) and Water Resources Commission.

Dewatering: It is common knowledge that pits that have been left to the mercy of the weather over time will be filled with either rain or ground water. These pits filled with water will be pumped out under controlled conditions before backfilling begins. The dewatering will be done by high-capacity hydraulic pumps.

Gravel Backfilling: The first stage of the backfilling phase will be to backfill with the washed-out gravel (from the mining activities) from the base and then compacted. This is to try and follow the sequence of the units before they were excavated. The processing plant tailings, which will be made up of washed boulders, cobbles, pebbles and sand will be constantly conveyed to the mined-out areas for backfilling purposes. This operation will be carried out *simultaneously* with mining. By that design, the tipper trucks, which carry the ore to the plant, will also carry washed gravels or tailings on their return to the mining face. This means that suitable machines will have to be placed at the tailings site to load the tipper trucks on their return to the mining face.

The duty of the tailings handling machine will be more than just loading trucks; the machine will be responsible for de-silting the sandpit as well. Hence, an excavator with a somewhat long boom and light to medium weight will be ideal for the job.

Overburden Backfilling: This is the zone that overlies between the gravel layer and the topsoil. For agricultural reasons, this zone usually separates the life supporting topsoil from the subsurface. This zone which may be composed of high percentage of clay will be filled after the gravel filling is completed.

Top Soil Backfilling: The final stage of the backfilling stage will be to fill and spread the top soil from the "topsoil stockpile". The top soil will be filled after the filling of the overburden.

Re-vegetation: Re-vegetation would begin with the planting of leguminous plant after which local and economic (mixed species of ofram, mahogany, teak and manzonia, all endangered wood species) would be planted over the entire area. Plant species common in the area would be planted to enable the land gain its fertility.

Monitoring: Monitoring of post restoration will be done by respective stakeholders mandated by laws of the industry. In the event of any unsatisfactory result from the reclamation and rehabilitation, Time Concept will be committed to finding lasting solution to such undesirable situation. Much help will be needed from the Agric extension officers stationed at the district assemblies (during the re-vegetation stage); such officers will be part and oversee the final stages of reclamation as well as the post reclamation in monitoring and evaluation.

Figure 10.2: Mine Closure Overview.

CHAPTER 11

11.0 CONCLUSION

The management of Ignite Machinery Rentals and Mining Ltd, Oworam concession has followed detailed mineral exploration procedure in evaluating its gold deposit for the Oworam alluvial gold concession. The company employed bankable due diligence procedures to the mineral resource estimation of its 32.57 sqkm property in the West Akim municipal of the Eastern Region. The exploration team have been able to acquire sufficient data on which the economic assessment and gold delineation of the concession has been based.

The evaluation of the exploration data obtained shows that alluvial gold gravel beds occur in the valleys and old terraces within the concession. The buried gravel beds have an average thickness of 1.60 m and have overburden with average thickness of 1.20 m. The estimated reserve of gold deposits in the gravel beds in the concession is approximately 127,965.23 oz.

The alluvial mineral deposits can conveniently be exploited through open-pit mining and treatment of the un-consolidated ore-bearing gravels with water-driven washing plant having a capacity of 160m^3/h. The company intends to operate one stationary site and haul the deposit from various mining pits simultaneously (concurrently) from the respective delineated gold zones. The projected mine life is estimated to be approximately 7 years.

The total revenues that Ignite Machinery Rentals and Mining Ltd expect to derive from the sales of gold from the Oworam concession is US$ 383,895,675.66 as against the initial investment of US$71,246,245 The economic viability of the proposed mining project in Ignite Machinery Rentals and Mining Ltd, Oworam Gold Concession is considered to be positive obtained from cost benefit analysis conducted. This means that the mining of alluvial mineral deposits in the IGNITE MACHINERY RENTALS AND MINING LTD Concession by open pit method could be implemented as it will be an economically profitable venture. The CEO and board of directors of Ignite Machinery Rentals and Mining Ltd therefore applying for a mining lease to enable them to commercialize the alluvial gold mining at Oworam concession by the Hon. minister of Ministry of Lands and Natural Resources through the Minerals Commission.

APPENDICES

Database of Pitting Program Conducted on the Oworam Concession

BH_TYPE	PIT_ID	UTM_EAST	UTM_NORTH	ELEV	MEDIUM	OVERBURDEN THICKNESS(m)	GRAVEL THICKNESS(m)	TOTAL PIT DEPTH	VISIBLE SPECKS	ASSAY (g/m3)	GRID
PIT	OWPT001	768402	656331	56	GRAVEL	0.40	1.60	2.00	1	0.81	AYENSU
PIT	OWPT002	768444	656211	64	GRAVEL	0.6	1.4	2.00	0	0.74	AYENSU
PIT	OWPT003	768410	656166	66	GRAVEL	1.00	1.60	2.60	0	0.53	AYENSU
PIT	OWPT004	768467	656265	72	GRAVEL	1.6	0.9	2.50	1	0.70	AYENSU
PIT	OWPT005	768426	656278	77	GRAVEL	1.00	0.80	1.80	0	0.47	AYENSU
PIT	OWPT006	768412	656233	82	GRAVEL	1.30	0.80	2.10	0	0.58	AYENSU
PIT	OWPT007	768368	656182	87	GRAVEL	1.50	0.90	2.40	1	0.85	AYENSU
PIT	OWPT008	768378	656318	77	GRAVEL	1.60	0.80	2.40	0	0.62	AYENSU
PIT	OWPT009	768412	656477	67	GRAVEL	1.40	0.60	2.00	1	0.76	AYENSU
PIT	OWPT010	768455	656474	72	GRAVEL	1.00	0.70	1.70	0	0.52	AYENSU
PIT	OWPT011	768427	656393	60	GRAVEL	1.30	0.60	1.90	0	0.46	AYENSU
PIT	OWPT012	768448	656559	63	GRAVEL	1.1	0.6	1.70	0	0.39	AYENSU
PIT	OWPT013	768368	656582	66	GRAVEL	1.00	0.70	1.70	0	0.59	AYENSU
PIT	OWPT014	768410	656546	69	GRAVEL	1.3	0.7	2.00	0	0.46	AYENSU
PIT	OWPT015	768402	656429	72	GRAVEL	1.50	0.50	2.00	1	0.64	AYENSU
PIT	OWPT016	768432	656808	75	GRAVEL	0.60	1.50	2.10	0	0.18	AYENSU
PIT	OWPT017	768387	656604	60	GRAVEL	0.70	2.40	3.10	0	0.20	AYENSU
PIT	OWPT018	768311	656637	70	GRAVEL	1.40	1.30	2.70	0	0.19	AYENSU
PIT	OWPT019	768332	656682	63	GRAVEL	1.30	1.50	2.80	2	0.81	AYENSU
PIT	OWPT020	768375	656647	61	GRAVEL	1.50	0.50	2.00	0	0.19	AYENSU
PIT	OWPT021	768455	656614	58	GRAVEL	1.00	1.60	2.60	0	0.20	AYENSU

PIT	OWPT022	768463	656667	65	GRAVEL	1.00	1.40	2.40	0	0.17	AYENSU
PIT	OWPT023	768470	656731	62	GRAVEL	1.30	1.50	2.80	2	1.20	AYENSU
PIT	OWPT024	768486	656795	70	GRAVEL	2.00	0.90	2.90	0	0.19	AYENSU
PIT	OWPT025	768412	656775	60	GRAVEL	1.20	1.60	2.80	0	0.76	AYENSU
PIT	OWPT026	768460	656782	63	GRAVEL	1.30	2.00	3.30	0	0.21	AYENSU
PIT	OWPT027	768441	656729	62	GRAVEL	1.50	2.40	3.90	0	0.20	AYENSU
PIT	OWPT028	768442	656671	61	GRAVEL	2.00	1.30	3.30	0	0.90	AYENSU
PIT	OWPT029	768435	656636	60	GRAVEL	0.60	2.60	3.20	0	0.98	AYENSU
PIT	OWPT030	768388	656661	60	GRAVEL	0.3	2	2.30	1	1.10	AYENSU
PIT	OWPT031	768341	656711	59	GRAVEL	0.2	1.6	1.80	0	0.00	AYENSU
PIT	OWPT032	768280	656641	58	GRAVEL	0.4	1.1	1.50	0	0.00	AYENSU
PIT	OWPT033	768257	655974	58	GRAVEL	1.20	1.80	3.00	1	1.51	AYENSU
PIT	OWPT034	768275	656026	57	GRAVEL	0.60	1.60	2.20	0	1.01	AYENSU
PIT	OWPT035	768328	656029	56	GRAVEL	0.70	1.40	2.10	0	0.00	AYENSU
PIT	OWPT036	768403	656007	56	GRAVEL	0.40	1.20	1.60	2	1.12	AYENSU
PIT	OWPT037	768420	656104	55	GRAVEL	0.60	1.30	1.90	0	1.03	AYENSU
PIT	OWPT038	768383	656098	54	GRAVEL	0.80	1.70	2.50	1	1.46	AYENSU
PIT	OWPT039	768378	656049	53	GRAVEL	1.90	1.70	3.60	0	0.81	AYENSU
PIT	OWPT040	768345	656011	53	GRAVEL	0.50	1.20	1.70	0	0.74	AYENSU
PIT	OWPT041	768247	656071	52	GRAVEL	1.10	1.52	2.62	0	0.96	AYENSU
PIT	OWPT042	768226	655985	68	GRAVEL	1.80	1.40	3.20	0	0.00	AYENSU
PIT	OWPT043	768122	655707	73	GRAVEL	1.60	1.90	3.50	0	0.47	AYENSU
PIT	OWPT044	768112	655448	78	GRAVEL	1.20	2.00	3.20	0	0.00	AYENSU
PIT	OWPT045	768098	655522	83	GRAVEL	1.30	1.90	3.20	0	0.81	AYENSU
PIT	OWPT046	768096	655619	88	GRAVEL	1.50	1.50	3.00	0	0.72	AYENSU

PIT	OWPT047	768188	655742	93	GRAVEL	1.70	1.60	3.30	0	0.00	AYENSU
PIT	OWPT048	768226	655808	86	GRAVEL	1.20	1.70	2.90	1	0.54	AYENSU
PIT	OWPT049	768143	655841	79	GRAVEL	1.1	1.5	2.60	0	0.94	AYENSU
PIT	OWPT050	768150	655898	72	GRAVEL	1.40	1.80	3.20	1	0.63	AYENSU
PIT	OWPT051	768219	655865	65	GRAVEL	1.00	1.90	2.90	1	0.60	AYENSU
PIT	OWPT052	768275	655896	82	GRAVEL	2.20	1.50	3.70	1	0.80	AYENSU
PIT	OWPT053	768241	655923	90	GRAVEL	1.20	1.20	2.40	1	0.90	AYENSU
PIT	OWPT054	768214	655899	88	GRAVEL	1.80	1.10	2.90	0	0.16	AYENSU
PIT	OWPT055	768138	655934	86	GRAVEL	1.40	1.00	2.40	0	0.60	AYENSU
PIT	OWPT056	768137	655803	84	GRAVEL	1.00	1.90	2.90	1	0.70	AYENSU
PIT	OWPT057	768203	655798	82	GRAVEL	1.20	1.60	2.80	0	0.87	AYENSU
PIT	OWPT058	768180	655779	80	GRAVEL	0.50	2.00	2.50	0	0.93	AYENSU
PIT	OWPT059	768105	655749	78	GRAVEL	0.60	2.40	3.00	0	0.91	AYENSU
PIT	OWPT060	768116	655380	76	GRAVEL	0.70	1.30	2.00	2	1.02	AYENSU
PIT	OWPT061	768087	655405	66	GRAVEL	0.90	2.60	3.50	1	0.82	AYENSU
PIT	OWPT062	767937	654942	61	GRAVEL	2.00	1.51	3.51	0	1.22	AYENSU
PIT	OWPT063	767875	654989	65	GRAVEL	2.00	1.00	3.00	1	0.91	AYENSU
PIT	OWPT064	767866	655062	66	GRAVEL	2.40	1.50	3.90	0	0.67	AYENSU
PIT	OWPT065	767797	655287	64	GRAVEL	0.6	2.4	3.00	0	0.72	AYENSU
PIT	OWPT066	767866	655325	75	GRAVEL	0.70	1.30	2.00	0	0.86	AYENSU
PIT	OWPT067	767987	655321	70	GRAVEL	0.90	2.60	3.50	1	0.29	AYENSU
PIT	OWPT068	767795	655326	68	GRAVEL	2.00	1.50	3.50	0	0.82	AYENSU
PIT	OWPT069	767856	655037	70	GRAVEL	1.10	1.80	2.90	1	0.51	AYENSU
PIT	OWPT070	767843	654975	68	GRAVEL	2.00	2.00	4.00	2	1.60	AYENSU
PIT	OWPT071	767910	654929	74	GRAVEL	1.20	1.80	3.00	1	0.61	AYENSU

PIT	OWPT072	767707	655358	75	GRAVEL	1.30	1.50	2.80	1	0.80	AYENSU
PIT	OWPT073	768060	655704	77	GRAVEL	1.50	1.90	3.40	0	0.29	AYENSU
PIT	OWPT074	768017	655715	65	GRAVEL	0.60	2.40	3.00	0	0.10	AYENSU
PIT	OWPT075	768043	655619	72	GRAVEL	0.80	1.70	2.50	1	0.62	AYENSU
PIT	OWPT076	768035	655537	68	GRAVEL	0.90	2.00	2.90	0	0.71	AYENSU
PIT	OWPT077	768065	655453	70	GRAVEL	1.00	1.50	2.50	1	1.00	AYENSU
PIT	OWPT078	767968	655423	68	GRAVEL	1.00	1.50	2.50	1	0.77	AYENSU
PIT	OWPT079	767980	655345	67	GRAVEL	1.10	1.50	2.60	0	0.50	AYENSU
PIT	OWPT080	767850	655361	66	GRAVEL	1.30	2.00	3.30	0	1.00	AYENSU
PIT	OWPT081	767731	655385	72	GRAVEL	1.50	1.60	3.10	0	0.00	AYENSU
PIT	OWPT082	767996	655402	76	GRAVEL	0.30	1.50	1.80	0	0.00	AYENSU
PIT	OWPT083	767774	655093	63	GRAVEL	1.00	1.90	2.90	1	0.78	AYENSU
PIT	OWPT084	767667	655079	68	GRAVEL	1.20	1.60	2.80	0	0.11	AYENSU
PIT	OWPT085	767639	655145	72	GRAVEL	0.50	2.00	2.50	0	0.82	AYENSU
PIT	OWPT086	767631	655242	68	GRAVEL	0.60	2.40	3.00	0	0.07	AYENSU
PIT	OWPT087	767665	655328	65	GRAVEL	0.70	1.30	2.00	1	0.93	AYENSU
PIT	OWPT088	767628	655341	62	GRAVEL	0.90	2.60	3.50	1	0.95	AYENSU
PIT	OWPT089	767604	655258	76	GRAVEL	2.00	0.51	2.51	0	1.15	AYENSU
PIT	OWPT090	767629	655051	67	GRAVEL	2.00	1.50	3.50	1	1.05	AYENSU
PIT	OWPT091	767746	655062	64	GRAVEL	2.20	1.50	3.70	0	0.64	AYENSU
PIT	OWPT092	767662	655119	68	GRAVEL	0.60	2.50	3.10	0	0.00	AYENSU
PIT	OWPT093	767578	655136	66	GRAVEL	0.80	1.70	2.50	1	0.82	AYENSU
PIT	OWPT094	767584	655181	68	GRAVEL	0.90	2.60	3.50	0	0.00	AYENSU